TCS CORPORATE WEBSITE

Architecture Guide

Role / audience	Architecture — Solution architects, technical leads, developers, AI agents
Version	1.0
Status	Draft for review
Owner	Solution Architect (proposed)
Last validated	2026-11-08
Documents	11
Classification	INTERNAL — classification to be confirmed

Generated from the Markdown sources of docs/handover (branch migrated-page-content-edam); do not edit this PDF.

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- Security boundaries
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- Screenshots
- Known limitations
- Open items
- Change history

Architecture Guide

Attribute	Value
Document ID	ARC-000
Audience	Solution architects, technical leads, developers, AI agents
Purpose	Entry point of the architecture package: the system, its delivery streams, the implementation status of every capability and where each architecture view is described
Scope	The TCS Corporate Website on AEM Sites as a Cloud Service with Universal Editor (xwalk) and Edge Delivery Services; the repository sradha-tcs/tcs-corp (main)
Prerequisites	None
Owner	Solution Architect (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	AGENTS.md, scripts/scripts.js, utils/site.js, paths.json, helix-query.yaml, component-models.json
Related documents	Solution architecture, Content architecture, Technical architecture, Integration architecture, Security architecture, Deployment architecture, Performance architecture, Localization architecture, Architecture decisions, Risks and open items
Related AEM paths	/content/tcs-corp/language-masters/en, /content/tcs-corp/language-master/en, /content/tcs-corp/design-block-master/en, /content/tcs-corp/us/en … /content/tcs-corp/kr/ko-kr

1. System context

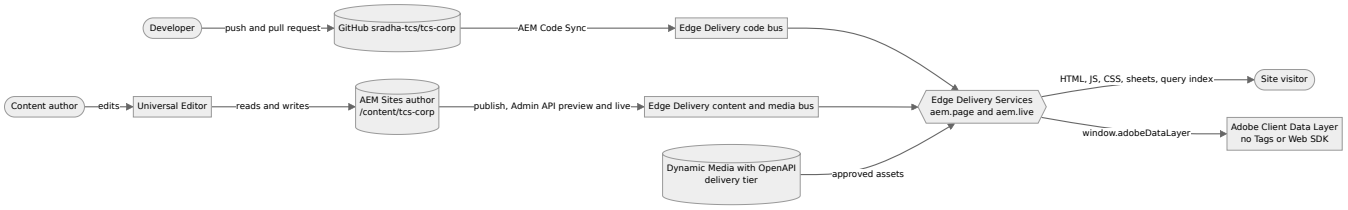


Diagram: system context (source: diagrams/system-context.mmd).

2. Delivery streams

Stream	Source	Delivered by	Environments
Code	GitHub sradha-tcs/tcs-corp (feature branches, main)	AEM Code Sync	<branch>--tcs-corp--sradha-tcs.aem.page / .aem.live

Stream	Source	Delivered by	Environments
Content	AEM Sites author /content/tcs-corp (Universal Editor)	Publish from AEM (technical account) or the Admin API	main--tcs-corp--sradha-tcs.aem.page (preview), .aem.live (live)
Configuration	Site configuration (Config Service: public.json paths, metadata source), fstab.yaml at onboarding only	Admin API /config	all branches
Assets	AEM Assets of the Sites environment (/content/dam/tcs-corp)	Media bus (images published with pages) or Dynamic Media with OpenAPI (approved assets, KB-D-067)	delivery tier of the environment

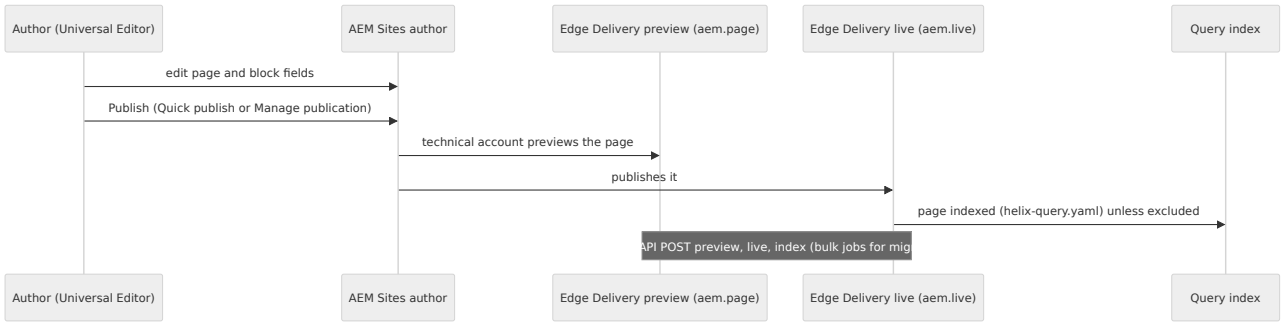


Diagram: authoring to delivery flow (source: diagrams/delivery-flow.mmd).

3. Site roots

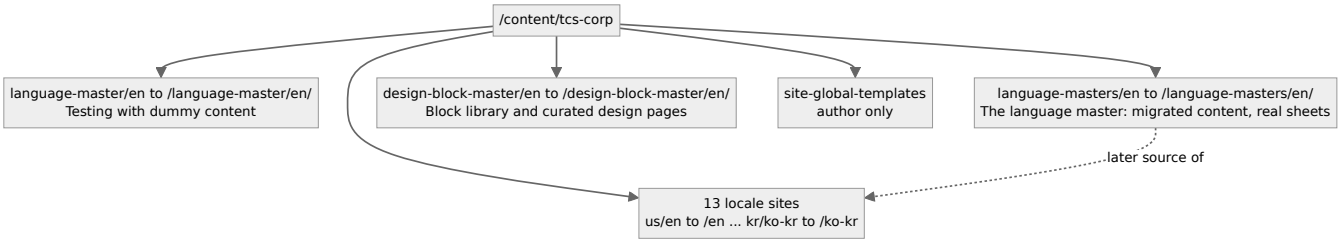


Diagram: site roots under /content/tcs-corp (source: diagrams/site-roots.mmd).

4. Implementation status

Status vocabulary: Implemented, Partially implemented, Configured but not validated, Planned, Not implemented, Blocked, To be confirmed.

Capability	Status	Evidence
Universal Editor authoring (xwalk) of 32 blocks and 5 default-content components	Implemented	component-models.json, component-definition.json, component-filters.json
Edge Delivery Services rendering, three-phase loading, module preloading	Implemented	scripts/scripts.js, scripts/module-graph.js, head.html
13 locale sites by path	Implemented (code and path mapping); content Partially implemented	utils/site.js LOCALES, paths.json
Three review roots /language-master/en, /language-masters/en, /design-block-master/en (KB-D-068)	Implemented	utils/site.js REVIEW_ROOTS, paths.json, site configuration

Capability	Status	Evidence
Query index and listing (List, Filter List, Cards from metadata)	Implemented	helix-query.yaml, utils/query-index.js, blocks/list, blocks/filter-list
Placeholders, taxonomy, data sheets, redirects sheets per locale	Implemented (reading); redirects Not applied by the platform	utils/placeholders.js, utils/taxonomy.js, utils/data.js; redirects need a CDN rule (KB-D-055, on hold)
Adobe Client Data Layer (page, components, interactions)	Implemented	decorators/data-layer-decorator.js, utils/analytics.js
Adobe Tags, Web SDK, Analytics, CJA, Target, consent platform	Not implemented	decision log; AGENTS.md
Dynamic Media with OpenAPI delivery of all assets (KB-D-067)	Partially implemented: rendering Implemented, page rewrite Configured but not validated, asset picker Planned (PR #200 open)	decorators/responsive-image-decorator.js, utils/media.js, aem-sites.sh on the delivery branch
Separate AEM Assets environment (KB-D-064)	Not implemented (not adopted)	decision log KB-D-064
Forms (Form Variations) submission back end	Not implemented (no back end connected)	blocks/form-variations; finding F-01
Personalization	Not implemented	docs: personalization (knowledge base)
CDN redirects / root redirect / to /en/	Planned (on hold)	KB-D-055
Production custom domain and Adobe Managed CDN	To be confirmed	not recorded in the repository
Monitoring and alerting beyond Edge Delivery RUM	To be confirmed	not recorded in the repository

5. Architecture views

- Solution architecture
- Content architecture
- Technical architecture
- Integration architecture
- Security architecture
- Deployment architecture
- Performance architecture
- Localization architecture
- Architecture decisions
- Risks and open items

6. Page rendering lifecycle

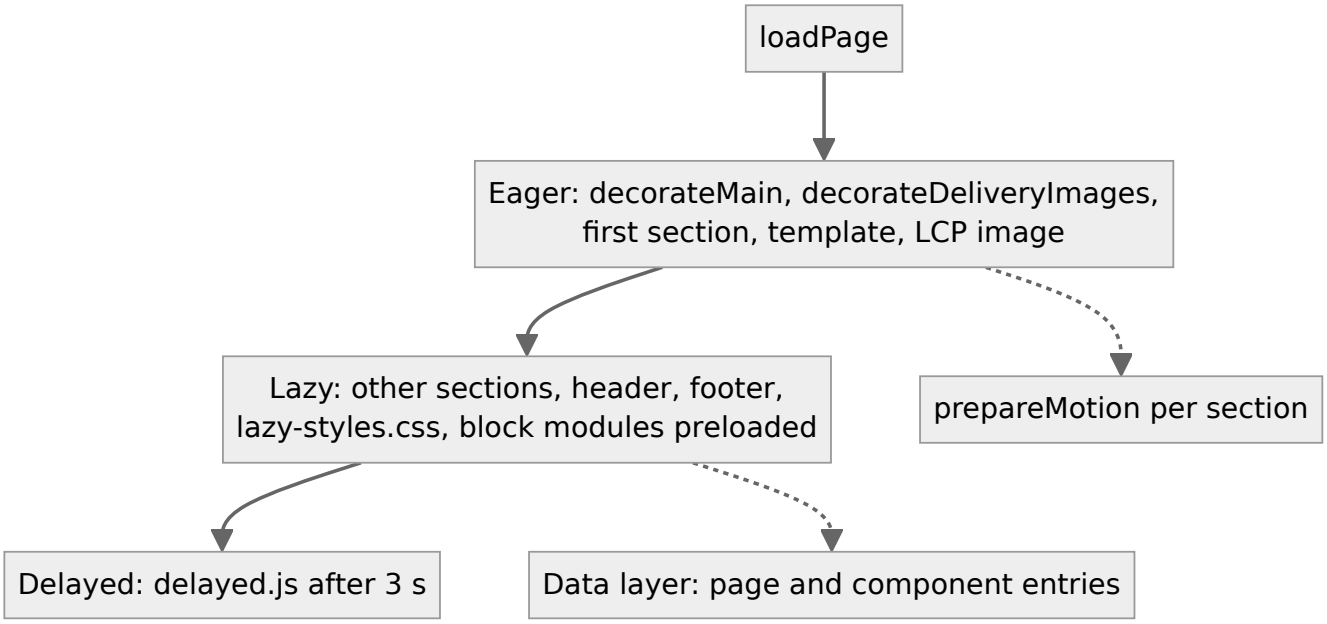


Diagram: three-phase page loading (source: diagrams/page-loading.mmd).

7. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

8. Known limitations

- Production domain, CDN and monitoring are not recorded in the repository (To be confirmed).

9. Open items

- Confirm the production domain and CDN (owner: Solution Architect, to be confirmed).
- Approve or reject the asset picker PR #200 after the Extension Manager set-up.

10. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Architecture Decisions

Attribute	Value
Document ID	ARC-009
Audience	Solution architects, technical leads, developers, AI agents
Purpose	The decision log (KB-D-xxx) as maintained in the code repository knowledge base
Scope	—
Prerequisites	—
Owner	Solution Architect (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/decision-log.md
Related documents	—
Related AEM paths	—

The authoritative decision log is docs/handover/knowledge-base/decision-log.md (code repository); this copy is generated from it.

1. How to Use This Log

- **Status:** Approved (implemented and recorded), Proposed (open, not implemented), Superseded (replaced by a later entry, named in the row). Entries derived by the knowledge base authors, not decided by the project, are listed separately under Recommendations (not approved).
- **Date:** the date of the handover revision or the merge date of the pull request that recorded the decision; "Not recorded" when the source has none.
- **References:** "rev 2.xx" is a row of the handover's Revision History, "F-xx" a finding and "O-xx" an open item of its Part 16, "PR #nn" a merged pull request of `sradha-tcs/tcs-corp`.
- **Adding a decision:** append the next id (`KB-D-0nn`), status Proposed until the user approves it; when it replaces an entry, set the old one to Superseded and name the new id. Never rewrite an Approved entry's history.

2. Site, Content and Configuration

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-001	11 October 2026	Code structure	Blocks repeated buttons, links, headings, images and property readers	Shared behaviour lives in single-purpose modules: `decorators/` build or decorate elements, `utils/` build no UI; decorators may import utils, never the reverse	Per-block implementations (replaced)	Blocks import shared modules; new shared behaviour goes to these folders with their READMEs	PR #23; AGENTS.md "Shared code"; decorators/README.md, utils/README.md	Approved
KB-D-002	11 October 2026	Site data	TCS Custom Properties held site data outside the standard Edge Delivery sources	Site data comes from standard sources: page properties and bulk metadata, placeholders, taxonomy and per-locale data sheets	TCS Custom Properties (replaced)	`utils/metadata.js`, `placeholders.js`, `taxonomy.js`, `data.js`; all UI text from placeholders	PR #24; AGENTS.md "Site structure and site data"	Approved
KB-D-003	11 October 2026	Site structure	13 markets and languages	13 locale sites (`/content/tcs-corp/<country>/<lang>`), each with references, error pages, block library, sheets and data, published at language-only paths (`/en`, `/de-de`, ...); Site Global Templates selected by Page Type	Not recorded	`utils/site.js` LOCALES; templates in `templates/`	PR #25, PR #26; AGENTS.md; templates/README.md	Approved
KB-D-004	11 October 2026	Configuration	Locale mappings and the code's locale list could drift	`paths.json` is generated from LOCALES (`npm run build:paths`); every locale sheet and the taxonomy page are mapped	A hand-maintained `paths.json` (replaced)	After a locale change: regenerate, commit both, update the site configuration's `public.json` paths (open item O-04)	PR #22, PR #27; tools/paths/generate-paths.mjs	Approved
KB-D-005	11 October 2026	Redirects	The platform applies only a root `/redirects.json`	Redirects are kept per locale (`/{locale}/redirects.json`), by the user's choice; List links follow them, the platform doesn't apply them	A single root `/redirects.json` (applied by the platform)	Redirects, including `/` → `/en/`, need another mechanism such as CDN rules (F-36, O-03)	AGENTS.md "Site structure and site data" (added in the commit of PR #27); handover 3.8	Approved
KB-D-006	11 October 2026	Navigation	Sections of the site need their own header and footer	Navigation, Footer and 404 Page are inherited like page properties: the page's value, else the nearest parent page's up to Home, else the locale's `references/nav` / `references/footer` / `error-pages/404`; cached per session	Not recorded	`utils/inherited-metadata.js`; changes show in a new session (F-35)	PR #29; AGENTS.md	Approved
KB-D-007	12 October 2026	Site structure	The locale root needed a Home page	Each locale page is published as its Home (`/de-de/` serves `/de-de/index`) and `paths.json` stays in sync with the site configuration	Not recorded	Folder mappings before Home mappings in `paths.json`	PR #32	Approved
KB-D-008	12 October 2026	SEO	No production domain is configured	The sitemap uses the aem.live address as its origin until a production domain is configured	Not recorded	Replace the origin in `helix-sitemap.yaml` at go-live (O-03)	PR #33; helix-sitemap.yaml	Approved

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-009	Not recorded	List	Lists could be built from Content Fragments	Content Fragment lists aren't supported in List: lists are built from pages (query index) or List Items	Content Fragment lists	Use pages with metadata, or Cards	Handover 16.4 Not Implemented, 6.16	Approved
KB-D-010	17 October 2026	Language master	The language master had to be reviewed before its rollout to /en	Temporary review publication at `/language-master/en/`: its own lists, references, sheets, inheritance up to its root page (`REVIEW_ROOTS`, `REVIEW_SHEETS`, `REVIEW_PAGES` in `utils/site.js`; `REVIEW` in `tools/paths/generate-paths.mjs`)	Not recorded	Temporary: must be removed after the rollout: unpublish, remove the code and paths, regenerate `paths.json`, re-apply the configuration (F-88)	rev 2.11, rev 2.14; F-88, F-91; PR #91–#102	Approved
KB-D-011	20 October 2026	Block library	Block library pages are test content	Block library pages stay out of the query index (by design): the locale block libraries are excluded from publishing in `paths.json` and from the index in `helix-query.yaml` (`/*/block-library`); the language master's library is published for the review with noindex (rev 2.20)	Not recorded	Lists don't show library pages; the index pattern doesn't match `/language-master/en/block-library` (requires validation)	rev 2.20, rev 2.35; helix-query.yaml, paths.json	Approved

3. Authoring Model and Universal Editor

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-012	13 October 2026	Additional tab	Authors needed anchors, hiding and spacing on blocks	Every block model except header and footer ends with an Additional tab (Anchor ID, Hide block, Padding / Margin as `classes_*` fields) applied by `decorators/block-options-decorator.js`	Not recorded	Shared classes in `styles/styles.css`; copy the tab into new blocks	PR #61; AGENTS.md "Additional tab"	Approved
KB-D-013	17 October 2026	Universal Editor dialogs	Universal Editor collapses tabs that don't fit on one line into a dropdown (List, Filter List, Separator, Video and others did)	At most three short tabs per dialog, the last one Additional; fields and stored values unchanged	The earlier 4–7 tab dialogs (replaced)	The tab names in the handover's Part 6 are those before 2.8 (F-84, F-107)	rev 2.8; F-84; PR #88	Approved

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-065	8 November 2026	Language master path	The language master under review lived at `/content/tcs-corp/language-master/en`, published at `/language-master/en/` (REVIEW_ROOTS, `paths.json`, `helix-query.yaml`); the product owner moved it to the AEM convention `language-masters`	The language master is `/content/tcs-corp/language-masters/en`, published at **`/language-masters/en/`**: `REVIEW_ROOTS = ['/language-masters/en']` (`utils/site.js`), the `paths.json` mappings (`npm run build:paths`), the query index exclusions; the migration's page packages 1.3.0 are below the new root. Transition (product owner, 8 November 2026): both roots are served until the tree is moved and edam-main is merged to main: `REVIEW_ROOTS` lists `/language-master/en` and `/language-masters/en`, `paths.json` and the site configuration map both, the query index excludes the references, data and block library of both; the old root is removed afterwards	Keep `language-master`	On author: move the tree (Sites console > Move, references adjusted); the site configuration's `public.json` paths must be updated by an administrator; the pages published at `/language-master/en/` stay live until unpublished (redirects: on hold, redirects-cdn.md)	`utils/site.js`, `paths.json`, `helix-query.yaml`, delivery branch `migrated-page-content` (`consolidated/packages-1.3.0-RENAME-REPORT.md`)	Approved (product owner, 8 November 2026)
KB-D-066	8 November 2026	Project documents	The project documents (the handover Markdown, PDF and its 593 screenshots, the author / developer / admin guides, the design proposals, the audit, animation and block documentation reports) lived in the code repository under `docs/` (88 MB, 698 files)	Approved (product owner): they move to the branch `migrated-page-content-edam` under `docs/` (same paths, https://github.com/sradha-tcs/tcs-corp/tree/migrated-page-content-edam/docs); only this knowledge base stays in the code (AGENTS.md makes it the starting point of every task). Links in the code repository point to the branch; the block READMEs embed the screenshots from it (`tools/docs/handover-screenshots.json`, refreshed with `HANDOVER=<handover file> npm run build:docs`); handover revisions are made on the branch	Keep everything in `docs/`; move the knowledge base too	Smaller code repository; the handover and reports are versioned with the migration deliverables; a documentation change in a code PR names the branch change	AGENTS.md, README.md	Approved (product owner, 8 November 2026)

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-068	8 November 2026	Site roots	Two language master roots were served during the move (KB-D-065); the product owner gave each tree a purpose and added a third	Approved (product owner): three review roots, each a site of its own with its references, error pages and sheets: `/content/tcs-corp/language-master/en` (testing: page functionality with dummy content, content limitations), `/content/tcs-corp/language-masters/en` (the language master: the migrated tcs.com content, validated with its real sheet data; later the source of the global site and the regions and languages, MSM or separate content to be decided once the content is final, as tcs.com's regions have no language master content today) and `/content/tcs-corp/design-block-master/en` (the block library and the curated design pages); `REVIEW_ROOTS`, `paths.json` (48 mappings) and the site configuration map all three, the query index leaves out their references, data and block libraries; the block READMEs' library examples and usage refer to `/design-block-master/en`	One root (KB-D-065)	Content, tests and library validations kept apart; the old root is no longer removed	AGENTS.md, content-architecture.md	Approved (product owner, 8 November 2026)
KB-D-069	8 November 2026	Documentation	The project documentation was one consolidated handover (Markdown 2.53, PDF 2.44) plus guides and reports on the documents branch	Approved (product owner): a role-specific documentation set generated from the repository and the published reference pages (master documentation prompt) replaces it in `docs/handover/` of `migrated-page-content-edam`: 01 architecture, 02 development (a folder per block and curated page), 03 authoring (a guide per component), 04 business analysis (inferred EPICs, features, stories), 05 QA (test cases, execution results with evidence), 06 operations, 07 administration and governance (Dynamic Media set-up), 08 release and DevOps, 09 knowledge base (humans, agents, LLM retrieval, machine-readable), 10 training, 11 audit and remediation; Markdown with paired PDFs; its generator in `docs-tools/` on that branch; every code change updates it; the knowledge base of this repository stays the source for code work; the former documents are kept in the tag `archive/docs-before-role-docs-2026-11-08`	Keep the consolidated handover	Each role reads its own documents; documentation follows the code through the generator; the handover PDF no longer lags	AGENTS.md, README.md	Approved (product owner, 8 November 2026)
KB-D-014	17 October 2026	List, Filter List	Universal Editor didn't save `multi: true` aem-content fields (`pages`, `featuredPages`)	Pages as composite multifields (`pageItems`, `featuredItems`, a Page per item)	Multi-value aem-content fields (still read)	Superseded for new content by item children (KB-D-015); the multifields stay in the dialogs and are still read	rev 2.9; F-86; PR #89	Superseded (KB-D-015)

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-015	17 October 2026	Content model	Composite multifields written by content packages weren't rendered on published pages (CTAs, stats, links, filters empty)	Repeating content is item children added with Add (List Items, Cards, `hero-cta`, `page-header-cta`, `teaser-link`, `teaser-stat`, `testimonial-stat`, `anchor-list-item`, `filter-list-tag`, `filter-list-category`, `filter-list-page`, `footer-link`); `itemRows` (`utils/cells.js`) finds them	Composite multifields (`container`, `multi: true`), still read for content authored before	New repeating fields must be item models; tests in `test/items/`	rev 2.10; F-87; PR #90; AGENTS.md "Item children"	Approved
KB-D-016	17 October 2026	Editor experience	Items just added had no content and seemed not to appear	In Universal Editor an item without content is a dashed placeholder (`itemEditorHint`, `decorators/editor-item-decorator.js`); on AEM author (no query index) lists say their pages show on the published page	Not recorded	Placeholders `itemEditorHint`, `listEditorPages`, `filterListEditor`	rev 2.8, rev 2.10, rev 2.12; F-85, F-89	Approved
KB-D-017	19 October 2026	Delivery rules	The code and local preview assumed markup AEM doesn't deliver (forms empty, tables and accordions with empty rows, no video poster)	Code and the library preview assume what AEM really delivers: tab rows, `name, classes` first cells, no grouped fields in key-value blocks, collapse suffixes, template values stored	The earlier assumptions (replaced)	Rules in AGENTS.md; options AEM doesn't deliver removed (Columns Additional tab, Teaser personalization, Hero and Table List Motion Type, Quote / Testimonial Style, Separator Auto / Custom)	rev 2.27; F-101; PR #118; docs/audit/BLOCK-ECOSYSTEM-AUDIT.md	Approved
KB-D-018	20 October 2026	Delivery rules	The re-audit found item names lower-cased, sample text on every new item, Lottie alt as link text, DAM JSON / video 404s	Case-sensitive values get their own field (Form Variations `fieldName`); no sample content in templates or model defaults; non-image DAM assets are published as an operations step	Not recorded	Content migrations and the publication step (package 1.0.9)	rev 2.29; F-102; PR #120; docs/audit/REAUDIT-2026-10-20.md	Approved

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-019	20 October 2026	Section containers	AEM doesn't deliver a section's `classes_*` fields, so the Additional tab of Tabs Container, Columns Container and Gen Carousel had no effect	Their Additional tab fields are `container*` (delivered as data attributes) applied by `scripts.js` `applySectionBlockOptions`	`classes_*` on sections (not delivered)	`.eslintrc.js` `max-cells` limits per container	rev 2.30; F-102; PR #121; AGENTS.md	Approved
KB-D-020	20 October 2026	Field naming	`hideTitle` would collapse into the title (Title suffix rule)	The List Item's hide toggle is `titleHidden`, the name the Teaser uses; values that must be read avoid the Text, Title, Type, Alt and MimeType suffixes (`buttonVariant`)	`hideTitle`	Same naming in List, Teaser and Testimonial models	rev 2.34; F-105; PR #125	Approved

4. Components

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-021	13 October 2026	Teaser, Hero	The "Lotto" image type is a Lottie animation	The Lottie player (lottie-web light) is vendored in `scripts/lib/lottie` and loaded once per page by `utils/lottie.js`, with a pause / play button	Not recorded	No external script dependency; Lottie JSON must be published as a DAM asset (KB-D-018)	PR #52; utils/lottie.js	Approved
KB-D-022	14 October 2026	Cards	Authors needed hand-made cards with the List's look	The Cards block is the manual authoring alternative to List: each Card holds its own content, with the List's presentation and shared code (`models/_card-list.json`, `utils/card-list-config.js`, `decorators/card-list-decorator.js`, `styles/card-list.css`)	Not recorded	Cards made before the redesign have no block model and changed look (F-34, F-41, F-42; proposal KB-D-046...051)	PR #62; rev 1.1; blocks/cards/README.md	Approved
KB-D-023	16 October 2026	Header	Figma's desktop header is a logo bar	Desktop (>= 1200px) shows the Figma logo bar only (TCS left, TATA right), a product decision; desktop navigation comes from Conversation Canvas blocks; tablet and mobile open the menu full screen one level at a time	Show the tablet menu on desktop by changing the 1200px breakpoint (recorded in F-72 as the way to revisit)	A page without a Canvas has no desktop navigation (F-72, P2)	rev 2.2; F-64, F-72; PR #81	Approved

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-024	16 October 2026	Conversation Canvas	Several Canvases per page (header, body, footer) floated independently	One floating journey for all Canvases of a page (`canvas-journey.js`, pure and unit tested; `canvas-controller.js` applies it); the header announces its menu with `header:menu` and the journey hides while it is open; header and Canvas don't import each other	The PR's "Don't float" boolean (replaced by the Floating journey field)	No Canvas logic in `header.js` or `footer.js`	rev 2.3; F-61 to F-63, F-73; PR #71; AGENTS.md "Floating Canvas"	Approved
KB-D-025	16 October 2026	Header	`/en/home` named an unpublished Navigation page and showed no header	The header falls back to the locale's `references/nav` when the configured Navigation page can't be loaded	Not recorded	Header always renders	rev 2.6; F-80; PR #83	Approved
KB-D-026	17 October 2026	List, Cards	The option class `teaser` collided with the Teaser block's styles	Display as teaser adds `display-teaser`; option classes must not be block names	`teaser` (replaced)	Lists next to a Teaser render	rev 2.7; F-83; PR #87	Approved
KB-D-027	18 October 2026	List	Pages needed their own card look	Card style and Card colour precedence: the List Item's value, else the List's (other than Mixed / Default), else the page's `card-style` / `card-theme` metadata, else automatic	Not recorded	New bulk metadata columns indexed as `cardStyle`, `cardTheme`	rev 2.16; PR #104; blocks/list/README.md	Approved
KB-D-028	19 October 2026	Columns Container	Edge Delivery has no blocks inside blocks; the Columns block holds default content only	Columns are a section container (Columns Container, like Tabs Container and Gen Carousel) whose columns hold default content and any block except Page Header, Footer and TCS Header; Columns per row (Automatic, 1–4)	Blocks inside the Columns block (not possible)	New content uses the container; the Columns block stays for existing content	rev 2.22, rev 2.23; PR #112, PR #113; blocks/columns-container/README.md	Approved
KB-D-029	19 October 2026	Columns Container	AEM renders a block with the class of its template name: the "Column" item came out as `column`	The column item's template name is "Column Break"; columns are plain wrappers, so editor updates of a component in a column work in place; `column` blocks authored before are read as Column Breaks	"Column" (replaced)	`test/models/block-names.test.mjs` checks template names against block folders	rev 2.24; F-99; PR #114	Approved
KB-D-030	19 October 2026	Columns Container	Column widths only applied to rows of two columns	Column widths removed: columns are of equal width	Column widths (rev 2.22, removed)	Simpler Layout tab	rev 2.26; PR #117	Approved
KB-D-031	19 October 2026	Button	AEM's built-in Button delivers only its link and Primary / Secondary (as bold / italic): Views, Icon, Alignment and Tertiary never reached the page	The Button authors add is the Button Block (`blocks/button-block`, title "Button", template name "Button Block"), a key-value block whose every field is delivered (`buttonVariant` for the type)	The built-in Button (now "Button (basic)")	Buttons authored before keep Primary / Secondary	rev 2.26; F-100; PR #117; AGENTS.md "Button"	Approved

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-032	20 October 2026	Columns, Button	The built-in Columns and Button were still offered next to their replacements	The built-in Columns and Button are no longer offered in any filter; their definitions stay so content authored before remains editable; inside a Tabs Container a tab's Column Breaks lay its content out in columns (Tab field Columns per row)	Keep offering both	Library content migrated (package 1.0.8); `blocks/columns` is Compatibility-only	rev 2.28; PR #119; blocks/columns/README.md	Approved
KB-D-033	20 October 2026	Teaser	Title Type changed only the level, Title size the size; BG Color added padding; Image position Right only from 900px	One Title Type sets level and size (Title size removed); Image position Left / Right / Top / Bottom, side by side from 600px; BG Color is only the colour (spacing from the Additional tab); Buttons side by side; List arrow on every link	Title size and the BG Color padding (removed)	Content action: Teasers that relied on Title size or the padding (F-103)	rev 2.32; F-103; PR #123	Approved
KB-D-034	20 October 2026	Backgrounds	Section offered only Highlight (grey); dark bands needed a block with its own colour	Section Style is one choice of Default, Grey, Dark, Dark Green; Teaser BG Color, Columns Container `containerBgColor` and Section Style share labels, classes (`bg-grey`, `bg-dark`, `bg-dark-green`) and tokens and set only the colour; Highlight no longer offered	Highlight only (F-21; sections authored with it keep their grey band)	Shared token `--color-bg-dark-green`	rev 2.32, rev 2.33; F-21, F-104; PR #124; AGENTS.md "Backgrounds and theme"	Approved
KB-D-035	20 October 2026	Theme	Content inside dark backgrounds could be unreadable (contrast down to 1.05:1)	Theme contract of the dark Section and Columns Container backgrounds: light text, link and focus tokens, underlined links in running text, inverted buttons; Grey Teasers and Columns Containers inside reset the tokens	Not recorded	New blocks use the semantic tokens so they follow	rev 2.34; F-105; PR #125; styles/styles.css	Approved
KB-D-036	20 October 2026	List Item	The request's "Link Item" needed per-item display control	The "Link Item" is the List Item (decided with the product owner); its Display tab adds Hide Title (`titleHidden`), Show Description, Show Image, Show Eyebrow, Show Label, Show Action Label (on by default)	Not recorded	Items authored before are unchanged	rev 2.34; F-105; PR #125; docs/audit/DIALOG-OPTIONS-2026-10-20.md	Approved
KB-D-037	19 October 2026	Accessibility	Secondary text, brand text and fill tokens failed WCAG 1.4.3	Contrast tokens changed to pass WCAG 1.4.3, subject to design sign-off	The earlier tokens (F-27, F-28)	Design sign-off is open (O-14)	rev 2.27; F-28; O-14	Approved

5. Animation, Performance and Loading

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-038	17 October 2026	Page loading	The footer appeared late on throttled phones	The footer loads alongside the main sections (hidden until they have loaded); the header loads nav and placeholders in parallel; shared list data starts with the lazy phase	Loading the footer after all sections (replaced)	Footer 1.2–2.8 s earlier on a throttled phone (as recorded)	rev 2.13; F-90; PR #100	Approved

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-039	18 October 2026	Header layout shift	The header and its Canvas pushed the page down when they arrived (CLS 0.14–0.25)	The header reserves its space until it loads (`--header-reserved` or the height remembered per breakpoint, `utils/header-space.js`); from rev 2.21 the reserve is the logo bar and the header Canvas row lies on the bottom of the hero (two zones)	Not recorded	CLS below 0.01 (as recorded)	rev 2.15, rev 2.21; F-82, F-93, F-98; PR #103, PR #92	Approved
KB-D-040	18 October 2026	Animation	Authors needed block motion without hurting loading	Additional tab Animation Type (content) and Background animation, Motion Speed and Animation Trigger (rev 2.19); run once in view, never in Universal Editor edit mode or for reduced motion; Preview plays them (rev 2.20); `utils/motion.js`, `decorators/motion-decorator.js`, `styles/motion.css`	Hero and Table List Motion Type (removed from the dialogs in rev 2.27; values authored before play as the new animations)	Background animation delays the LCP image above the fold (F-94)	rev 2.18–2.20; F-94 to F-97; PR #106–#108; docs/animation/README.md	Approved
KB-D-041	18 October 2026	Skeletons	Query index data held back sections	Blocks built from the query index render through `renderWhenLoaded`: data later than 150ms shows a block-shaped skeleton	Waiting for the data	Small shift below the block when content is taller (F-95, accepted)	rev 2.18; F-95; decorators/skeleton-decorator.js	Approved
KB-D-042	20 October 2026	Module loading	Without a build step, modules loaded level by level (up to 4 round trips before the first section, up to 7 more per block)	Module preloading: `head.html` modulepreload links for the `scripts.js` modules and `preloadBlockModules` from the generated `scripts/module-graph.js` (`npm run build:modules`, checked by `npm test`); no forced layouts while decorating	Not recorded (the project keeps "no build step")	Every import change requires `npm run build:modules`	rev 2.36; F-106; PR #127; AGENTS.md "Module preloading"	Approved
KB-D-043	20 October 2026	Fonts	Text reflowed when Inter replaced an unscaled system font on Android / Linux (CLS 0.33)	Fallback faces scaled to Inter per system font (`inter-fallback`, `inter-fallback-roboto`, `inter-fallback-dejavu`)	A single Arial-based fallback (replaced)	Keep the faces when changing the font	rev 2.36; F-106; PR #128; styles/styles.css	Approved

6. Documentation and Process

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-044	21 October 2026	Block documentation	The handover's component chapters lagged behind the code	Every block folder has a README in the structure of `blocks/README.md`; its dialog fields, screenshots, files and library usage are generated (`npm run build:docs`) and validated by `npm test`; the READMEs are current where Part 6 differs	Part 6 only	Part 6 needs an alignment pass (F-107, open)	rev 2.37; F-107; PR #129; docs/blocks/BLOCK-DOCUMENTATION-REPORT.md	Approved
KB-D-045	Not recorded	Publishing	Code changes must be reviewable on real content	Feature branch, PR with a feature preview test URL, `gh pr checks` (lint, Code Sync, PageSpeed), human review and merge; Code Sync deploys `main`	Not recorded	PRs without a test URL fail the PageSpeed check	AGENTS.md "Publishing Process"; handover 12.2, 13.15	Approved

7. Proposed Decisions

The Cards and Card refactoring proposal (14 October 2026, "For review — decisions D-1 to D-6 open") lists six product owner decisions; the proposal's recommendation is quoted, nothing is implemented.

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-046	14 October 2026	Cards (D-1)	Whether Universal Editor offers Card in the + of a new key-value Cards block (open item O-20)	Proposed: confirm by test before Phase 0	Not recorded	Blocks Phase 0 of the proposal	cards-card-refactoring.md section 21	Proposed
KB-D-047	14 October 2026	Cards (D-2)	Cards made before the redesign changed look (F-41)	Proposed: restore the original look (Classic) for those blocks, Figma design for new blocks	Move them to the Figma design	Content migration of G1 instances	Same, section 21; F-41	Proposed
KB-D-048	14 October 2026	Cards (D-3)	Overlap with List and Filter List	Proposed: dynamic sources in Cards through shared code; review in 6 months whether List can be retired	Keep Cards manual only	Overlapping listing blocks	Same, section 21	Proposed
KB-D-049	14 October 2026	Cards (D-4)	The Card item's cell limit	Proposed: raise the Card cell limit to 22	Merge the tracking attributes into one field	`.eslintrc.js max-cells`	Same, section 21	Proposed
KB-D-050	14 October 2026	Cards (D-5)	Personalization of cards	Proposed: attributes now; behaviour when Target is connected	Hide / show by audience	Requires a personalization architecture decision (none implemented)	Same, section 21	Proposed

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-051	14 October 2026	Cards (D-6)	Filtering in Cards	Proposed: client-side filtering in Cards for up to 50 cards; Filter List beyond	Filter List for every filtered set	Not recorded	Same, section 21	Proposed
KB-D-052	22 October 2026	Header over the first block	KB-D-039 (the header reserves its space, CLS below 0.01) assumes the layout is known before the header loads; an author option on the TCS Header (on the nav page, loaded in the lazy phase) isn't	Proposed: the TCS Header's Header over the first block (Auto: over an opening light Hero, as before; Always: over any first block; Never: the page starts below the header), at every width (desktop only at first), as <html data-header-overlay>; remembered per browser (utils/header-space.js, header-overlay-v1) and applied before the header loads on the next pages, as KB-D-039's header height	A page metadata column read in the head (no first-page shift, set per page); automatic only (the Hero rule)	A visitor's first page whose mode isn't Auto shifts once when the header arrives (measured locally: CLS 0.086 for Never over a light Hero, 0 on the next page); a block other than the Hero needs its own top space under the logos (Padding top, --header-overlay-space)	blocks/tcs-header, styles/styles.css, blocks/hero, templates/home	Proposed
KB-D-053	22 October 2026	Conversation Canvas journey options	KB-D-024's Floating journey field offered Floats and docks, Docks only and Decorative for every Canvas Style: on Primary the field did nothing or hid the dock for good, and Decorative let the floating row pass over a Canvas in place (two rows on screen)	Proposed: three behaviours only, Primary (sticky: floats from the top of the page to the footer), Floats and docks, Stays in place (nofloat, the former Docks only; decorative is read the same way); Floating journey and Quick actions shown only for Default / Secondary, Navigation style and BG Style only for Primary (dialog conditions); while a row floats it is the only Canvas row in use (the others are inert placeholders), with none floating every row is in use; Canvases with Hide block or not shown aren't active	Keep Decorative as a separate option	No existing content uses Docks only, Decorative or Primary on /en (scan of the published pages and the block library, 22 October 2026)	blocks/conversation-canvas; canvas-journey.js focused; README "Floating journey: the three behaviours"	Proposed

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-0054	22 October 2026	Media contacts	The Press Release template appended "Media contacts" (Media Contacts sheet) at the end of every press release in the browser: not in the content tree, not authorable, always after the last section	Proposed (requested by the user): a **Media Contacts** block (`blocks/media-contacts`: Title, Title Type, Additional tab) renders the sheet's contacts where it is authored, grouped by designation and region; the template only adds the Page Header. `mailto:` / `tel:` clicks are recorded in the data layer by kind (email / phone), never with the address (`utils/privacy.js` `contactLinkKind`)	Keep the template section and add only the grouping; the template inserts the block when none is authored	Press releases that relied on the template show no contacts until the block is added (the block library press release template page, the Site Global Templates press-release page in AEM); `decorators/contact-list-decorator.js` now exports `buildContactGroups` (replaces `buildContactList`)	blocks/media-contacts, templates/press-release, decorators/contact-list-decorator.js, decorators/data-layer-decorator.js	Proposed
KB-D-0058	23 October 2026	List, List Item	KB-D-036 gave the List Item Show Action Label (on by default) on top of the List's Show call to action, and Hide Title kept the action label; authors asked for the call to action to be decided per List Item only	Proposed (requested by the user): the List has no Show call to action (removed from its dialog; a value saved on Lists authored before is ignored); a List Item shows its call to action (`p.list-card-action`, the Call to action label, else Explore / Download) only with its own Show Action Label on (off by default), each item independently, rendered or left out by the existing card list code (`showsCta` in `utils/card-list-config.js`); with Hide Title on and no call to action the title stays visible as the link. The Cards block keeps its Show call to action. List Item rows delivered without their tab cells (19 or 13 cells) are read field by field	Keep the List toggle with an item override	Existing Lists lose their calls to action until each List Item's Show Action Label is turned on; cards from the query index (Child pages, Tagged pages) have no List Item and show none	Changes KB-D-036; blocks/list/README.md; branch `feature/list-item-action-label-refactor`	Proposed

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-059	29 October 2026	List Show call to action: Child pages with Panel cards	KB-D-058 removed the List's Show call to action, so cards from the query index (no List Item) never showed a call to action; the request: a Show call to action for Child pages with Panel cards only	Proposed (requested by the user): the List's own Show call to action is in the dialog only for List source Child pages, Display as teaser on and Card style Panel, and shows or hides Explore on every fetched card; for every other source and card style a value saved before isn't read and the cards keep their call to action (`ctaToggleApplies`); List Items keep KB-D-058 (their Show Action Label alone decides); the Cards block keeps its toggle	Keep KB-D-058's no call to action on the other query-index cards	Query-index cards other than Child pages + Panel show their default call to action again (Explore on panels, Download on reports), whatever was saved	blocks/list (_list.json, list-data.js), utils/card-list-config.js	Proposed
KB-D-060	29 October 2026	List Item Show Action Label on by default	KB-D-058 made a List Item's Show Action Label off by default, so a newly added List Item (Fixed list, e.g. Panel cards) showed no Explore call to action until the author turned it on	Proposed (requested by the user): Show Action Label (List Item Display tab) is on by default (model value `true`; `itemDisplay` reads it on unless authored `false`); Off renders no action element and no default label; each item decides for its own card only, the List's Show call to action never overrides it (KB-D-058, KB-D-059 otherwise unchanged). The toggle is on every List Item: Universal Editor item conditions can't read the List's source or Card style, and List Items render only in a Fixed list	Keep KB-D-058's off by default	List Items never changed (no value stored) show their call to action again: Explore on panels, Download on reports, the authored label otherwise; items with Show Action Label authored off are unchanged	blocks/list (_list.json, list-data.js)	Proposed

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-061	29 October 2026	Form Variations Additional tab	Commit 9aae6f7 ("block changes", 27 October 2026) rewrote Form Variations and removed its Additional tab fields: Hide block, Padding / Margin and the animation fields (Animation Type, Motion Speed, Animation Trigger, Background animation). AGENTS.md requires the Additional tab on every block but header and footer, and `test/motion/motion.test.mjs` expects the animation fields on Form Variations (2 failing tests since)	Proposed: restore the Additional tab of Form Variations (the fields of the other blocks, Background animation included: the block has an authored background)	Keep the block without the tab: then AGENTS.md, the Animation section, `CONTENT_TARGETS` / `BACKGROUNDS` and the motion tests drop Form Variations	Restoring: forms get Hide block, spacing and animation again; keeping: the rule gets an exception	9aae6f7; test/motion	Proposed
KB-D-055	24 October 2026	Redirects at the CDN	The platform applies only a root `/redirects.json`; the redirects are kept per locale (KB-D-005), so nothing redirects: `/` returns 404 (F-36) and the old tcs.com URLs of the 3,191 migrated pages have no locale prefix	Proposed: the production CDN applies the rules of redirects-cdn.md: `/` → `/en/`, the Redirects sheet rows of the path's locale (read from the published sheets, refreshed every 5 minutes), old paths without a locale → the default locale + the path in lower case; decisions R-D1 (302 until go-live), R-D2 (`/` always `/en/`), R-D3 (pattern rule only for the migrated paths), R-D4 (at the CDN) open	A root `/redirects.json` (exact paths only, replaces the per-locale sheets: a change to KB-D-005); redirects in the browser from the 404 page (no 301 for search engines)	Authors keep maintaining one sheet per locale; the CDN Owner implements the rules once; O-03 (domain, CDN) must be resolved first	F-36; O-03; handover 3.8 (rev 2.44)	Proposed

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-056	24 October 2026	Author host in configuration files	AGENTS.md allows no AEM program / environment ids or author hosts in any file; two files need the author host to work: `fstab.yaml` (the content source: preview and publish read the pages from it) and `.migration/project.json` (managed by the workspace tooling, which can't be changed or untracked)	Proposed: both files are a recorded exception to the rule; no other file may hold the host or the ids (the handover since 2.43, F-112; the block README and knowledge base validators)	Move the content source to the site configuration (`content.source`) and remove `fstab.yaml` (needs the Site Administrator and a test of preview and publish); keep the rule without exceptions (not possible while the files are needed)	AGENTS.md's privacy rule gets the exception once approved	F-112; rev 2.44	Proposed
KB-D-062	3 November 2026	Hero Secondary CTA colours in the CTA List	hero.css styled the Hero's Secondary CTAs from block classes (secondary-bg-light-grey, secondary-bg-black-40, secondary-border-black, secondary-border-white) that no field set; a Style tab select for the whole Hero was proposed, but the colours are wanted per CTA in the CTA List. This conflicts with the AGENTS.md rule "item children instead of composite multifields" (the CTA List is a composite multifield kept for Heroes authored before; CTA items are the current way to add CTAs)	Proposed: the CTA List gets Primary CTA BG Color (`ctaPrimaryBackground`: Blue-cyan (azure), #216AC3, `primary-bg-azure`, Primary only; no Primary border colour), Secondary CTA BG Color (`ctaBackground`), Secondary CTA Border color (`ctaBorder`) and, for every CTA view, CTA Text color (`ctaTextColor`: Default (Black) `cta-text-black`, the field default, or White `cta-text-white`: the chosen colour applies whatever the CTA view and background; Figma: white text on a black background), labelled "(Secondary only)" and always shown (conditions inside multi containers aren't supported by Universal Editor: `var` resolves absolutely, not per row); `hero.js` applies them only to a Secondary CTA, treats their values as CTA values and adds them to that CTA's button as classes, `hero.css` styles the button; no Style tab fields	The same fields on the CTA item (`hero-cta`, follows the rule); a Style tab select for all Secondary CTAs of the Hero	Only CTAs in the CTA List can have the colours, CTA items can't; composite multifields didn't render from content packages (AGENTS.md); authors see the two fields on Default and Primary CTAs too, where they have no effect	blocks/hero/README.md, blocks/hero/hero.js, blocks/hero/hero.css, blocks/hero/_hero.json	Proposed

Decision ID	Date	Area	Context	Decision	Alternatives	Impact	References	Status
KB-D-063	29 October 2026	Image Collage block	The Teaser has a Layout Image collage with Collage image items (`teaser-collage-image`, PR #167); a separate Image Collage block (`blocks/image-collage`, items `image-collage-item`) was requested for the same design. AGENTS.md: reuse before create (extend an existing block)	Proposed: add the Image Collage block next to the Teaser layout; decide later which one authors use and deprecate the other (content using it would move)	Keep only the Teaser layout; keep only the new block and remove the Teaser layout	Two ways to author the same design until one is retired; the code and styles are duplicated	blocks/image-collage/README.md, blocks/teaser/README.md	Proposed
KB-D-064	8 November 2026	Assets (EDAM)	The site's assets live in the Sites author's own DAM (`/content/dam/tcs-corp/`, "AEM Assets: asset picker; images on the media bus", architecture.md); the product owner asked to use a separate AEM Assets environment (EDAM) for the project, including the migrated pages' assets (338 asset packages, 22 GB)	Proposed: a separate AEM Assets as a Cloud Service environment is the source of all site assets (rendering of delivery URLs in the code: `decorateDeliveryImages`, PR "EDAM delivery images"); authors pick from it with the Universal Editor Custom Content Advisor; pages store Dynamic Media with OpenAPI delivery URLs (images through the Edge Delivery media bus); the migration uploads the assets there as EDAM zips with a Metadata Import file (content packages cannot be installed on EDAM) and rebuilds the page packages (1.3.0) with the delivery URLs (E-D1 answered 8 November 2026: Dynamic Media with OpenAPI is licensed). **Decided 8 November 2026 (product owner): not adopted.** The delivery host given for E-D2 is the Sites environment's own delivery tier: there is no separate AEM Assets environment. The assets stay in the Sites author's DAM and the pages reference them by DAM path (the alternative "Keep the Sites author's DAM"): the migration installs `consolidated/packages/` (asset packages 1.2.0, page packages 1.3.0) with `install-packages.*`; no approval, asset ID export, delivery URL rewrite or Custom Content Advisor (the asset picker PR #195 closed unmerged). The rendering of delivery URLs (PR #193) stays: it changes nothing for DAM paths and renders a delivery URL an author pastes; E-D3 to E-D5 no longer apply	Keep the Sites author's DAM (today); Dynamic Media on the Sites author	New integration: Custom Content Advisor extension and configuration, 20 asset fields in 15 models, decorators for delivery URLs, sheet image columns; order: tags and EDAM zips with metadata on EDAM, approval, asset ID export, page packages 1.3.0 on Sites	remote-assets-edam.md	Not adopted (product owner, 8 November 2026)

8. Recommendations (not approved)

Recommendations recorded in the handover (Part 16 recommendations, 11.7, 12.3) and those derived for this knowledge base. None is a decision: each needs an owner's approval and a decision log entry before implementation.

Recommendation	Source	Status
Integrate Adobe Tags with the Web SDK and the ACDL extension in `scripts/delayed.js`, with consent	Handover 11.7; F-06; O-07 (blocked: measurement plan, Tags property, datastream, consent platform)	Recommendation (not approved)
Run `npm test` in the Build workflow; tag releases; replace the boilerplate values of `package.json`, the PR template and the Sidekick project name	F-37; handover 12.3	Recommendation (not approved)
Configure a CDN or root redirect for `/` → `/en/` and apply the per-locale redirects sheets	F-36; O-03	Specified in redirects-cdn.md (KB-D-055, Proposed)
Add hreflang alternates and a language switcher	F-39	Recommendation (not approved)
Connect a form back end before publishing any form, or hide forms	F-01; O-08	Recommendation (not approved)
Align the handover's Part 6 component chapters with the block READMEs	F-107	Recommendation (not approved)
Reference agent-instructions.md from the root AGENTS.md and place copilot-instructions.md in `.github/`	Derived for this knowledge base	Recommendation (not approved)
Remove the language master review code and paths after the rollout to /en (KB-D-010)	F-88	Recommendation (not approved): planned, the timing isn't recorded

9. Related

- Knowledge base index
- Agent guidelines
- Agent instructions (for AGENTS.md)
- Copilot instructions
- Repository inventory
- Topic files: architecture, content architecture, Universal Editor, personalization, deployment and release
- Project handover (Revision History; Part 16 — Handover Findings and Open Items Register)
- AGENTS.md, docs/proposals/README.md, docs/audit/README.md

10. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

11. Known limitations

- None recorded for this document.

12. Open items

- None.

13. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Architecture Risks and Open Items

Attribute	Value
Document ID	ARC-010
Audience	Solution architects, technical leads, developers, AI agents
Purpose	Risks and open architecture items with their status
Scope	—
Prerequisites	—
Owner	Solution Architect (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/decision-log.md, docs/handover/knowledge-base/reports/discrepancy-report.md
Related documents	—
Related AEM paths	—

1. Open architecture items

ID	Item	Status	Owner (proposed)
AR-01	Redirects: per-locale sheets aren't applied by the platform; / to /en/ needs a CDN rule (KB-D-055)	Planned (on hold)	Solution Architect
AR-02	Dynamic Media for all assets (KB-D-067): reprocessing, approval, asset IDs, page rewrite and reinstall pending; picker PR #200 needs the Extension Manager set-up	Partially implemented	AEM Administrator
AR-03	Adobe Tags / Web SDK / consent not configured: data layer only	Not implemented	Product Owner
AR-04	Forms have no submission back end (F-01)	Not implemented	Product Owner
AR-05	MSM or separate regional content for the language master rollout	To be confirmed	Product Owner
AR-06	/en locale content was unpublished and deleted from Edge Delivery on 8 November 2026 16:41–16:48 UTC (author-side action)	Open: to confirm intent	Content Lead
AR-07	/design-block-master/en content links to /language-master/en (1,281 links) and to unmapped /content/tcs-corp/design-block-master/english paths (358 links)	Open	Content Lead
AR-08	Production domain, CDN, monitoring not recorded	To be confirmed	Solution Architect

2. Recommendations (not approved)

3. Recommendations (not approved)

Recommendations recorded in the handover (Part 16 recommendations, 11.7, 12.3) and those derived for this knowledge base. None is a decision: each needs an owner's approval and a decision log entry before implementation.

Recommendation	Source	Status
Integrate Adobe Tags with the Web SDK and the ACDL extension in `scripts/delayed.js`, with consent	Handover 11.7; F-06; O-07 (blocked: measurement plan, Tags property, datastream, consent platform)	Recommendation (not approved)
Run `npm test` in the Build workflow; tag releases; replace the boilerplate values of `package.json`, the PR template and the Sidekick project name	F-37; handover 12.3	Recommendation (not approved)
Configure a CDN or root redirect for `/` → `/en/` and apply the per-locale redirects sheets	F-36; O-03	Specified in redirects-cdn.md (KB-D-055, Proposed)
Add hreflang alternates and a language switcher	F-39	Recommendation (not approved)
Connect a form back end before publishing any form, or hide forms	F-01; O-08	Recommendation (not approved)
Align the handover's Part 6 component chapters with the block READMEs	F-107	Recommendation (not approved)
Reference agent-instructions.md from the root AGENTS.md and place copilot-instructions.md in `.github/`	Derived for this knowledge base	Recommendation (not approved)
Remove the language master review code and paths after the rollout to /en (KB-D-010)	F-88	Recommendation (not approved): planned, the timing isn't recorded

4. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

5. Known limitations

- None recorded for this document.

6. Open items

- None.

7. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Content Architecture

Attribute	Value
Document ID	ARC-002
Audience	Solution architects, technical leads, developers, AI agents
Purpose	The site and locale hierarchy, review roots, page structure, metadata, taxonomy, placeholders, references, data sheets, redirects and publishing dependencies
Scope	—
Prerequisites	—
Owner	Solution Architect (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/content-architecture.md, utils/site.js, paths.json, helix-query.yaml
Related documents	—
Related AEM paths	/content/tcs-corp/language-master/en, /content/tcs-corp/language-masters/en, /content/tcs-corp/design-block-master/en

1. The three review roots (KB-D-068)

Author path	Published at	Purpose
/content/tcs-corp/language-master/en	/language-master/en/	Testing page functionality with dummy content, content limitations and other test aspects
/content/tcs-corp/language-masters/en	/language-masters/en/	The language master: the 8,432 migrated tcs.com pages, validated with real sheet data; later the source of the global site and the regions (MSM or separate content to be decided once the content is final)
/content/tcs-corp/design-block-master/en	/design-block-master/en/	The block library and the curated design pages

Each root has its own references (nav, footer), error pages and sheets and lists its own pages from the query index (`REVIEW_ROOTS`, `REVIEW_SHEETS`, `REVIEW_PAGES` in utils/site.js).

2. Site and locale hierarchy

Purpose. One Edge Delivery site serves 13 locale sites by path. Each locale owns its pages and its own references, error pages, sheets, data and block library.

Structure. The AEM tree `/content/tcs-corp/{country}/{language}` is published at a language-only path. The locales are defined once in `LOCALES` (`utils/site.js`); the first entry is the default (`DEFAULT_LOCALE`, `en`).

AEM folder	Published root	lang		AEM folder	Published root	lang
us/en	/en	en-US		se/sv-se	/sv-se	sv-SE
de/de-de	/de-de	de-DE		dk/da-dk	/da-dk	da-DK
fr/fr-fr	/fr-fr	fr-FR		no/no-no	/no-no	no-NO
ca/fr-ca	/fr-ca	fr-CA		cn/zh-cn	/zh-cn	zh-CN
es/es-es	/es-es	es-ES		jp/ja-jp	/ja-jp	ja-JP
pt/pt-pt	/pt-pt	pt-PT		kr/ko-kr	/ko-kr	ko-KR
nl/nl-nl	/nl-nl	nl-NL				

```
/content/tcs-corp/
├── {country}/{language}/   a locale site; the locale page itself is Home (published as /{locale}/)
│   ├── …                   content pages
│   ├── references/         nav, footer, alternative navs / footers, footer Canvas page, modals,
fragments
│   ├── error-pages/        404 (used by 404.html); 403 / 500 for a CDN
│   ├── block-library/      block validation pages; excluded from publishing
│   ├── placeholders        sheet: Key, Text, Description, Category
│   ├── metadata            sheet: bulk metadata
│   ├── taxonomy            taxonomy page (tag, title)
│   ├── redirects           sheet: Source, Destination
│   └── data/               sheets: authors, locations, partners, social-links, media-contacts
├── language-masters/en/     temporary: published at /language-masters/en for review
└── site-global-templates/  template pages authors copy; excluded from publishing
```

Path mapping (`paths.json`, generated by `npm run build:paths` from `LOCALES` and `REVIEW_ROOTS`). One generic set per site root: the folder mapping (`/content/tcs-corp/de/de-de/:/de-de/`: child pages and sheets), the Home page mapping (`/content/tcs-corp/de/de-de:/de-de/`) and the taxonomy page (`…/taxonomy:/de-de/taxonomy.json`): 39 mappings for the locales, 3 temporarily for the language master. The sheets (`placeholders`, `metadata`, `redirects`, `data/*`) need no mapping of their own: the folder mapping publishes them at the same path and spreadsheets get `.json` appended; the former 112 per-sheet mappings are kept commented in the generator. `includes`: the 13 locale pages, `/content/dam/tcs-corp/` and `/content/tcs-corp/language-masters/en`; `excludes`: each locale's `block-library` and `block-library/**`, and `site-global-templates` with its children. AEM applies the site configuration's `public.json` → `paths`, not the repository file (AGENTS.md "Notes").

Locale resolution (code). `getLocale(pathname)` reads the AEM path when editing (`/content/tcs-corp/de/de-de/…`) and the published path on the site (`/de-de/…`); pages outside a locale (Site Global Templates, drafts) get `en`. Helpers: `localePagePath`, `resolvePagePath`, `localeSheetPath`, `toPublishedPath`, `toAemPath`, `pageLink`, `getParentPages`, `sheetRows`.

Review roots (KB-D-068). There are three review roots, each published as a site of its own (KB-D-068): `/language-master/en` (testing page functionality with dummy content, content limitations), `/language-masters/en` (the language master: the migrated tcs.com content, validated with its real sheet data, later the source of the global site and the regions and languages; MSM or not is open until the content is final) and `/design-block-master/en` (the block library and the curated design pages). Pages below each (`REVIEW_ROOTS`) list their own pages from the query index (`indexRoot`), read their own placeholders, redirects, taxonomy and data sheets (`REVIEW_SHEETS`), use their own `references/` and `error-pages/` (`REVIEW_PAGES`) and inherit up to their root page. The temporary review of the language master ends with its rollout (handover Part 3.9, finding F-88).

Language master URLs (validated 23 October 2026). The mappings follow Adobe's path mapping rules (https://www.aem.live/developer/authoring-path-mapping): the folder mapping `/content/tcs-corp/language-masters/en/:/language-masters/en/` covers every child page, and the page mapping `/content/tcs-corp/language-masters/en:/language-masters/en/` is the special case that publishes the root page as `/language-masters/en/index`, served at `/language-masters/en/` (the same pattern as `/en/`). No other mapping

is needed. The mappings of every language master come from `REVIEW_ROOTS` (`utils/site.js`): `tools/paths/generate-paths.mjs` writes this set for each root, so another language master (e.g. `/language-masters/de`) is added to the code and to `paths.json` with one entry and `npm run build:paths`; `test/site/paths.test.mjs` checks every locale and language master is mapped this way and that `paths.json` is in step.

AEM page	Preview	Live
`/content/tcs-corp/language-masters/en` (Home)	`https://main--tcs-corp--sradha-tcs.aem.page/language-master/en/`	`https://main--tcs-corp--sradha-tcs.aem.live/language-master/en/`
`/content/tcs-corp/language-masters/en/<path>` (child pages)	`https://main--tcs-corp--sradha-tcs.aem.page/language-master/en/<path>`	`https://main--tcs-corp--sradha-tcs.aem.live/language-master/en/<path>`

The Home URL ends with `/` (`/language-masters/en` without it is a 404, as `/en`). On 23 October 2026 the child pages, sheets, references and error pages resolved, but the Home page was a 404 and wasn't in the query index: it has to be previewed and published from AEM (Manage Publication on `/content/tcs-corp/language-masters/en`). If it still returns a 404 after publishing, check that the site configuration's `public.json` → `paths` has the page mapping (AEM applies that configuration, not the repository file).

Rules.

- Add or change a locale only in `LOCALES`, then run `npm run build:paths` and commit both files; the site administrator re-applies `public.json` and `metadata.source`.
- Nothing is published at `/`: a `/` → `/en/` redirect needs another mechanism (handover F-36).

Common mistakes. Editing `paths.json` by hand (it is overwritten by the generator); expecting a repository `paths.json` change to take effect without the site configuration; creating pages directly under `/content/tcs-corp` (not published).

3. Page structure

Purpose. A page is a sequence of sections; each section holds default content (text, headings, images, links) and blocks. The structure is the contract between author and code (AGENTS.md "Blocks").

Page properties (`models/_page.json`, model `page-metadata`):

Field	Name	Effect
Title	`jcr:title`	`<title>`, `og:title`
Description	`jcr:description`	`description` meta
Keywords	`keywords`	`keywords` meta
Page Type	`page-type`	Default (empty), Home `home`, Article `article`, Blog `blog`, Press Release `press-release`: body class and template
Navigation	`nav`	Header page, inherited by child pages
Footer	`footer`	Footer page, inherited by child pages
404 Page	`error-page`	404 page of the subtree, inherited

Page tags are not a field of this model; they come from the standard AEM page properties (`cq:tags`, published as `cq-tags`) or a `tags` column of the bulk metadata sheet (handover Part 7.1.1).

Section properties (`models/_section.json`), applied by `decorateSectionsWithOptions` in `scripts/scripts.js`:

Field	Name	Delivered as	Result
Section Name	`name`	—	Not one of the section options read by `scripts.js`
Style	`style`	section classes	Default, Grey `bg-grey`, Dark `bg-dark`, Dark Green `bg-dark-green` (colour only; dark ones apply the theme contract in `styles/styles.css`)
Width Type	`sectionWidth`	`data-sectionwidth`	`fixed-width` class (full width is the default)
Rounded edge	`roundedEdge`	`data-roundededge`	`rounded` class
Background Image / Image alt text	`backgroundImage`, `backgroundImageAlt`	data attributes	`.section-background` with a responsive picture, `has-background`
Anchor name	`anchorName`	`data-anchorname`	section `id` (suffixed `-2`, `-3` when taken)

Documents that use a `section-metadata` block are read the same way (rows `Section Width`, `Rounded Edge`, `Background Image`, `Background Image Alt`, `Anchor Name`).

Container sections. Gen Carousel, Tabs Container and Columns Container are sections whose content `scripts.js` wraps in a synthetic block (`gen-carousel`, `tabs`, `columns-container`). Sections don't get `classes_*` fields delivered, so their Additional tab fields are `container*` data attributes turned into block classes by `applySectionBlockOptions` (AGENTS.md "AEM delivery rules"; columns-container, tabs, gen-carousel).

Blocks and default content. The `section` filter allows 26 components (`component-filters.json`); the inventory and each block's content contract are in blocks/README.md. Repeating content is item children added with Add, not composite multifields (AGENTS.md "Item children").

Common mistakes. Putting a section inside a Tabs Container (a section can't hold a section: use Column Breaks); relying on a section `classes_*` field (not delivered); adding sample text to templates or model defaults (stored on every new component).

4. Templates

Purpose. A Page Type gives recurring page kinds their structure. Template pages that authors copy live in `/content/tcs-corp/site-global-templates/` (not published).

Structure. `loadTemplate` (`scripts/scripts.js`) loads `templates/{name}/{name}.js` and `.css` when the Page Type (`page-type`, else `template` metadata) is in `TEMPLATES`:

Page Type	Adds	Data
`home`	Layout styles only	—
`article`	A Page Header from the page metadata (unless authored), "About the author"	Authors sheet (`author-id`)
`blog`	The article structure and styles	Authors sheet
`press-release`	A Page Header (unless authored)	Page metadata (the media contacts are the Media Contacts block, KB-D-054)

Source: templates/README.md (including how to add a template).

Rules. The Page Type page property wins over the metadata sheet's `page-type` column; Page Type also becomes a body class. Changing a template page doesn't change pages already copied from it (handover

Part 4.8).

Common mistakes. A `page-type` value that isn't in `TEMPLATES` (only the body class is added); expecting template content in the query index (templates build DOM in the browser).

5. Metadata model

Purpose. Page-level values (SEO, social, card content, page type, author, dates, flags) that code, the query index and the data layer read. Every value is a `<meta>` in the server-delivered head.

Sources and precedence.

Source	Where authored	Published as
Page properties (`models/_page.json`)	Universal Editor page properties	`<title>`, `description`, `keywords`, `page-type`, `nav`, `footer`, `error-page` metas
Standard AEM page properties	Sites console (e.g. Tags `cq:tags`)	`cq-tags` (one comma-separated meta)
Bulk metadata sheet	`/content/tcs-corp/{country}/{language}/metadata`	`/{locale}/metadata.json`; each column becomes `<meta name="{column}">` on pages matching the row's `url` pattern

Page-level values win over bulk metadata; bulk rows apply top to bottom (handover Part 8.1.2). The bulk sheets apply only once all 13 are listed in the site configuration's `metadata.source`, and each sheet's URL patterns must start with its locale (`/de-de/**`) (AGENTS.md "Notes").

Bulk metadata columns (AGENTS.md, 41 columns): `url`, `page-type`, `meta-title`, `meta-description`, `meta-keywords`, `author-id`, `publish-date`, `modified-date`, `reading-time`, `og-title`, `og-description`, `og-image`, `og-type`, `twitter-card`, `twitter-title`, `twitter-description`, `twitter-image`, `canonical-url`, `robots`, `region`, `country`, `language`, `persona`, `thumbnail-image`, `featured-image`, `campaign-id`, `content-type`, `featured`, `priority`, `listen-link`, `author`, `card-title`, `card-description`, `card-image`, `card-image-alt`, `image-alt`, `eyebrow`, `category`, `hide-in-nav`, `card-style`, `card-theme`.

Consumers.

Consumer	Reads	Source
SEO decorator (eager)	`meta-title` → `<title>`; `meta-description`, `meta-keywords`; `og-*` (falling back to `meta-*`); `twitter-*` (falling back to `og-*`, `meta-*`); `canonical-url` → `<link rel="canonical">` and `og:url`; `language` → `<html lang>`	`decorators/seo-decorator.js`
`getPageMetadata()`	title, description, image (`featured-image`, else `og:image`), `page-type` (else `template`), `content-type`, `category`, tags, `author-id`, `author`, `publish-date` (else `publication-date`), `modified-date`, `reading-time`, `listen-link`, `thumbnail-image`, `featured-image`, `image-alt`, `region`, `country` (else the locale's), language, `persona`, `campaign-id`, `featured` (yes / true / 1 / on / y), `priority` (number)	`utils/metadata.js`
`getPageTags()`	`article:tag` metas and `cq-tags`, else a comma-separated `tags` meta	`utils/metadata.js`
`getPageLanguage()`	`language` when it is a language code, else the locale's `lang`	`utils/metadata.js`
Page Header, templates, data layer page entry	`getPageMetadata()`	`page-header`, `decorators/data-layer-decorator.js`
Query index	the server head (see indexing-and-search.md)	`helix-query.yaml`

Rules. The query index reads the server-delivered head, so values set by JavaScript (the SEO decorator's) aren't indexed (handover Part 8.1.3). `robots` is a standard name and works server side. Use `card-*`, `eyebrow`, `hide-in-nav`, `card-style`, `card-theme` for List cards (List README "Page metadata contract").

Example.

```
url                   | page-type      | hide-in-nav
/de-de/newsroom/**    | press-release  |
/de-de/insights/**    | article        |
/de-de/legal/imprint  |                | yes
```

Common mistakes. A sheet row pattern without the locale prefix; adding columns to a sheet that isn't listed in `metadata.source` (no effect); expecting a metadata change to reach lists before the page is republished and indexed. Field reference per column: handover Part 8 — Metadata Documentation (8.2, 8.4).

6. Taxonomy

Purpose. Tag IDs on pages and in block fields are turned into localised labels at runtime.

Structure. Tags are maintained in AEM Tagging; a taxonomy page per locale (`/content/tcs-corp/{country}/{language}/taxonomy`) is published as `/{locale}/taxonomy.json` with columns `tag` and `title`. With translated titles the JSON is multi-sheet: `loadTaxonomy()` reads the sheet named after the locale's ISO 639-1 language (`de`), else `default`, else the first sheet (`sheetRows`).

Consumers. `resolveTags(values)` (`utils/taxonomy.js`): a value with a taxonomy entry gets its title; a value containing `:` or `/` without an entry gets a label derived from the ID (`parseTag`, `utils/tags.js`); any other value is shown as written. Used by Teaser, Page Header, List, Cards, Filter List and Conversation Canvas.

Rules. The published namespace is `tcs-corp:` (handover Part 7.1.2; finding F-08: some help texts and examples use `tcs:`). Pick tags with the tag picker; the `aem-tag` fields don't restrict the namespace (handover Part 7.1.1). A selected tag also matches its child tags in List (`hasTag`, handover Part 7.2.2).

Example. `tcs-corp:region/europe` → `Europe` from `/en/taxonomy.json`; without a taxonomy entry, an ID-derived label.

Common mistakes. Typing tag IDs by hand; not publishing the locale's taxonomy page (labels fall back to IDs; handover F-09 records that only `/en/taxonomy.json` was published when checked); translating tag titles without republishing the taxonomy page. Governance and lifecycle: handover Part 7 — Taxonomy Documentation.

7. Placeholders

Purpose. All user-facing UI text in code is authorable and translated per locale (AGENTS.md "Accessibility": all UI text comes from placeholders).

Structure. `/{locale}/placeholders.json` with columns Key, Text, Description, Category; code uses Key and Text. Keys are compared without case and separators (`Pause Slides`, `pause-slides` and `pauseSlides` match `pauseSlides`). `{name}` variables are filled by `fillPlaceholder`.

Consumers. `const t = await getPlaceholders(); t('nextSlide', 'Next slide')` in every block with UI text, the header, footer and `scripts.js` (404 "Go back", key `goBack`). Source: `utils/placeholders.js`; key lists per block in the block READMEs ("For Site Administrators").

Rules. Code always passes an English fallback. There is no fallback from a locale's sheet to the English sheet: a key missing in `/de-de/placeholders.json` shows the code's English text (handover Part 4.9.3).

Common mistakes. Editing the sheet without publishing it; adding a new key in code without adding it to the 13 sheets. Full key list: handover Part 4.5.3.

8. References, fragments and inherited metadata

Purpose. Pages that aren't content pages but are included in other pages: the header navigation, the footer, alternatives per site area, the footer Canvas page, modals and fragments.

Structure (in each locale's `references/`):

Page	Used by
`nav`	Header (default Navigation of the locale)
`footer`	Footer (default Footer of the locale)
`nav-{name}`, `footer-{name}`	Header / Footer when selected in a page's Navigation / Footer (or the `nav` / `footer` sheet column)
Footer Canvas page	The Footer block's Conversation Canvas Page
`modals/{name}`	Links whose path contains `/modals/` open as a modal (`autolinkModals`); the Modal block
Fragments	Fragment block; Tab fragments

Inheritance (`getInheritedMetadata(name)`, `utils/inherited-metadata.js`, for `nav`, `footer`, `error-page`):

1. The page's own value (page property or bulk sheet column).
2. Else the nearest parent page's value up to the locale's Home page (`getParentPages`; for the language master up to `/language-masters/en`): the parents' published HTML heads are fetched in parallel.
3. Else the default: `references/nav`, `references/footer`, or for 404 `error-pages/404`.

Values may be AEM paths, published paths or paths relative to the locale root (`references/nav-careers`); `resolvePagePath` returns the AEM path in the editor and the published path on the site. Parent values are cached in `sessionStorage` (`tcs-inherited-metadata`) for the browser session.

Consumers. header (falls back to the locale's `references/nav` when a configured page can't load; with none published the header stays empty), footer (adds the Social Links sheet), `loadErrorPage` (404), fragment, modal, tabs.

Rules. Keep reference pages in `references/` (excluded from the query index and the sitemap); `fragments/` and `modals/` folders anywhere are excluded too. A fragment change shows on every including page once the fragment is published. `loadFragment(path, from)` fetches only paths starting with `/` and stops recursion by the chain of including fragments.

Example. Navigation = `references/nav-careers` on `/en/careers` gives every page below it the Careers header unless a page sets its own.

Common mistakes. Testing a changed Navigation in the same browser session (cached); leaving `nav` unpublished in a new locale (empty header); placing a fragment outside `references/`, `fragments/` or `modals/` (it gets indexed and listed).

9. Data sheets

Purpose. Shared structured data per locale, maintained by authors as AEM spreadsheets.

Structure and consumers (`utils/data.js`; each sheet loaded once per page view, values trimmed, missing sheets give empty lists):

Sheet (/{locale}/data/…json)	Columns	Read by
authors	author-id, name, designation, bio, image, linkedin, region	Page Header (getAuthor, case-insensitive id), article / blog templates, List feature card and Filter List spotlights (getAuthors)
social-links	platform, url, icon, sort-order	Footer (in sort-order, rows without one last)
media-contacts	id, name, designation, email, phone, region	Media Contacts block (blocks/media-contacts: inRegion by the page's region, else all; grouped by designation and region)
locations	location-id, location-name, country, region, latitude, longitude	getLocations() only; no component reads it (handover F-40)
partners	partner-id, partner-name, logo, website, tier, description	getPartners() only; no component reads it (handover F-40)

Rules. The query index excludes /*/data/**. Sheet JSON is public: the Authors and Media Contacts sheets publish names, e-mail addresses and phone numbers by design. Column reference and authoring steps: handover Part 9 — Data Sheets Documentation.

Common mistakes. An author-id on a page that differs from the sheet in case (Filter List and List match exactly; Page Header case-insensitively, handover F-30); unpublished sheets in non-English locales.

10. Shared content

Mechanism	What is shared	Included by
Header / Footer	references/nav, references/footer or inherited alternatives	Every page, automatically
Fragment block	A reference page	Fragment block Reference field
Footer Canvas	A page with a Conversation Canvas block	Footer block field
Tabs	A fragment per tab	Tab Fragment field
Modals	Pages under a /modals/ path	Any link, or the Modal block
Placeholders, taxonomy, data sheets	UI text, tag labels, structured data	Code, per locale
Block library	Every block variation per locale	Not included in pages; excluded from publishing (the language master's is published for the review, handover Part 3.9)
Site Global Templates	Starter pages	Copied by authors; not published

All Conversation Canvases of a page (header, body, footer) share one floating journey (AGENTS.md "Floating Canvas", conversation-canvas).

11. Navigation model

Area	Model	Source
Header	The Navigation page (inherited) has the brand (TCS logo), the sections (the menu, any depth) and the tools (TATA logo). Desktop (1200px and wider) shows the logo bar without navigation; tablet and mobile open the menu full screen one level at a time	AGENTS.md "Header", blocks/header/README.md

Area	Model	Source
Site navigation on desktop	The Conversation Canvas: its pop-up menu and the Primary dock list pages from the query index below a Menu root path	conversation-canvas README, indexing-and-search.md
Footer	The footer page's Footer block: Conversation Canvas Page, Copyright Text (`{year}`), Footer Link items; social icons from the Social Links sheet	blocks/footer/README.md
In page	Anchor List links to section Anchor names and block Anchor IDs (`anchor-…`); the URL hash scrolls after the sections load; `tabs:reveal` opens the tab holding a target	anchor-list, `scripts/scripts.js`
Per site area	Set Navigation / Footer on the area's top page	inheritance above

Not implemented: a breadcrumb, a language selector and `hreflang` alternates (handover Parts 4.9.6 and 16.4).

12. Error pages

Purpose. An authorable, locale-specific 404 page.

Structure. Edge Delivery serves `404.html` for a missing path; `loadErrorPage` (`scripts/scripts.js`) tries, in order: the inherited **404 Page** (`error-page`) of the missing page's nearest parent; the locale's `/{locale}/error-pages/404`; `/en/error-pages/404`. A page without content (no text, image, picture, video or iframe) is skipped; when none loads the built-in content ("404", "Page Not Found", "Go home" to the locale's Home) stays. Content from another locale gets its `lang` on `<main>`; "Go back" (placeholder `goBack`) is added for same-origin referrers.

Rules. 403 and 500 aren't served by Edge Delivery; the authored `error-pages/403` and `500` are for a CDN (not configured: Requires validation). Error pages are excluded from the query index (`/*/error-pages/**`). Details: handover Part 10 — Error Pages.

Common mistakes. An empty `error-pages/404` (skipped, the next candidate shows); expecting the 404 Page property of the missing page itself (the lookup starts at its nearest existing parent).

13. Redirects

Purpose. Per-locale lists of moved pages.

Structure. `/{locale}/redirects.json` with Source and Destination columns (either case).

Rules and status.

- The platform applies only a root `/redirects.json`; the per-locale sheets are **not applied** automatically. Redirects, including `/` → `/en/`, need another mechanism such as CDN rules (AGENTS.md "Notes"; handover Part 3.8; findings F-36, 16.4). Status: Requires validation (no mechanism configured).
- In code only the List block uses the sheet: `resolveRedirect` points card links to the final target, at most 5 hops, stopping at loops and external destinations (`utils/redirects.js`). Filter List, navigation and running text links aren't rewritten.
- Sources must start with `/` (rows without a destination are ignored); paths are compared without query, hash, `.html` and trailing slash.

Common mistakes. Expecting a sheet row to redirect visitors; chains of redirects (point sources at final destinations).

14. Localization

Topic	Implementation	Status
Locale structure	13 separate page trees; locale resolved from the path	Active
`<html lang>`	`language` metadata when valid, else the locale's tag	Active
UI text	Placeholders sheet per locale; English fallback in code	Active
Tag labels	Taxonomy page per locale, language sheet when translated	Active (only `/en` taxonomy published when checked, F-09)
Bulk metadata	One sheet per locale; all 13 in `metadata.source`	Requires validation (F-11, open item O-04)
Data sheets	One set per locale	Active
Lists	Each locale lists only its own pages (`indexRoot`)	Active
Dates	`utils/dates.js` `formatDate`; English month abbreviations in every locale (handover Part 4.9.7)	Active
Translation process, MSM / language copies, language selector, `hreflang`	Not in code	Not recorded

15. Content governance

Verified rules (enforced by code or configuration):

- Locale resources sit at fixed locations (`references/nav`, `references/footer`, `error-pages/404`, the sheets, `data/*`): code resolves them per locale.
- Navigation, Footer and 404 Page are inherited from parent pages up to Home.
- `hide-in-nav = yes` keeps a page out of List results (not Filter List or the Canvas menu: F-30, F-33).
- Reference, error, data, nav / footer, fragments and modals paths are excluded from the query index.
- Sheets and reference pages take effect only once published.
- No sample content in templates or model defaults (AEM stores definition template values on every new component).

Ownership, review cadences and approval flows: Not recorded in the repository (handover Parts 7.4, 14 and 17 describe proposed processes and open owners).

16. Publishing and indexing dependencies

Change	Must also be published / done	Effect
New or changed page	Publish the page	Index row and sitemap update; lists show it (see indexing-and-search.md)
Page metadata used by cards	Republish the listed page	List / Filter List cards change; the List page itself needs no republish
Sheet (placeholders, metadata, taxonomy, redirects, data)	Publish the sheet	Next page load (each loaded once per page view)
Bulk metadata	Sheet listed in `metadata.source`	Metas on matching pages
Navigation / footer / fragment	Publish the reference page	Every including page; inherited values in a new browser session
Non-image DAM assets (Lottie JSON, videos, captions)	Publish the assets	Served only once published; images travel with the page (AGENTS.md)
`helix-query.yaml`	Reindex (Admin API index operation) or republish	Existing rows unchanged until then
`LOCALES`	`npm run build:paths`, re-apply `public.json` and `metadata.source`	New paths published

17. Related

- architecture.md — runtime, delivery, integrations, caching
- indexing-and-search.md — query index, listings, sitemap, reindexing
- blocks.md, project-overview.md, repository-structure.md — related knowledge base files
- AGENTS.md — "Site structure and site data", "AEM delivery rules"
- utils/README.md, templates/README.md, blocks/README.md
- Handover (TCS-CORP-HANDOVER.md): Part 3 — Site Structure, Part 4 — Content Architecture, Part 7 — Taxonomy Documentation, Part 8 — Metadata Documentation, Part 9 — Data Sheets Documentation, Part 10 — Error Pages

18. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

19. Known limitations

- None recorded for this document.

20. Open items

- /design-block-master/en content still links to /language-master/en and to /content/tcs-corp/design-block-master/english (see the curated design page audit).

21. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Deployment Architecture

Attribute	Value
Document ID	ARC-006
Audience	Solution architects, technical leads, developers, AI agents
Purpose	Delivery streams, environments, promotion, content packages, publishing, reindexing and rollback
Scope	—
Prerequisites	—
Owner	Solution Architect (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/deployment-and-release.md, .github/workflows/main.yaml
Related documents	—
Related AEM paths	—

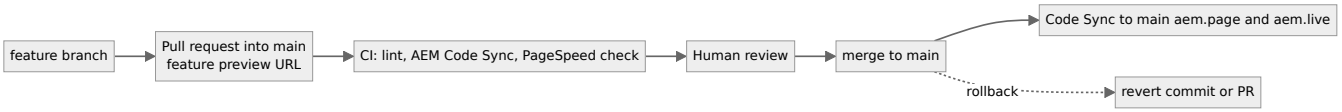
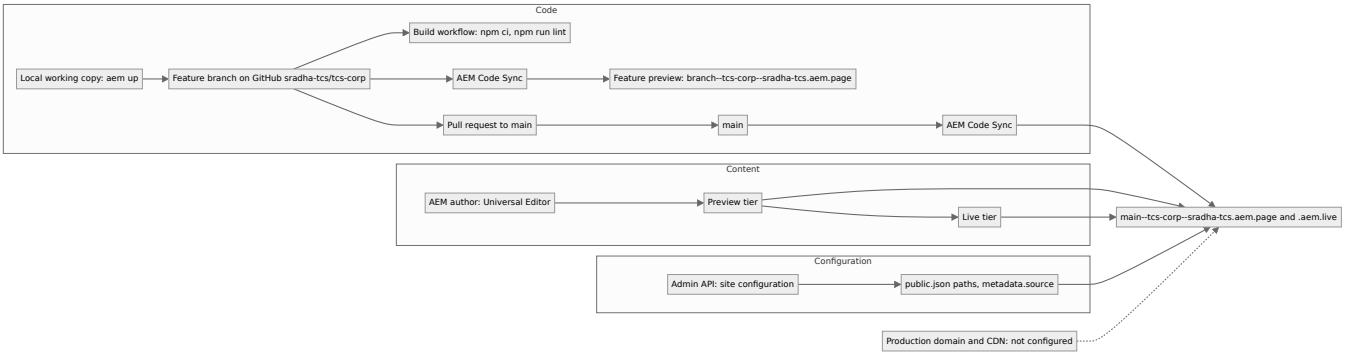


Diagram: code release flow (source: diagrams/release-flow.mmd).

1. Delivery streams

Content, code and configuration are released independently: a code merge doesn't publish content and publishing content doesn't deploy code. A release that needs both goes **code first, then content** (handover Part 12, introduction).



2. Local development

Task	Command	Notes
Install dependencies	`npm install`	Also installs the Husky pre-commit hook (`prepare: husky`)
Start the dev server	`npx -y @adobe/aem-cli up --no-open --forward-browser-logs` (or `aem up` with the CLI installed globally)	`http://localhost:3000`: local code, previewed content
Serve static test content	Add `--html-folder drafts`	Files in `drafts/` (`.html` or `.plain.html`); 14 files are tracked
Lint	`npm run lint` (`lint:js`: ESLint on `.json`, `.js`, `.mjs`; `lint:css`: Stylelint on `blocks/**/*.css`, `styles/*.css`)	`npm run lint:fix` fixes what it can
Unit tests	`npm test` (`node --test 'test/**/*.test.mjs'`)	Pure modules, model layout, docs, module graph
Rebuild the Universal Editor files	`npm run build:json`	`component-models.json`, `component-definition.json`, `component-filters.json` from the partials
Regenerate the path mapping	`npm run build:paths`	After changing LOCALES in utils/site.js; commit both
Regenerate the module graph	`npm run build:modules`	tools/modules/generate-module-graph.mjs, used for module preloading
Regenerate block READMEs	`npm run build:docs`	Generated sections of the block READMEs (blocks/README.md)

Inspect delivered content with `curl http://localhost:3000/{path}`, `.md` and `.plain.html` (AGENTS.md). Node.js: README.md states 20 or newer; the workflows install Node 24. The local preview renderer used by the audits (`.home-work/render-lib.mjs`, named in AGENTS.md) is not tracked in Git.

3. Branches and feature previews

- `main` is production. Changes go through short-lived branches merged with pull requests (130 merge commits on `main` at 2d06316, latest #129).
- Branch prefixes in use: `feature/`, `feat/`, `fix/`, `perf/`, `docs/` (for example `fix/block-reaudit`, `perf/module-preload`, `docs/block-readmes`). The handover's `feature/tcs-<topic>` convention (12.2.2) is not what recent history uses; no naming rule is enforced.
- AEM Code Sync serves every pushed branch at `https://{branch}--tcs-corp--sradha-tcs.aem.page/` (slashes become hyphens; keep names short and lower case: branch, repository and owner form the host name).

4. Pull requests

The publishing process of AGENTS.md:

1. Push the feature branch; Code Sync makes it available on the feature preview.
2. Run PageSpeed Insights on the feature preview URL and fix issues; target a score of 100.
3. Open a pull request to `main`. The description **must** link `https://{branch}--tcs-corp--sradha-tcs.aem.page/{path}` showing the change, or the PR is rejected. Without a page, create test content in `drafts/` and ask an author to create a CMS page to link.
4. Check the status of code sync, linting and performance tests (`gh pr checks`).
5. A human reviewer reviews the code, opens the URL and merges; Code Sync updates `main`.

.github/pull_request_template.md asks for the GitHub issue (`Fix #<gh-issue-id>`) and Before / After test URLs. It still contains the boilerplate placeholders `{repo}` and `{owner}`: replace them with `tcs-corp` and `sradha-tcs` (the `cleanup-on-create` workflow that would replace them only runs on repository creation or manual dispatch).

Approval rules: the number of required reviews, required checks and branch protection on `main` are not recorded (handover 12.3; open item O-05).

5. Automated checks

Check	Defined in	When	What it does	Status at 2d06316
Husky pre-commit	.husky/pre-commit → .husky/pre-commit.mjs	Every local commit	If staged files include a model partial (`_*.json`), runs `npm run build:json` and stages the three aggregated files	Aggregated files match the partials: Passed
Build	.github/workflows/main.yaml	Every push, every branch (`on: [push]`)	`npm ci`, then `npm run lint` (ESLint with `airbnb-base`, `plugin:json`, `plugin:xwalk/recommended`; Stylelint `stylelint-config-standard`)	`npm run lint` locally: Passed
Unit tests	package.json `test`	Manual only: not in any workflow	311 tests in 53 suites (`test/`)	`npm test` locally: Passed (311 / 311)
cleanup-on-create	.github/workflows/cleanup-on-create.yaml	Repository creation or manual dispatch	Boilerplate clean-up; no role in releases	Not applicable
Renovate	.renovaterc.json	Renovate schedule	Dependency PRs; devDependency PRs labelled `ignore-psi-check` and auto-merged when tests pass; ESLint pinned to `^8.0.0`	Not applicable
AEM Code Sync	GitHub app (not a repository file)	Every push	Syncs the branch to the code bus and reports a check	Not verified here (no GitHub access)
PageSpeed (`aem-psi-check`)	AEM GitHub app (not a repository file)	Pull requests	PageSpeed Insights on the URLs in the PR description	Not verified here

The CI gate is lint only. Run `npm test` before every PR; adding it to `main.yaml` is a recommendation of the handover (14.9 #19, F-37), not a decision.

6. Content packages

The block library, the curated design pages and sample pages are delivered to AEM as content packages built **outside the repository**:

- The build and migration tooling lives in the delivery workspace (`.home-work/bl/`, for example `bl-migrate.mjs`, which maps old fields to new ones when the pages and the package are built); it is not tracked in Git (BLOCK-ECOSYSTEM-AUDIT.md, section 11).
- The packages are distributed on the branch `home-content-package` of the repository (files only, no site code; its README describes each package). For example `block-library/tcs-corp-block-library-content-1.0.13.zip` (current), the curated design pages, Home and Columns Container packages and the matching sheet rows (placeholders, metadata, taxonomy, data CSVs).
- Install with AEM **Package Manager** (Upload Package, Install). Installing replaces each page of the package: changes made in AEM to those pages since the last install are overwritten; pages outside the package are kept.
- Tags `home-content-package_<date>` mark earlier states of that branch ("Downloads branch before the handover PDF …"); they are not code releases.

What the handover records (revision history):

Package	Change	Publication recorded
1.0.3, 1.0.4	Block library Animation folder	Published with the language master review (2.20)
1.0.7	Library migrated after the block audit	Not recorded (2.27)
1.0.8	Columns → Columns Containers / Column Breaks, buttons → Button blocks	Not recorded (2.28)
1.0.9	Re-audit fixes; language-master Placeholders (109) and Metadata (311 rows) imported	312 / 312 previewed and published; defect R-L-1 found on live (2.30)
1.0.10	Container sections' `container*` fields	312 / 312 previewed, published and reindexed (2.31)
1.0.12	Section Style colours (with the 1.0.11 Teaser and Columns Container options)	313 / 313 published and reindexed (2.35)
1.0.13	List Item display options, dark-background theme	315 / 315 previewed, published and reindexed (2.35)

7. Environment promotion

Step	From → to	How	Owner
1	Local → feature preview	Push the branch; Code Sync	Developer
2	Feature preview → `main` (preview and live code)	Reviewed PR merged; Code Sync updates `main--tcs-corp--sradha-tcs.aem.page` and `.aem.live`	Reviewer
3	Universal Editor	Authors reload the editor to get new definitions, models and filters	Content team
4	Configuration	Apply `public.json` paths, `metadata.source`, sitemap or CDN settings through the configuration Admin API, saving the current JSON first	Site Administrator
5	Content: author → preview → live	Publish sheets, then fragments, then pages	Content team

There is no separate staging environment in the repository: feature previews and the `main` preview tier play that role. The author environment reads the code of `main` (fstab.yaml mount point). The production domain and CDN are not configured (open item O-03).

8. Installation validation

After a code release or a package install, validate on the preview tier first, then on live:

- ☐ Code Sync check green on the merge commit; a changed file on `main--tcs-corp--sradha-tcs.aem.page/blocks/…` matches the commit (Part 17, R-06).
- ☐ The Universal Editor shows the new fields and components after a reload (R-08).
- ☐ After a package install: every package page previewed, published and, where listed, reindexed, with the count recorded (as in 312 / 312 for 1.0.10).
- ☐ Changed pages and one page per changed block render without `failed to load module for …` or `failed to load template …` (V-02).
- ☐ Header, footer and 404 page correct per locale; sheets reachable (V-03, V-08).
- ☐ Non-image DAM assets used by the pages published (Lottie, video, captions).
- ☐ `aem.page` compared with the local preview block by block (docs/audit/tools/compare.mjs): this comparison found R-L-1 and R-L-2 after 1.0.9.

9. Publishing

Content publishing (Universal Editor, Sites console, sheets, references, assets, scheduling) is described in authoring.md; Admin API operations in administration-and-operations.md. In a release, publish in this order: sheets (placeholders, taxonomy, data, metadata) → fragments (nav, footer, other references) → pages (handover 12.4.2, Part 17 R-09). Content that depends on new code is prepared in AEM but published only after Code Sync of the merge.

10. Reindexing

- Publishing updates the page's row in `/query-index.json` automatically.
- After a change to helix-query.yaml, rows are rewritten only when pages are republished or re-indexed: `POST https://admin.hlx.page/index/sradha-tcs/tcs-corp/main/{path}` for the affected paths (bulk paths agreed with the site owner), then check the columns (Part 17, R-10, W-01).
- The packages 1.0.10, 1.0.12 and 1.0.13 were reindexed page by page (Admin API `preview`, `live`, `index` per page); after 1.0.10 the query index listed 307 language-master pages (handover revision 2.31).
- The sitemap (helix-sitemap.yaml) is built from the index.

11. Smoke and regression testing

Test	Command / tool	Scope	Result at 2d06316
Unit tests	`npm test`	`test/` (cards, conversation-canvas, docs, items, list, media, models, modules, motion, privacy, separator, site)	Passed (311 / 311)
Lint	`npm run lint`	JS, JSON, models (xwalk rules), CSS	Passed
Generated files	`npm run build:json` then check that nothing changed (Part 17, R-03)	Aggregated component files	Passed (compared with a fresh merge)
Runtime harness	docs/audit/tools/harness.mjs	Every block library and curated page of the local preview at 1440, 1024, 768, 600 and 390 px: console and page errors, failed requests, block status, overflow, duplicate ids, broken images, data layer, axe (WCAG A / AA) at 1440 and 390, a click test	Not run for this page
Published vs local	docs/audit/tools/compare.mjs	Block-by-block signature of `aem.page` against the local preview	Not run for this page
Inventory	docs/audit/tools/inventory.mjs	Definitions, models, filters, fields, options and their usage in the content package	Not run for this page
Defect register	docs/audit/tools/register.mjs	The audit's defects and outcomes	Not applicable
Manual	Part 17 V-01 … V-09	Content, code, configuration, data layer, SEO, accessibility, locale launch, cache	Not recorded

The audit tools hard-code paths of the delivery workspace (a Playwright install, an axe copy under `(a local workspace path)`, `drafts/en/block-library`, the unpacked content package under `.home-work/`), which are not in Git: re-running them from a clean clone is Blocked until those inputs are provided. Their recorded runs: 940 page loads before and after the first audit, and the re-audit runs (BLOCK-ECOSYSTEM-AUDIT.md, REAUDIT-2026-10-20.md, data/runtime-before-after.csv). The block libraries in each locale (`block-library/`) are the manual regression pages for every variation.

12. Rollback

No rollback has been recorded; the procedures below are the handover's (Part 12, 12.6) and Require validation.

Stream	Procedure
Code	Revert the merged PR in GitHub (a revert PR, checks, merge) or `git revert -m 1 <merge-commit>` on a branch, newest first; Code Sync deploys the previous code. Never force-push or rewrite `main`
Models	The revert restores partials and aggregated files; authors reload the editor; values saved with removed fields stay stored but aren't shown
Content	Sites console **Timeline > Revert to this Version** (or Restore Version), check, publish again; sheets the same way; unpublish when no good version exists
Configuration	Re-apply the saved JSON (`public.json` paths from a previous `paths.json`: `git show <commit>:paths.json`; the previous `metadata.source` list); repository configuration files by reverting the PR, then re-index or regenerate the sitemap
Content package	Reinstall the previous package version from the `home-content-package` branch (installing replaces the package's pages)

Emergency change approval and the hotfix path are not recorded (handover 12.6.1, Part 17 R-15).

13. Release evidence

Evidence	Where	State
Change history	Merge commits of the pull requests on `main`; the handover revision history (versions 0.1 to 2.37)	Available
Validation reports	docs/audit/ (audit, re-audit, dialog options; CSV registers and runtime before / after)	Available
Publication records	Handover revisions 2.31 and 2.35 (counts of pages previewed, published, reindexed)	Available
Release tags	None on `main`; `package.json` keeps the boilerplate version `1.3.0`; no `CHANGELOG.md`	Not recorded
Change records	Change management process	Not recorded
Release checklists	Handover 12.4 and Part 17, 17.5 (R-01 … R-15) define what to attach	Defined, no completed record in the repository

14. Production readiness

Item	State
Code on `main`: lint, unit tests, generated files	Passed at 2d06316
Preview and live on `*.aem.page` / `*.aem.live`	Active
Production domain, CDN, push invalidation, robots.txt	Blocked: not configured (O-03)
Sitemap origin	`.aem.live` host until the production domain exists
Redirects (per-locale sheets, `/` → `/en/`)	Blocked: no mechanism applies them
Site configuration `public.json` paths and `metadata.source`	Requires validation (O-04)
Language master review (`REVIEW_*`, `/language-masters/en`)	Temporary; remove after the rollout to `/en`
Adobe Tags / Web SDK, consent	Blocked: measurement plan and consent model
Form back end, search service	Blocked (O-08, O-09)
Non-image DAM assets published	Requires validation (re-audit operations action)
Branch protection, required checks, support model and SLAs	Not recorded (O-05, O-13)
Universal Editor dialog checks on the real author	Blocked: author access (audit section 9)

15. Related

- authoring.md, administration-and-operations.md
- architecture.md, repository-structure.md, indexing-and-search.md, decision-log.md
- AGENTS.md: environments and publishing process
- package.json, .github/workflows/main.yaml, .github/pull_request_template.md, .husky/pre-commit.mjs, .renovaterc.json
- docs/audit/README.md and its tools
- Handover (TCS-CORP-HANDOVER.md): Part 12 — Publishing, Deployment & Release, Part 13 — Global Troubleshooting Guide (13.15 lint and PageSpeed failures, 13.22 Code Sync), Part 17 — Operations Guide (17.5 Release Activities, 17.6 Validation Activities)

16. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

17. Known limitations

- None recorded for this document.

18. Open items

- None.

19. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Integration Architecture

Attribute	Value
Document ID	ARC-004
Audience	Solution architects, technical leads, developers, AI agents
Purpose	Integration points: query index and search, Adobe Client Data Layer, Dynamic Media, forms, personalization, external embeds
Scope	—
Prerequisites	—
Owner	Solution Architect (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/indexing-and-search.md, docs/handover/knowledge-base/personalization.md, docs/handover/knowledge-base/data-layer-and-analytics.md, helix-query.yaml
Related documents	—
Related AEM paths	—

1. Integration status

Integration	Status
Query index and listing (List, Filter List, Cards from metadata)	Implemented
Adobe Client Data Layer (page, components, interactions)	Implemented
Adobe Tags, Web SDK, Analytics, CJA, Target, consent platform	Not implemented
Dynamic Media with OpenAPI delivery of all assets (KB-D-067)	Partially implemented: rendering Implemented, page rewrite Configured but not validated, asset picker Planned (PR #200 open)
Forms (Form Variations) submission back end	Not implemented (no back end connected)
Personalization	Not implemented
CDN redirects / root redirect / to /en/	Planned (on hold)
Production custom domain and Adobe Managed CDN	To be confirmed

2. Query index and search (section)

3. Query index definition

`helix-query.yaml` defines **four indexes** (8 November 2026): `pages`, published as `/query-index.json`, for the 13 locales, and one per review root (KB-D-068): `language_masters` (`/language-masters/en/query-index.json`), `language_master` (`/language-master/en/query-index.json`) and `design_block_master` (`/design-block-master/en/query-index.json`), with the same exclusions and columns. A page reads only its root's index (`indexUrlOf` in `utils/query-index.js`); the sitemap (`helix-sitemap.yaml`) is built from `/query-index.json`, so it lists the locales only.

Included: `/**`.

Excluded:

Pattern	Why (from the file's comments)
`/**.json`	Sheets and other JSON resources
`/*/references/**`	Fragments, nav and footer pages, modals of each locale
`/*/error-pages/**`	Error pages
`/*/data/**`	Data sheets
`/*/nav`, `/*/footer`	Nav and footer pages at the locale root
`/*/block-library`, `/*/block-library/**`	Block libraries (also excluded from publishing in `paths.json`; safeguard)
`/site-global-templates/**`	Template pages (not published; safeguard)
`/language-masters/*/references/**`, `/language-masters/*/error-pages/**`, `/language-masters/*/data/**`, `/language-masters/*/nav`, `/language-masters/*/footer`, `/language-masters/*/block-library`, `/language-masters/*/block-library/**`	The same for the language master under review, a folder level deeper (#131, F-108; to remove with the REVIEW settings after the rollout)
`/**/fragments/**`, `/**/modals/**`	Fragment and modal folders anywhere

The `/*/…` patterns name one path segment before the folder (e.g. `/en/block-library`); the language master (`/language-masters/en/…`) is a level deeper and has its own patterns. Until 21 October 2026 it had none: 166 block library pages, 6 reference pages and the 404 page of the language master were in the live index (F-108). After the fix (#131) a reindex reports "requested path does not match index configuration" for these pages but doesn't remove their rows: they were removed with the Admin API index DELETE (173 / 173; the pages stay published). The index went from 354 to 181 rows (137 language-master content pages, 44 `/en` pages), and the Lists and Filter Lists of the library and the curated Insights page still filled.

Columns (37 properties, all read from the server-delivered `<head>`):

Group	Index columns (meta source)
Platform / standard	`lastModified` (`last-modified` header, parsed timestamp), `robots`, `title` (`og:title`), `description`, `image` (`og:image`, path only), `tags` (all `article:tag`, multi-value), `cqTags` (`cq-tags`, AEM page Tags as one comma-separated meta)
SEO	`metaTitle`, `metaDescription`
Classification	`pageType`, `contentType`, `category`, `eyebrow`
Author and dates	`author`, `authorId`, `publishDate`, `modifiedDate`, `readingTime`
Images	`thumbnailImage`, `featuredImage`, `imageAlt`
Audience	`region`, `country`, `language`, `persona`, `campaignId`
Featuring	`featured`, `priority`, `featureCard` and `featuredPriority` (earlier names, read as fallbacks)

Group	Index columns (meta source)
List cards	`cardTitle`, `cardDescription`, `cardImage`, `cardImageAlt`, `cardStyle`, `cardTheme`, `hideInNav`

Rules:

- Values set by JavaScript (e.g. the SEO decorator) aren't indexed: only the delivered head counts (handover Part 8.1.3).
- `/query-index.json` is public: don't put internal-only values in metadata (handover Part 2.10.7).
- A new metadata column used by a block needs a property here, then a reindex (List README "Extending the Block", Filter List README step 3).

4. Query index use in code

`utils/query-index.js` is the only reader of `/query-index.json`.

Export	Behaviour
`getPages()`	Fetches `/query-index.json?offset={n}&limit=500` until `total` is reached (or a request fails or returns no rows); keeps one promise per page view (module variable); filters rows to `indexRoot()`; maps each row to a typed page (`path`, `title`, `description`, `image`, `tags`, `pageType`, `contentType`, `publishDate`, `readingTime`, `thumbnailImage`, `featuredImage`, `author`, `authorId`, `featured`, `priority`, `lastModified`, `row`). A failed load gives `[]`
`tagsOf(row)`	Merges `tags` (JSON list or comma list) and `cqTags`, without duplicates (`test/list/query-index-tags.test.mjs`)
`columnOf(page, name)`	A column by its metadata name (`featured-image` → `featuredImage`)
`listOf(value)`	A multi-value column: JSON array or comma-separated
`getPagesAt(paths)`	Pages the index doesn't have (AEM author, or not indexed yet), read from their own HTML in parallel, once per path per page view (`rowOfDocument`); failures are left out

Typed values: `featured` is true for `yes`, `true`, `1`, `on`, `y` in `featured` or `featureCard`; `priority` is the number in `priority`, else `featuredPriority`, else `null`.

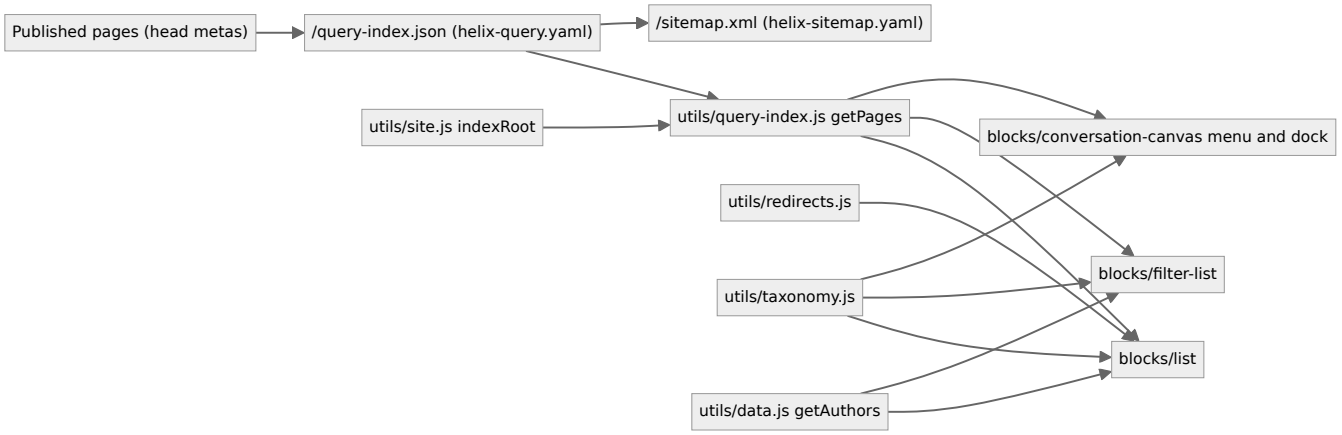
Locale scoping. `indexRoot()` (`utils/site.js`) returns the review root (`/language-master/en`, `/language-masters/en` or `/design-block-master/en`, `REVIEW_ROOTS`, KB-D-068) for pages below it, else the page's locale root (`/de-de`). `getPages()` keeps rows whose path is that root or below it (`test/site/index-root.test.mjs`). Every page view still downloads the rows of all locales.

AEM author. AEM author has no query index (`isAuthor()` in `utils/environment.js`: the path starts with `/content/`); `getPages()` then returns `[]` and blocks show editor hints (below). Fixed-list pages are read with `getPagesAt`.

5. Consumers

Consumer	Uses the index for	Source
List	Child pages, Tagged pages, card metadata of Fixed list pages	blocks/list/README.md
Filter List	All candidate pages (Query Index source) or the Manual List pages' metadata	blocks/filter-list/README.md
Conversation Canvas	The pop-up site menu and the Primary dock navigation (`path`, `title`)	blocks/conversation-canvas/README.md
Sitemap	`/sitemap.xml` source	`helix-sitemap.yaml`

Consumer	Uses the index for	Source
Cards	**Not** a consumer: authored fields only (tag labels from the taxonomy, UI text from placeholders)	blocks/cards/README.md



List

- **Sources:** Child pages (below the Parent page or the current page, direct children or down to Child depth), Tagged pages (below the Search root, Any / All of the Tags, never the current page; a tag matches its child tags, `hasTag`), Fixed list (List Items; pages, paths or external URLs).
- **Data rules:** pages with `hide-in-nav = yes` are not listed (`isListable`); card links point to redirect targets (`utils/redirects.js`, at most 5 hops); card values follow the page metadata contract (`card-title` → `og:title`, `card-description` → description, `card-image` → `og:image` / `thumbnail-image` / `featured-image`, …) and a List Item's authored fields win field by field.
- **Sorting:** Order by Manual (authored order for Fixed lists, path order for pages), Title (`Intl.Collator`, numeric, page language) or Last modified (`modified-date`, else last published); Sort order ascending / descending; ties by path. Deduplication by destination path; Maximum items applies last.

Filter List

- **Sources:** Query Index (pages below Root path, of the selected Page types and Content types, never the current page) or Manual List (Filter Page items in authored order; Sort by ignored).
- **Sorting:** `latest` (`publishDate`, then `lastModified`), `featured`, `priority` (lower first, then latest), `alphabetical` (`Intl.Collator` with the page language).
- **Filtering:** filter pills from Filter Tag items (single or multi select), an advanced filter panel of categories below taxonomy roots (AND / OR), Clear all and Reset filters; filters run in memory and aren't in the URL.
- **Spotlights:** hero (Latest, Featured, Manual) and featured cards (Marked as featured, Priority 1 to 3, Manual).
- **Pagination:** Cards per page (default 9), Load more adds Load more count (default 6) and moves focus to the first new card; "Showing {n} of {total}".
- **Known differences from List:** `hide-in-nav` and redirects aren't applied; the Authors lookup is exact (handover F-30; audit D-A-29 deferred).

Conversation Canvas

- **Menu:** pages below the Menu root path (default: the locale root) up to Menu depth levels, sorted by level then path (alphabetical), titled by `title`, else the last path segment (`loadMenuTree`). `hide-in-nav` isn't applied (F-33); the menu order follows the path, not a navigation order column (F-78).
- **Search:** submitting dispatches the bubbling, cancelable `conversation-canvas:search` event with `detail.query`; an integration that handles it calls `preventDefault()`; else the authored Search page opens with `?q={query}`; else a polite status "Search isn't available yet." (`searchUnavailable`). Voice input

uses the browser's `SpeechRecognition` where available. The data layer records `cmp:search` with the term passed through `redactSearchTerm` (`utils/privacy.js`).
- **No search index or results page exists in the repository:** No verified search service was identified.

6. Empty, editor and error states

Block	Situation	Shown
List	No cards on the site	Empty behaviour: `message` (default; the authored message, else `listEmpty` "There are no items to show."), `hide` (block hidden; not in the editor or on AEM author), `fallback` (the List Items as a Fixed list)
List	AEM author / Universal Editor, Child or Tagged pages	`listEditorPages` "The pages show on the published page (from the query index)."
List	Configuration incomplete	Editor hint (warnings `tags-missing`, `parent-missing`, `root-missing`, `items-missing`)
Filter List	No matching pages or failed index	`noResources` "No resources found.", hint and Reset filters
Filter List	AEM author / Universal Editor	`filterListEditor` "The results show on the published page, from the pages' metadata."; items as chips
Conversation Canvas	No pages (including the editor)	`noPagesFound` "No pages found" in the menu
All	Index request fails	`getPages()` resolves to `[]`; the block renders its empty state; a failing deferred render is logged and the skeleton removed

7. Performance

- **Prefetch:** `prefetchData(main)` (`scripts/scripts.js`) starts `getPages()` and `loadTaxonomy()` when the lazy phase begins on pages with a List, Filter List or Conversation Canvas, and `loadRedirects()` with a List; not in the eager phase, so it doesn't compete with the LCP image. The blocks reuse the same cached promises.
- **Skeletons:** List and Filter List render through `renderWhenLoaded` (`decorators/skeleton-decorator.js`): data within 150ms renders directly; later data shows a block-shaped skeleton (at least 250ms) so the section and those below aren't held back. The Canvas menus use `skeletonWhile`.
- **Transfer:** a page downloads only its root's index (8 November 2026: before, every page downloaded the shared index of all roots, 11,821 rows in 24 sequential requests of 500 rows, about 1.4 MB compressed, finished after about 18 s on a migrated page): requests of 1,000 rows, the ones after the first in parallel; a review root whose index doesn't exist yet falls back to `/query-index.json`. The language master index is still about 8,300 rows (about 1 MB compressed): per-locale indexes of the locales and a lighter navigation index remain recommendations (F-30).
- **Placement:** avoid index-based blocks in the first section (LCP, handover Part 13.16).

8. Sitemap

`helix-sitemap.yaml` defines one sitemap, `default`: source `/query-index.json`, destination `/sitemap.xml`, `lastmod: YYYY-MM-DD`, origin `https://main--tcs-corp--sradha-tcs.aem.live` (to be replaced by the production domain once `cdn.prod.host` is configured, per the file's comment).

- It covers every indexed page of all locales; there are no per-locale sitemaps and no `hreflang` alternates (handover Part 4.9.6).
- Handover Part 2.10.7 states that pages with `robots noindex` are left out of the sitemap while they stay in the index and in List / Filter List (no code checks `robots`); the sitemap behaviour is platform

behaviour (Requires validation).
- The sitemap is regenerated when the index changes; the Admin API `sitemap` operation regenerates it on demand (handover Part 12.1.9).

9. Reindex dependencies

Event	Index effect	Source
Page published	Row added or updated	handover Part 12.1.7 (platform)
Page unpublished	Row removed	handover Part 12.1.7 (platform)
`helix-query.yaml` changed (new column, changed exclude)	Existing rows unchanged until the pages are republished or reindexed	handover Part 12.1.7
Page metadata changed (card title, tags, featured, priority, content type)	Lists change after the page is republished (and reindexed)	List and Filter List READMEs
Bulk metadata	Indexed only if the sheet applies (`metadata.source`) and the page is republished	handover Part 8

Verified reindex operation. The Admin API index operation `POST https://admin.hlx.page/index/{org}/{site}/{ref}/{path}` (org `sradha-tcs`, site `tcs-corp`, ref `main`) re-indexes a published path (handover Part 12.1.9). It was used per page, after `preview` and `live`, for the language master packages (handover revisions 2.31: 312 / 312 pages; 2.35: 313 / 313 and 315 / 315 pages). Calls need an authorised user or API key; no credentials are recorded. Bulk (`/*`) operations act on all locales and must be agreed with the site owner (handover Part 2.10.13).

Excluded by design. Block library pages stay out of the query index (handover revision 2.35; `helix-query.yaml` and `paths.json` excludes for the locale block libraries).

Open finding. F-10 (handover Part 16): the live index didn't match `helix-query.yaml` (card, eyebrow, category, hide-in-nav, author and feature columns missing) until the site is reindexed. The register still lists it as open; the language master pages were reindexed in revisions 2.31 and 2.35. Whether the 13 locale trees were reindexed is Not recorded.

10. Post-reindex validation

A checklist derived from the configuration and code (results of a full run are Not recorded; revision 2.31 recorded 307 language-master pages in the index, 306 with an image):

1. `GET /query-index.json?offset=0&limit=500` on the live host: `total` matches the expected number of published, non-excluded pages; page through until `total`.
2. A row of each locale has the 37 columns of `helix-query.yaml` (e.g. `cardTitle`, `hideInNav`, `cardStyle`, `cardTheme` present, even if empty).
3. No row below `references/`, `error-pages/`, `data/`, `block-library`, `fragments/`, `modals/`, `site-global-templates`, and no `.json` path.
4. Tagged pages have `tags` or `cqTags` values (F-07: AEM may publish page Tags only as `cq-tags`; code merges both).
5. On preview / live (not AEM author): a List (Child pages and Tagged pages), a Filter List and a Conversation Canvas menu of each changed locale show the expected pages; a page with `hide-in-nav = yes` is absent from the List.
6. `/sitemap.xml` lists the pages with the configured origin.

11. Related

- architecture.md — runtime phases, caching, integrations
- content-architecture.md — metadata model, taxonomy, locales, references
- blocks.md, data-layer-and-analytics.md — related knowledge base files
- blocks/list/README.md, blocks/filter-list/README.md, blocks/conversation-canvas/README.md, blocks/cards/README.md
- utils/README.md, decorators/README.md, docs/animation/README.md (skeletons)
- AGENTS.md — "Site structure and site data" notes on the query index
- Handover (TCS-CORP-HANDOVER.md): Part 2 — Solution Architecture (2.10.7, 2.11.4), Part 12 — Publishing, Deployment & Release (12.1.7, 12.1.9), Part 13 — Global Troubleshooting Guide (13.9–13.11, 13.25), Part 16 — Handover Findings and Open Items Register

12. Personalization (section)

13. Verified state

Item	State	Evidence
Adobe Target (at.js, Web SDK decisions, mboxes)	Not implemented	No Target, `alloy` or decision code in `blocks/`, `decorators/`, `utils/`, `scripts/`, `templates/`
Adobe Journey Optimizer (web channel, offers)	Not implemented	No AJO code
AEP Web SDK / Edge Network	Not configured (blocked: measurement plan)	`scripts/delayed.js` holds only a comment; see Data layer and analytics
Audiences, segments, visitor profiles	None	No audience source, no identity, no consent (Security, privacy and consent)
Decision scopes, activities, offers	None	—
Edge or server-side personalization	None	Edge Delivery serves the same cached HTML to every visitor
Experimentation (A/B tests)	None	No experimentation plugin or code
Personalized impressions in the data layer	None	No personalization events or fields in `decorators/data-layer-decorator.js`
Personalization fields in block dialogs	None	No `personalization` field in `models/` or `blocks/*/_*.json`

14. Decisions found

Decision	Source
The Teaser's Personalization option (a class `personalization` and `data-personalization="true"` that nothing read) was removed from the dialog and the code; library content was migrated; Teasers authored with it still render	Handover revision 2.27 (19 October 2026), audit defect D-B-3 (`docs/audit/BLOCK-ECOSYSTEM-AUDIT.md`), re-audit section 1 (`docs/audit/REAUDIT-2026-10-20.md`), `blocks/teaser/README.md`
Every block is classified "Not personalizable"; personalized rendering, fallback, flicker, cache-key and analytics tests are blocked until an approved personalization design exists	Audit section 8
Area-specific navigation and footers are content choices (inherited Navigation / Footer pages), not audience personalization	`blocks/header/README.md`, `blocks/footer/README.md`

Several handover chapters written before 2.27 (for example 1.6.2, 2.9.8, the Teaser chapter 6.11 Variation 66 and 16.4) still describe the Teaser marker; the code and revision 2.27 are authoritative.

15. Block support

No block supports personalization: every block (see the inventory in `blocks/README.md`) renders its authored default content, and each block README states "This block does not currently expose verified personalization behavior".

Script hooks that exist but have no caller in the repository:

Hook	Block	Default without a caller
`setConversationCanvasSuggestions(block, suggestions)`	Conversation Canvas (Primary, sticky)	The page's tag labels as suggestion pills (`blocks/conversation-canvas/README.md`)
`setConversationCanvasFilters(block, tags)`	Conversation Canvas (Secondary)	No filter tags (the list stays hidden)
`conversation-canvas:search` DOM event (cancelable)	Conversation Canvas	The Search page with ?q=, else a "not available" status
`form-variations:submit` DOM event with `detail.respondWith(promise)`	Form Variations	POST to the Submit action, else "not connected"

These are integration points (code comments name a personalisation service as one possible caller), not personalization features.

16. What a future implementation can build on

Foundation	What it offers	Source
Adobe Client Data Layer	Page entry (language, Page Type, tags, path), one component entry per block with a stable id (Anchor ID / Component ID, else `{block}-{n}`), `cmp:*` events	Data layer and analytics
Default content of every block	The authored content renders for everyone and is the natural fallback	Block READMEs; Edge Delivery cached HTML
Additional tab	Anchor ID (a stable block id and component id for targeting), Hide block, spacing and animation classes on every block model except header and footer	`AGENTS.md` (Additional tab), `decorators/block-options-decorator.js`
Page metadata	Page properties and the bulk metadata columns, read by `getPageMetadata()`	`utils/metadata.js`
Query index	Metadata columns indexed per page (below)	`helix-query.yaml`, `utils/query-index.js`
Delayed phase	`scripts/delayed.js`, imported 3 seconds after load, where the planned Tags / Web SDK embed goes	`scripts/scripts.js` `loadDelayed`
Shared DOM events	`tabs:change`, `tabs:reveal`, `header:menu`, `conversation-canvas:search`, `form-variations:submit`	`AGENTS.md` (Shared events and helpers)

17. Data that could feed personalization

Data	Where it is available today	Current use
`persona`	Bulk metadata column, `getPageMetadata().persona`, query index `persona`	None in blocks (handover 14.9, item 17)
`region`, `country`, `language`	Metadata columns, query index; `xdm:language` in the data layer page entry	`region` selects the contacts of the Media Contacts block (`blocks/media-contacts/media-contacts.js`, `inRegion`): page metadata, not the visitor
`campaign-id`	Metadata column, `getPageMetadata().campaignId`, query index `campaignId`	None

Data	Where it is available today	Current use
`page-type`, `content-type`, `category`, tags	Metadata, query index, data layer (`xdm:template`, `xdm:tags`)	Templates, Page Header, List, Filter List
`featured`, `priority`	Metadata, query index	Filter List spotlight and featured cards
Data sheets (`authors`, `locations`, `partners`, `social-links`, `media-contacts`; `region` columns)	`utils/data.js`	Page Header, footer, Press Release

All of these describe the page, not the visitor. No visitor attribute (segment, behaviour, geolocation, login) is collected.

18. Requirements for any future personalization

Requirements recorded for a future design; none is implemented, and the design itself is a blocked dependency (audit section 9: activities, audiences, the personalization design).

Area	Requirement	Source
Default content and fallback	The authored content stays the default and renders when no decision arrives, the service fails or consent is denied	Handover 2.9.8 (failure handling), audit section 7 (Personalization: denied → default content)
Consent	No decision request before consent; the consent model and categories must be approved first	Handover 11.7.6, audit section 7
Identity	An identity strategy (Web SDK ECID or other) must be decided; none exists	Not recorded (blocked)
Caching	Edge Delivery serves the same cached default content to every visitor; the design must define where variants are applied (browser or edge) and how they interact with the cache (cache keys)	Audit section 8 (cache-key tests blocked)
Flicker and performance	Personalizing above-the-fold content conflicts with the eager LCP phase; any pre-hiding needs a performance review; keep the Core Web Vitals targets	Handover 2.9.8 and 11.7.7
Security	Offer content must respect the CSP (`strict-dynamic`) and be inserted with DOM APIs	Handover 2.9.8
Accessibility	Personalized content must meet the same WCAG 2.1 AA rules as the block it replaces (headings, names, focus, placeholders for UI text)	`AGENTS.md` (Accessibility)
Analytics	Personalized impressions and decisions must be measurable through the existing data layer (no parallel tracking)	Data layer and analytics
Authoring and QA modes	Nothing personalized in Universal Editor edit mode; preview, live and a forced-variant QA mode need defining	Blocked (personalization design)

19. Validation status

Test	Status
Default content of every block renders (no personalization)	Passed (audit runtime runs: 940 page loads, no block not loaded)
Personalized rendering, fallback, flicker, cache key, personalized analytics	Blocked: approved personalization design, Adobe Target / AJO access

20. Related

- Architecture, Blocks, Indexing and search, Decision log

- Data layer and analytics
- Security, privacy and consent
- `blocks/README.md` and the block READMEs (Personalization sections)
- `AGENTS.md` (Site structure and site data: metadata columns)
- `docs/audit/BLOCK-ECOSYSTEM-AUDIT.md` (sections 7, 8, 9)
- Handover Part 2 — Solution Architecture (2.9.8), Part 11 — Analytics & Data Layer (11.7.7), Part 16 — Handover Findings and Open Items Register (`TCS-CORP-HANDOVER.md`)

21. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

22. Known limitations

- Adobe Tags, Web SDK, Analytics, Target and a consent platform are not configured; nothing in this document implies they are.

23. Open items

- None.

24. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Localization Architecture

Attribute	Value
Document ID	ARC-008
Audience	Solution architects, technical leads, developers, AI agents
Purpose	The locales, their published paths, the language master and the review roots, and how code resolves the locale
Scope	—
Prerequisites	—
Owner	Solution Architect (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	utils/site.js, paths.json, tools/paths/generate-paths.mjs
Related documents	—
Related AEM paths	—

1. Locales

AEM path	Published	lang
/content/tcs-corp/us/en	/en	en-US
/content/tcs-corp/de/de-de	/de-de	de-DE
/content/tcs-corp/fr/fr-fr	/fr-fr	fr-FR
/content/tcs-corp/ca/fr-ca	/fr-ca	fr-CA
/content/tcs-corp/es/es-es	/es-es	es-ES
/content/tcs-corp/pt/pt-pt	/pt-pt	pt-PT
/content/tcs-corp/nl/nl-nl	/nl-nl	nl-NL
/content/tcs-corp/se/sv-se	/sv-se	sv-SE
/content/tcs-corp/dk/da-dk	/da-dk	da-DK
/content/tcs-corp/no/no-no	/no-no	no-NO
/content/tcs-corp/cn/zh-cn	/zh-cn	zh-CN
/content/tcs-corp/jp/ja-jp	/ja-jp	ja-JP
/content/tcs-corp/kr/ko-kr	/ko-kr	ko-KR

2. Language master and review roots

The language master `/content/tcs-corp/language-masters/en` is the future source of the global site and the regions; whether the regions are created with MSM or carry their own content is open until the content is final (tcs.com's regions don't carry language master content today). The testing root `/content/tcs-corp/language-master/en` and the design root `/content/tcs-corp/design-block-master/en` are not rollout sources.

3. Site and locale hierarchy

Purpose. One Edge Delivery site serves 13 locale sites by path. Each locale owns its pages and its own references, error pages, sheets, data and block library.

Structure. The AEM tree `/content/tcs-corp/{country}/{language}` is published at a language-only path. The locales are defined once in `LOCALES` (`utils/site.js`); the first entry is the default (`DEFAULT_LOCALE`, `en`).

AEM folder	Published root	lang		AEM folder	Published root	lang
us/en	/en	en-US		se/sv-se	/sv-se	sv-SE
de/de-de	/de-de	de-DE		dk/da-dk	/da-dk	da-DK
fr/fr-fr	/fr-fr	fr-FR		no/no-no	/no-no	no-NO
ca/fr-ca	/fr-ca	fr-CA		cn/zh-cn	/zh-cn	zh-CN
es/es-es	/es-es	es-ES		jp/ja-jp	/ja-jp	ja-JP
pt/pt-pt	/pt-pt	pt-PT		kr/ko-kr	/ko-kr	ko-KR
nl/nl-nl	/nl-nl	nl-NL				

```
/content/tcs-corp/
├── {country}/{language}/   a locale site; the locale page itself is Home (published as /{locale}/)
│   ├── …                   content pages
│   ├── references/         nav, footer, alternative navs / footers, footer Canvas page, modals,
fragments
│   ├── error-pages/        404 (used by 404.html); 403 / 500 for a CDN
│   ├── block-library/      block validation pages; excluded from publishing
│   ├── placeholders        sheet: Key, Text, Description, Category
│   ├── metadata            sheet: bulk metadata
│   ├── taxonomy            taxonomy page (tag, title)
│   ├── redirects           sheet: Source, Destination
│   └── data/               sheets: authors, locations, partners, social-links, media-contacts
├── language-masters/en/     temporary: published at /language-masters/en for review
└── site-global-templates/  template pages authors copy; excluded from publishing
```

Path mapping (`paths.json`, generated by `npm run build:paths` from `LOCALES` and `REVIEW_ROOTS`). One generic set per site root: the folder mapping (`/content/tcs-corp/de/de-de/:/de-de/`: child pages and sheets), the Home page mapping (`/content/tcs-corp/de/de-de:/de-de/`) and the taxonomy page (`…/taxonomy:/de-de/taxonomy.json`): 39 mappings for the locales, 3 temporarily for the language master. The sheets (`placeholders`, `metadata`, `redirects`, `data/*`) need no mapping of their own: the folder mapping publishes them at the same path and spreadsheets get `.json` appended; the former 112 per-sheet mappings are kept commented in the generator. `includes`: the 13 locale pages, `/content/dam/tcs-corp/` and `/content/tcs-corp/language-masters/en`; `excludes`: each locale's `block-library` and `block-library/**`, and `site-global-templates` with its children. AEM applies the site configuration's `public.json` → `paths`, not the repository file (AGENTS.md "Notes").

Locale resolution (code). `getLocale(pathname)` reads the AEM path when editing (`/content/tcs-corp/de/de-de/…`) and the published path on the site (`/de-de/…`); pages outside a locale (Site Global Templates, drafts) get `en`. Helpers: `localePagePath`, `resolvePagePath`, `localeSheetPath`, `toPublishedPath`, `toAemPath`, `pageLink`, `getParentPages`, `sheetRows`.

Review roots (KB-D-068). There are three review roots, each published as a site of its own (KB-D-068): `/language-master/en` (testing page functionality with dummy content, content limitations), `/language-masters/en` (the language master: the migrated tcs.com content, validated with its real sheet data, later the source of the global site and the regions and languages; MSM or not is open until the content is final) and `/design-block-master/en` (the block library and the curated design pages). Pages below each (`REVIEW_ROOTS`) list their own pages from the query index (`indexRoot`), read their own placeholders, redirects, taxonomy and data sheets (`REVIEW_SHEETS`), use their own `references/` and `error-pages/` (`REVIEW_PAGES`) and inherit up to their

root page. The temporary review of the language master ends with its rollout (handover Part 3.9, finding F-88).

Language master URLs (validated 23 October 2026). The mappings follow Adobe's path mapping rules (https://www.aem.live/developer/authoring-path-mapping): the folder mapping `/content/tcs-corp/language-masters/en/:/language-masters/en/` covers every child page, and the page mapping `/content/tcs-corp/language-masters/en:/language-masters/en/` is the special case that publishes the root page as `/language-masters/en/index`, served at `/language-masters/en/` (the same pattern as `/en/`). No other mapping is needed. The mappings of every language master come from `REVIEW_ROOTS` (`utils/site.js`): `tools/paths/generate-paths.mjs` writes this set for each root, so another language master (e.g. `/language-masters/de`) is added to the code and to `paths.json` with one entry and `npm run build:paths`; `test/site/paths.test.mjs` checks every locale and language master is mapped this way and that `paths.json` is in step.

AEM page	Preview	Live
`/content/tcs-corp/language-masters/en` (Home)	`https://main--tcs-corp--sradha-tcs.aem.page/language-master/en/`	`https://main--tcs-corp--sradha-tcs.aem.live/language-master/en/`
`/content/tcs-corp/language-masters/en/<path>` (child pages)	`https://main--tcs-corp--sradha-tcs.aem.page/language-master/en/<path>`	`https://main--tcs-corp--sradha-tcs.aem.live/language-master/en/<path>`

The Home URL ends with `/` (`/language-masters/en` without it is a 404, as `/en`). On 23 October 2026 the child pages, sheets, references and error pages resolved, but the Home page was a 404 and wasn't in the query index: it has to be previewed and published from AEM (Manage Publication on `/content/tcs-corp/language-masters/en`). If it still returns a 404 after publishing, check that the site configuration's `public.json` → `paths` has the page mapping (AEM applies that configuration, not the repository file).

Rules.

- Add or change a locale only in `LOCALES`, then run `npm run build:paths` and commit both files; the site administrator re-applies `public.json` and `metadata.source`.
- Nothing is published at `/`: a `/` → `/en/` redirect needs another mechanism (handover F-36).

Common mistakes. Editing `paths.json` by hand (it is overwritten by the generator); expecting a repository `paths.json` change to take effect without the site configuration; creating pages directly under `/content/tcs-corp` (not published).

4. Localization

Topic	Implementation	Status
Locale structure	13 separate page trees; locale resolved from the path	Active
`<html lang>`	`language` metadata when valid, else the locale's tag	Active
UI text	Placeholders sheet per locale; English fallback in code	Active
Tag labels	Taxonomy page per locale, language sheet when translated	Active (only `/en` taxonomy published when checked, F-09)
Bulk metadata	One sheet per locale; all 13 in `metadata.source`	Requires validation (F-11, open item O-04)
Data sheets	One set per locale	Active
Lists	Each locale lists only its own pages (`indexRoot`)	Active
Dates	`utils/dates.js` `formatDate`; English month abbreviations in every locale (handover Part 4.9.7)	Active
Translation process, MSM / language copies, language selector, `hreflang`	Not in code	Not recorded

5. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

6. Known limitations

- None recorded for this document.

7. Open items

- Decide MSM or separate regional content once the language master content is final (owner: Product Owner).

8. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Performance Architecture

Attribute	Value
Document ID	ARC-007
Audience	Solution architects, technical leads, developers, AI agents
Purpose	Core Web Vitals requirements, loading strategy, images, fonts, layout shift prevention and budgets
Scope	—
Prerequisites	—
Owner	Solution Architect (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/performance.md, scripts/scripts.js, styles/styles.css
Related documents	—
Related AEM paths	—

1. Requirements (Core Web Vitals and NFRs)

Requirement	Value	Source	Enforcement
PageSpeed Insights / Lighthouse score	100, mobile and desktop, on the feature preview URL before merge	AGENTS.md "Publishing Process" step 3, handover 1.7.1	Process: the aem-psi-check app on pull requests; CI (`.github/workflows/main.yaml`) runs lint only
Largest Contentful Paint	< 2.5 s (RUM, p75)	handover 13.16 (escalation thresholds), revision 2.36 ("the 2.5 s NFR")	Code: eager phase loads only the first section and its first image
Cumulative Layout Shift	≤ 0.1 (< 0.1 at p75 in RUM)	handover 13.16, revision 2.36 ("NFR ≤ 0.1")	Code: header space reservation, footer hidden until sections load, metric-matched fallback fonts
Interaction to Next Paint	escalate above 200 ms at p75	handover 13.16	No specific code rule; long tasks avoided while decorating
Three-phase loading, per-block code splitting, module preloading, no forced layout while decorating, image optimisation, no third-party scripts on load, shared data loaded once and early, RUM	as implemented below	handover 1.7.1 table	Code (module graph also checked by `npm test`)

2. Measured evidence

Recorded in handover revision **2.36** (20 October 2026) and finding **F-106** (closed in 2.36):

Measurement	Before	After
Lighthouse 12 on aem.live (curated Home, Insights, What we do, a block library page), mobile / desktop	96–100 / 100; LCP 1.2–1.4 s mobile; CLS ≤ 0.011; Total Blocking Time up to 220 ms	99–100 mobile, 100 desktop (Home and What we do)
Local preview, 4× CPU and mobile network	LCP 1.9–2.6 s; all sections 4.4–5.4 s; blocking 60–80 ms	LCP 1.7–1.95 s; all sections 3.5–3.8 s; blocking 25–48 ms
aem.live, branch against main, same throttling, 3 runs: Home	LCP 1.83–1.88 s; header 3.30–3.43 s; all sections 3.89–4.04 s	LCP 1.12–1.61 s; header 2.05–2.51 s; all sections 2.34–2.77 s (about 37 % sooner)
aem.live, same: What we do	LCP 2.36–2.43 s; all sections 3.42–3.50 s	LCP 0.94–1.00 s; all sections 1.60–1.68 s (about 52 % sooner)
Font swap on a block library page with a long intro (mobile, Android / Linux metrics)	CLS 0.33 (paragraph 230 → 205 px)	CLS 0.003 (205 → 205 px)
Regression	—	955 loads: 0 errors, 0 axe violations, 0 overflow; 791 / 791 test clicks

Causes found (F-106): no build step, so the 21 modules of `scripts.js` loaded over 4 network round trips before the first section and each block's (up to 42 modules) over up to 7 more; one 225 ms forced layout per List scroller; the Canvas controller rewrote `<html>` custom properties on every resize during loading (100–120 ms tasks); header block stylesheets were requested only after its fragments were decorated. Earlier layout-shift fixes: F-90 (footer, sticky Anchor List, CLS 0.35, 2.13), F-93 (header, CLS 0.14–0.25 → below 0.01, 2.15).

3. Lab and field data

Kind	Tool	Status
Lab	Lighthouse / PageSpeed Insights (pull requests, manual runs), throttled browser runs recorded in 2.36	Recorded for the pages above
Field (RUM)	`sampleRUM` in `scripts/aem.js` (Edge Delivery Operational Telemetry): about 1 in 100 page views (`weight` 100), checkpoints to `ot.aem.live`, the RUM enhancer loaded for sampled views after the first section (`sampleRUM.enhance()` in `loadMainSections`); `404.html` sends a `404` checkpoint	Implemented; **no field results are recorded** in the repository or handover; dashboard access and domain key "[To be confirmed]" (handover 2.10.14)

Lab figures don't replace field data: low-traffic locales give few samples (handover 2.10.14). Read RUM before and after a release when it becomes available.

4. JavaScript loading

Phase	What runs (`scripts/scripts.js`)
Eager (`loadEager`)	`decorateHead` (SEO tags, `<html lang>`), `trackInteractions` (data-layer listeners and page entry), template / theme / page-type classes, the 404 content or the Page Type template (`templates/{name}`), `decorateMain`, `reserveHeaderSpace`, `body.appear`, `preloadBlockModules(first section)`, then the first section with `prepareMotion` and `waitForFirstImage` (sets the first image `loading="eager"` and waits for it); `fonts.css` only on wide viewports or when cached
Lazy (`loadLazy`)	Modal auto-links, `preloadBlockModules` for the rest of `main` and the chrome blocks (`header`, `footer`, `fragment`, `tcs-header`, `conversation-canvas`), `prefetchData`, `loadHeader`, `loadFooter` (hidden), `loadMainSections` (each section in order, animations prepared before it shows), then footer shown, `trackBlocks(main)`, Go back on 404 pages, hash scroll, `lazy-styles.css`, `loadFonts`

Phase	What runs (scripts/scripts.js)
Delayed (loadDelayed)	After 3 s: `trackBlocks(document)` (blocks added later, such as header, footer, modals) and `import('./delayed.js')`, which is empty ("add delayed functionality here")

- **Per-block code splitting**: `loadBlock` (`scripts/aem.js`) loads a block's CSS and JS in parallel only when the block is on the page; blocks of a section load one after another, and sections load in order.
- **Module preloading**: there is no build step, so each import level is a network round trip. `head.html` has 20 `modulepreload` links (the modules `scripts.js` imports); `preloadBlockModules` adds `modulepreload` for each block's modules and a `preload as="style"` for its stylesheet from the generated `scripts/module-graph.js`. Stylesheet preloads go in the **body**, because `loadCSS` skips any stylesheet whose href is already linked in the head. Run `npm run build:modules` after any import change; `npm test` (`test/modules/module-graph.test.mjs`) fails when the graph or the head links are stale (AGENTS.md "Setup Commands").
- **Dynamic imports**: on-demand modules are imported only when needed: `utils/inherited-metadata.js` (404 pages), `utils/placeholders.js` (Go back), `utils/query-index.js`, `utils/taxonomy.js`, `utils/redirects.js` (`prefetchData`), `blocks/modal/modal.js` (modal links and the video pop-up), template modules; the Lottie player (`scripts/lib/lottie`, vendored) loads when an animation is near the viewport.

5. CSS loading

Stylesheet	When it loads
`styles/styles.css`	Render-blocking `<link>` in `head.html` (tokens, global styles, sections, buttons, theme contract, fallback font faces, skeletons, Additional tab classes)
`blocks/<name>/<name>.css`	With the block (`loadBlock`), preloaded by `preloadBlockModules` (`blockStylesheet`)
`styles/card-list.css`	`@import` from `blocks/list/list.css` and `blocks/cards/cards.css` (a chained request after the block stylesheet; not in the preload list)
`templates/<name>/<name>.css`	With the Page Type template in the eager phase (`templates/blog/blog.css` imports `article.css`)
`styles/motion.css`	Only on pages with an animated block (`prepareMotion`)
`styles/fonts.css`	Eager on viewports ≥ 900px or when the session flag is set, else end of the lazy phase
`styles/lazy-styles.css`	End of the lazy phase (currently only the Universal Editor item placeholder styles)

6. Images

- Authored images are optimised `/media_…` renditions served as `<picture>` with WebP and fallback sources (`createOptimizedPicture`: 2000px from 600px, 750px below; `format=webply&optimize=medium`).
- The first image of the first section is the LCP candidate: `waitForFirstImage` sets `loading="eager"` on it and waits for it before the lazy phase. The Hero sets `loading="eager"` and `fetchPriority="high"` on its image when it opens the page; the Page Header does the same for its featured image; List / Cards "Image loading" Eager gives only the first image a high fetch priority (`decorators/card-list-decorator.js`).
- `buildImage` (`decorators/responsive-image-decorator.js`) optimises same-origin media URLs and lazy-loads other URLs; section backgrounds use 2000px / 750px. Open: a featured image that isn't site media stays lazy (handover F-14).
- Icons are lazy `<img>` (16 × 16 attributes). Images committed to git must be optimised (AGENTS.md "Performance"); the only font file is 48,256 bytes.

7. Fonts

- One self-hosted variable font (`fonts/inter-variable.woff2`, `font-display: swap`, Latin `unicode-range`); no external font service.
- Eager only when `window.innerWidth >= 900` or `sessionStorage` `fonts-loaded` is set (set after the first load, except on localhost); otherwise loaded at the end of the lazy phase, so the font doesn't compete with the LCP image on small screens.
- Text first renders in a fallback face scaled to Inter: `inter-fallback` (Arial, Liberation Sans), `inter-fallback-roboto` (Android), `inter-fallback-dejavu` (Linux), in `styles/styles.css` (values in design-system.md). Keep the faces when the font changes, or text reflows when Inter arrives (2.36: CLS 0.33 → 0.003).

8. Layout shift prevention

Source of shift	Prevention
Header (lazy phase)	`header:not(:has(.nav-wrapper))` keeps `--header-reserved` (the logo bar `--nav-height`: 4rem, 7.75rem from 1200px) or the height remembered for the breakpoint (`utils/header-space.js`, `localStorage` key `header-height-v2-<breakpoint>`); from 1200px the header lies over a light first hero out of the flow; the header Canvas lies on the hero and takes no header space
Header / footer content	`header .header` and `footer .footer` stay `visibility: hidden` until `data-block-status="loaded"`
Footer	`display: none` until all main sections have loaded
Fonts	Metric-matched fallback faces
Data blocks (List, Filter List)	Skeletons shaped like the block (`renderWhenLoaded`) when data takes longer than 150 ms; shown at least 250 ms so they don't flash
Sticky Anchor List	Doesn't pin while its section is hidden (F-90)
Animations	Only `opacity`, `transform`, `clip-path`; first frames applied only on screen; Background animation on the first hero delays the LCP image paint (F-94: use it below the fold)

9. DOM size, listeners and observers

- DOM size: no measurement or limit is recorded. Skeletons add about 20–40 small spans per block, removed with the skeleton; animations add no wrapper elements (animation report, section I).
- Data-layer listeners are delegated: one capturing listener per event type on `document` (`click`, `toggle`, `close`, `play` / `pause` / `ended`, `submit`, `change`), registered once (`trackInteractions`).
- Observers: one IntersectionObserver for all animations (`decorators/motion-decorator.js`); `utils/viewport.js` `onVisible` / `watchVisibility` for deferred work (Embed, Video, Gen Carousel, Lottie in `decorators/animation-decorator.js`); own IntersectionObservers in the Separator (motion) and the Modal (inside the editor); ResizeObserver in scrollers, masonry, Table, Anchor List, Header and the Canvas controller (resizes measured once they settle).
- Scroll listeners are passive and scheduled (Anchor List, Canvas controller); the Canvas never measures while scrolling.
- Layout reads are batched: scrollers measure in one shared animation frame, reads before writes; the Canvas controller writes `<html>` custom properties only when they change (AGENTS.md module preloading paragraph).

10. Query and data requests

Data	Loading
Query index (`/query-index.json`)	`utils/query-index.js` `getPages()`: fetched once per page view in pages of 500 rows (`offset` / `limit`), kept as an in-memory promise, filtered to the locale (or the review root)
Taxonomy, redirects, placeholders, data sheets	Fetched once per page view, cached in memory
Inherited metadata (Navigation, Footer, 404 Page)	Parent pages' heads loaded in parallel, cached per browser session (`sessionStorage`)
Prefetch	`prefetchData` starts the query index and taxonomy (and, with a List, the redirects) when the lazy phase begins on pages with a List, Filter List or Conversation Canvas; not in the eager phase, so it doesn't compete with the LCP image
Late data	`renderWhenLoaded`: data within 150 ms renders directly; else a skeleton, so the section is shown and the sections below aren't held back (a 1.5 s index showed the sections below a List at 0.5 s instead of after the index)

11. Analytics dispatch

The Adobe Client Data Layer (`window.adobeDataLayer`) is filled by `decorators/data-layer-decorator.js`: the page entry and interaction listeners in the eager phase (`trackInteractions`), the blocks of `main` once its sections have loaded (`trackBlocks(main)`), and the rest (header, footer, later blocks) in the delayed phase after 3 s (`trackBlocks(document)`). No analytics library consumes it: Adobe Tags and the Web SDK are **not configured** (blocked: measurement plan), and `scripts/delayed.js` is empty, so there is no tag-manager cost today. When Tags is added, load it from the delayed phase (`scripts/delayed.js`) and re-measure.

12. Personalization

None: no Adobe Target, AJO or edge personalization runtime exists, so there is no flicker, anti-flicker snippet or personalization request. Every page is the cached default content.

13. Third-party code

Code	When
`head.html`	No external script; a CSP with `nonce-aem` and `strict-dynamic`
X (Twitter) widgets	`platform.twitter.com/widgets.js` only when an X embed is rendered (`blocks/embed/embed.js`)
YouTube / Vimeo	Behind a placeholder with a Play button when a placeholder image is authored, else loaded when the block is visible (`onVisible`); privacy-enhanced domains
Lottie player	Vendored (`scripts/lib/lottie/lottie_light.min.js`), loaded when an animation is near the viewport
RUM enhancer	Loaded for sampled views only
DOMPurify	`scripts/dompurify.min.js`, used by the Universal Editor support code

14. Budgets

No verified performance budget (bytes, requests, long tasks) is defined in the repository, CI or handover. The working targets are the NFRs above (PSI 100, LCP < 2.5 s, CLS ≤ 0.1, INP ≤ 200 ms at p75). Proposed budgets are a recommendation, not a decision.

15. Testing tools and evidence

Tool	Use	Where recorded
PageSpeed Insights (aem-psi-check on pull requests)	Score and diagnostics for the preview URL in the PR description; a missing or wrong URL fails it	handover 13.15; PR descriptions
Lighthouse (local, Chrome)	Lab scores with mobile / desktop presets	handover revision 2.36
Throttled browser runs (4× CPU, mobile network)	LCP, header time, all-sections time, blocking time on the local preview and on aem.live (branch vs main, 3 runs)	handover revision 2.36
Performance panel recording	Long tasks, forced layouts, the block that decorates last in the first section	handover 13.16 diagnosis
Audit harness (`docs/audit/tools/harness.mjs`)	Every block library and curated page at 1440 / 1024 / 768 / 600 / 390px: console errors, failed requests, block status, overflow, broken images, axe, data layer, clicks	audit section 2, re-audit section 2; uses delivery-workspace paths (Requires validation before re-use)
`npm test`	`test/modules/module-graph.test.mjs` keeps the preload graph and `head.html` current	repository
RUM (`sampleRUM`)	Field Core Web Vitals	No results recorded

No Lighthouse or performance script is committed to the repository; the 2.36 runs are documented in the handover only.

16. Required performance validation

Validate before merging (AGENTS.md "Publishing Process"; handover 12.4 release checklist and 13.16):

Change	Validate
A block's JS or CSS	PageSpeed Insights 100 on the feature preview of a page using the block (mobile and desktop); `npm run build:modules` if imports changed, then `npm test`; no forced layout while decorating (reads before writes in one frame); no new eager work in the first section; the block README's "Performance Considerations" updated
A dialog option or combination	The block library page of the option, first section and below the fold; options that load media or data (Image loading Eager / Lazy, Background animation, Lotto, autoplay video, List sources) checked for LCP and CLS; the dialog options report "Responsive and performance" section is the model (CSS-only options: no new requests, listeners or blocking resources)
Block library pages	The harness run (0 errors, 0 failed requests, 0 overflow) and Lighthouse on a representative library page
Curated design pages and key templates (Home, Insights, What we do, article)	Lighthouse mobile / desktop and a throttled run against main on aem.live; CLS from the font swap on Android-like metrics (long intro paragraphs)
A new third-party script	Loaded from `scripts/delayed.js` or on interaction / visibility, never in `head.html`; CSP compatible
Fonts	Keep the fallback faces; measure CLS on mobile

Author guidance that affects performance: keep the LCP image in the first block (Hero or Page Header); set List / Cards "Image loading" to Eager only for lists in the first section; avoid page lists from the query index and Background animation as the first block (handover 13.16, F-94).

17. Gaps and recommendations

Gap	Status	Proposed next step (not an approved decision)
Field data (RUM) results and dashboard access	Requires validation: RUM domain key and dashboard access "[To be confirmed]"	Review p75 LCP / CLS / INP per locale after each release
Performance budgets	Not recorded	Define budgets (JS / CSS bytes, requests, long tasks) per template
PSI is a process check, not enforced by CI	Recorded (handover 1.7.1)	Keep the PR preview URL mandatory
`styles/card-list.css` is an `@import` of the List and Cards stylesheets, outside the preload graph	Observation	Consider preloading it with the block stylesheet
Featured image from a non-site URL stays lazy (F-14)	Open	Apply loading attributes to plain `img` too
Harness and performance scripts tied to the delivery workspace	Requires validation	Commit runnable versions under `tools/`

18. Related

- design-system.md: fonts and fallback faces, tokens
- accessibility.md: skeleton and loading states, reduced motion
- Other knowledge base pages: architecture.md, testing-and-quality.md, troubleshooting.md, data-layer-and-analytics.md, security-privacy-consent.md, deployment-and-release.md
- AGENTS.md "Three-Phase Page Loading" and Performance
- `scripts/scripts.js`, `head.html`, `scripts/module-graph.js`, `tools/modules/generate-module-graph.mjs`, `decorators/skeleton-decorator.js`, `utils/query-index.js`
- Each block README's "Performance Considerations" (`blocks/README.md`)
- Animation report (I. Performance), block ecosystem audit
- Handover: Part 1 — 1.7.1 Performance, 2.10.14 Operational Telemetry (RUM), 13.15 and 13.16 (troubleshooting), Part 16 — F-106, revision 2.36

19. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

20. Known limitations

- No PageSpeed or field measurement was taken in this program (Not recorded).

21. Open items

- None.

22. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Security Architecture

Attribute	Value
Document ID	ARC-005
Audience	Solution architects, technical leads, developers, AI agents
Purpose	Security boundaries, sanitisation, link safety, CSP, secrets, storage, data minimisation and consent dependencies
Scope	—
Prerequisites	—
Owner	Solution Architect (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/security-privacy-consent.md, utils/privacy.js
Related documents	—
Related AEM paths	—

This file summarizes the security and privacy measures implemented in the front-end code of the TCS Corporate Website (content sanitization, link safety, external resources, the Content Security Policy, storage, cookies, query strings, logging and data minimisation) and records what is not implemented (consent, identity). It describes the code's behaviour only: no legal classifications, retention rules or approvals are recorded here, and decisions that need privacy or legal input are listed as blocked.

1. Status at a glance

Area	Status	Where
Universal Editor update sanitization (DOMPurify)	Active	`scripts/editor-support.js`
Same-origin fragment and error page loading	Active	`blocks/fragment/fragment.js`, `scripts/scripts.js` `loadPageContent`
Safe URL check for card links	Active	`utils/card-list-config.js` `isSafeUrl` (List, Cards)
New-tab links with `noopener noreferrer` and a screen reader hint	Active	`decorators/link-decorator.js`
Privacy-enhanced video players	Active	Embed (always), Video (Privacy mode, default on)
Content Security Policy (nonce, `strict-dynamic`)	Active	`head.html`, `404.html`
Secrets in the repository	None found	—
Search term redaction, field names only in form events	Active	`utils/privacy.js`, `decorators/data-layer-decorator.js`
Cookies set by project code	None	—

Area	Status	Where
Consent management, visitor identity	Not implemented (blocked: consent model)	—
Generic iframe embeds: host allow-list, `sandbox`	Not implemented (known limitation)	`decorators/media-decorator.js` `createFrame`

2. Content sanitization

Path	What happens	Code
Universal Editor updates	The HTML returned by the editor after an edit is sanitized with DOMPurify (`USE_PROFILES: { html: true }`) before it is parsed and re-decorated; DOMPurify is loaded on demand with `loadScript`	`scripts/editor-support.js` `applyChanges`
Rich text in the editor	`decorateRichtext` groups rich text instrumentation into editable wrappers; it moves attributes and does not sanitize	`scripts/editor-support-rte.js`
Fragments (header, footer, Modal, Tabs, Fragment block)	`loadFragment` fetches only paths that start with `/` and not `//` (`{path}.plain.html`, same origin) and sets the response as `innerHTML`; a fragment that includes itself or the current page isn't loaded again	`blocks/fragment/fragment.js`
Error pages	`loadPageContent` fetches same-origin `.plain.html` paths (the inherited 404 Page, else the locale's `error-pages/404`) into `main`	`scripts/scripts.js`
Block rendering	Blocks build elements with DOM APIs; players are built without writing authored URLs into HTML strings	`decorators/media-decorator.js`

DOMPurify is the only user of `scripts/dompurify.min.js`; it is not loaded on the public site. The `innerHTML` uses in `fragment.js` and `scripts.js` rely on Edge Delivery `.plain.html` responses from the same origin (authored content). `scripts/aem.js` (never modified) also uses `innerHTML` in `buildBlock` when a block is built from strings (auto-blocking).

3. Link safety

Measure	Behaviour	Code
`isSafeUrl(href)`	Allows `http:`, `https:`, `mailto:`, `tel:` and relative paths (`/`, `./`, `../`, `#`, `?`, a word); any other scheme (`javascript:`, `data:`, …) is rejected and the card is not rendered	`utils/card-list-config.js`, used by `blocks/list/list-data.js` and `blocks/cards/cards-data.js`
New tab	`setLinkTarget` sets `target="_blank"` with `rel="noopener noreferrer"`; `announceNewTab` adds the "opens in a new tab" hint from the placeholders	`decorators/link-decorator.js`
Logo links	A TCS Header logo link typed as a bare domain (`www.example.com`) gets `https://` so it isn't a relative link	`blocks/tcs-header/tcs-header.js` `buildLogo`
Fragment paths	Only same-origin absolute paths (`/…`, not `//…`)	`loadFragment`
AEM paths	`toSitePath` maps same-origin AEM paths to published paths; other origins pass through unchanged	`utils/urls.js`

Links outside List and Cards are rendered as AEM delivers them; no other block runs a scheme check (Requires validation: how AEM delivers a `javascript:` link authored in other fields).

4. External resources

Resource	Behaviour	Code
YouTube (Embed)	Always the privacy-enhanced host `www.youtube-nocookie.com`; only the video id is passed on (`rel=0`, `enablejsapi=1`)	`blocks/embed/embed.js`, `utils/media.js`
YouTube (Video)	`youtube-nocookie.com` when the Privacy mode field is on (default on), else `www.youtube.com`	`blocks/video/video.js`
Vimeo	`player.vimeo.com` with `dnt=1` (Embed always; Video with Privacy mode); the video id is URL-encoded	`utils/media.js`, `blocks/embed/embed.js`, `blocks/video/video.js`
X (Twitter) posts	`https://platform.twitter.com/widgets.js` is loaded only when an X embed renders; `x.com` links are rewritten to `twitter.com`	`blocks/embed/embed.js` `embedTwitter`
Any other Embed URL	A generic lazy iframe (`allow="encrypted-media"`, fullscreen allowed) with no host allow-list and no `sandbox` attribute: a known limitation; only approved, framable URLs should be authored	`decorators/media-decorator.js` `createFrame`, `blocks/embed/README.md`
Embed loading	With a placeholder image the player loads only after the visitor presses Play; without one, when the block scrolls into view	`blocks/embed/embed.js`
Form submissions	A valid Form Variations submission goes to a page integration (`form-variations:submit`), else is POSTed as JSON to the authored Submit action URL (no allow-list), else the visitor is told the form isn't connected	`blocks/form-variations/form-variations.js`
RUM	`sampleRUM` sends sampled checkpoints with `navigator.sendBeacon` to the RUM collector (`ot.aem.live`) and loads the RUM enhancer	`scripts/aem.js`
Fonts, Lottie player, DOMPurify	Served from the repository (no font or script CDN)	`fonts/`, `scripts/lib/lottie/`, `scripts/dompurify.min.js`

Third-party players and widgets set cookies and collect data as their providers do; no consent gate exists before they load (handover open item O-16: approved embed domains and consent for third-party players).

5. Content Security Policy

`head.html` (all pages) and `404.html` declare:

```
script-src 'nonce-aem' 'strict-dynamic' 'unsafe-inline' http: https:; base-uri 'self'; object-src 'none';
```

Directive	Effect
`'nonce-aem'` with `'strict-dynamic'`	Only the `<script nonce="aem">` tags (and `modulepreload` links with the nonce) and the scripts they load run; `'unsafe-inline' http: https:` are fallbacks ignored by browsers that support `strict-dynamic`
`base-uri 'self'`	Prevents <base> injection
`object-src 'none'`	Blocks plugins
`move-to-http-header="true"`	Edge Delivery sends the policy as a response header
Not set	`style-src`, `img-src`, `frame-src`, `connect-src` (embeds, form POSTs and RUM aren't restricted by the policy)

A future tag manager must be loaded by project code (`scripts/delayed.js`) so `strict-dynamic` trusts it (handover 2.7.1, 13.23 Content Security Policy Blocking Scripts).

6. Secrets and environment configuration

- No secrets were found in the repository (no API keys, tokens or passwords in the code or configuration); everything in it is public client-side code. Keep it that way: an integration that needs credentials needs a server-side or Adobe-managed component.
- Configuration is code and site configuration, not environment variables: `fstab.yaml` (the content source), `paths.json` (generated from `utils/site.js`), `helix-query.yaml`, `helix-sitemap.yaml`, and the site configuration in the Edge Delivery admin service.
- Runtime context checks: `isAuthor()` (path starts with `/content/`: AEM author, no query index), `isEditing()` (Universal Editor edit mode, `adobe-ue-edit`), `isInEditor(el)` (`utils/environment.js`); `localhost` skips the `fonts-loaded` flag.
- `.hlxignore` keeps dot-folders, `*.md`, `package.json`, `package-lock.json`, `test/*`, `_*` (model partials), `docs/*`, `LICENSE` and `karma.config.js` off the code bus; other repository folders are served. Block libraries and Site Global Templates are excluded from publishing in `paths.json`.
- Author access, Admin API roles and whether preview / live hosts are access-restricted are not recorded in the repository (handover 2.7.3: To be confirmed).

7. Storage and cookies

Key	Storage	Written by	Content
`header-height-v2-{mobile|tablet|desktop}`	`localStorage`	`utils/header-space.js`	The loaded header height in rem, so the next page reserves it (no layout shift)
`header-overlay-v1`	`localStorage`	`utils/header-space.js` `setHeaderOverlay` (from the TCS Header)	The nav page's Header over the first block mode (`always` / `never`), so the next page lays out for it; no personal data
`fonts-loaded`	`sessionStorage`	`scripts/scripts.js` `loadFonts` (not on `localhost`)	`true` once `fonts.css` has loaded
`tcs-inherited-metadata`	`sessionStorage`	`utils/inherited-metadata.js`	The Navigation, Footer and 404 Page values of parent pages, by URL

All storage access is wrapped in `try / catch` (private mode works). None of these values identify a visitor. Project code sets no cookies (`document.cookie` is not used), and `scripts/aem.js` (RUM) sets none. Third-party players (X widget, generic iframes, YouTube and Vimeo once played) may set their own cookies. The data layer code writes no storage.

8. Query strings and debug flags

Parameter	Read or written by	Effect
`?q=`	Written by the Conversation Canvas when a Search page is set	The raw (unredacted) query in the Search page URL; no code in the repository reads it (re-audit R-D-12: blocked until Tags exists)
`?debug=list`, `?debug=separator`	`blocks/list/list.js`, `blocks/separator/separator.js`	`console.debug` of the block's configuration and data
`?rum=`, `?optel=`	`scripts/aem.js`	RUM sampling rate (`on`, `off`, `high`, `low`)
`?lighthouse=on`	`scripts/aem.js`	Sets `window.hlx.lighthouse`

Other query strings pass through links unchanged (`toSitePath` keeps `search` and `hash`), and link URLs with their query strings are pushed to the data layer as `xdm:linkURL`.

9. Logging and debugging safeguards

- `console.error` only for failures (template, auto-blocking, skeleton render, motion, Lottie, Canvas navigation); `console.warn` in the editor rich text support; `console.debug` only behind the `?debug=` flags above. No visitor input is logged. No remote logging exists besides RUM.
- Editor-only hints (empty Fragment, Form without Submit action, list placeholders) render only in Universal Editor (`isEditing()` / `isAuthor()`).
- Analytics never breaks the page: data layer pushes are wrapped in `try / catch` (`utils/analytics.js`).

10. Data minimisation and personal information

Data	Handling	Code
Search terms	Redacted before the data layer: e-mail addresses → `[email]`, phone-like numbers → `[number]`, trimmed, at most 100 characters; empty searches not pushed (unit tested)	`utils/privacy.js`, `test/privacy/privacy.test.mjs`
Form values	Never in the data layer (`cmp:submit` carries field names only); sent only to the page integration or the authored Submit action	`decorators/data-layer-decorator.js`, `blocks/form-variations/form-variations.js`
Labels	`dc:title` values truncated to 100 characters	`decorators/data-layer-decorator.js`
`mailto:` / `tel:` links	Pushed in full as `xdm:linkURL` (exit links): open finding F-05	`decorators/data-layer-decorator.js`
Authored contact data	The Authors and Media Contacts sheets (names, designations, e-mail, phone, LinkedIn) are published as `/{locale}/data/*.json` by design and excluded from the query index (`/*/data/**`)	`utils/data.js`, `helix-query.yaml`
Authored ids and labels	Anchor IDs, Component IDs and Analytics labels are authored and pushed to the data layer: authors must not put personal data in them (handover 11.9)	—
404 RUM checkpoint	`sampleRUM('404', { source: document.referrer })` sends the referrer of sampled 404 views	`404.html`

11. Identity, consent and analytics / personalization dependencies

- **Identity**: no login, no gated content, no visitor id read or set by project code.
- **Consent**: no consent banner, consent management platform or consent check is implemented. The data layer is filled without consent because nothing leaves the page; third-party embeds load without a consent gate. The consent model, categories and per-locale requirements are blocked dependencies (handover 11.7.6 and open item O-07; audit section 7).
- **Analytics**: Adobe Tags and the AEP Web SDK are not configured (blocked: measurement plan); a future embed belongs in `scripts/delayed.js`, never in `head.html` or Universal Editor (Data layer and analytics).
- **Personalization**: no Adobe Target or AJO integration (Personalization).

12. Third-party dependencies

Dependency	Kind	Version	Notes
DOMPurify	Vendored runtime script	3.2.4 (Apache 2.0 / MPL 2.0, header in the file)	`scripts/dompurify.min.js`; Universal Editor only
Lottie (lottie_light)	Vendored runtime script	5.13.0 (MIT, `LICENSE.md`)	`scripts/lib/lottie/`; loaded by `utils/lottie.js` when an animation renders

Dependency	Kind	Version	Notes
npm packages	Development only	See `package.json` devDependencies (ESLint, Stylelint, husky, merge-json-cli, npm-run-all, eslint-plugin-xwalk)	No runtime npm dependencies
Renovate	Dependency updates	`.renovaterc.json`	`config:recommended`; dev dependencies automerge; ESLint pinned to v8
CI	GitHub Actions	`.github/workflows/main.yaml`	`npm ci` and `npm run lint` on push (no `npm test`, no dependency audit step)

Third-party services loaded at runtime: the X widget script, the YouTube / Vimeo players and generic iframe hosts (only when authored), and the RUM collector.

13. Known limitations and open items

Item	Status	Source
Generic iframes: no host allow-list, no `sandbox`	Known limitation; approved embed domains not recorded	`decorators/media-decorator.js`, Embed README, handover O-16
Third-party players and the X widget load without consent	Blocked: consent model	Audit section 7
Form Submit action accepts any URL; no form back end defined	Requires validation (handover O-08: form back end and privacy notices)	`blocks/form-variations/form-variations.js`
`mailto:` / `tel:` links in the data layer	Open (F-05)	Handover Part 16
Raw query in the Search page URL (?q=)	Blocked until Tags / Web SDK exists (R-D-12)	`docs/audit/REAUDIT-2026-10-20.md`
CSP leaves `frame-src`, `connect-src`, `img-src`, `style-src` unrestricted	By design for embeds; Requires validation by a security review	`head.html`
Preview / live access restriction, Admin API roles	Not recorded	Handover 2.7.3

14. Related

- Architecture, Blocks, Indexing and search, Decision log
- Data layer and analytics
- Personalization
- `AGENTS.md` (Security Considerations, Accessibility)
- `utils/README.md`, `decorators/README.md`
- `blocks/embed/README.md`, `blocks/video/README.md`, `blocks/form-variations/README.md`
- `docs/audit/BLOCK-ECOSYSTEM-AUDIT.md` (section 7), `docs/audit/REAUDIT-2026-10-20.md`
- Handover Part 2 — Solution Architecture (2.7), Part 11 — Analytics & Data Layer (11.7.6, 11.9), Part 13 — Global Troubleshooting Guide (13.23, 13.24), Part 16 — Handover Findings and Open Items Register (`TCS-CORP-HANDOVER.md`)

15. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

16. Known limitations

- None recorded for this document.

17. Open items

- None.

18. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Solution Architecture

Attribute	Value
Document ID	ARC-001
Audience	Solution architects, technical leads, developers, AI agents
Purpose	Business and technical context, scope, technology stack and constraints of the solution
Scope	—
Prerequisites	—
Owner	Solution Architect (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/project-overview.md, AGENTS.md
Related documents	—
Related AEM paths	/content/tcs-corp/language-masters/en

This page gives anyone joining the TCS Corporate Website project, human or AI agent, a verified summary of what the project is: its objectives and business value, who it serves, the brand, the 13 locale sites and the temporary language master review, the site structure, how content and code are delivered, the technology stack, what is in and out of scope, and the assumptions, dependencies and constraints it works under. It summarizes and links the detailed sources (the handover Part 1, `AGENTS.md`, `utils/site.js`) rather than copying them.

1. Project at a Glance

Attribute	Value
Project name	TCS Corporate Website (code name `tcs-corp`)
Purpose	The multi-locale corporate web presence of TCS (Tata Consultancy Services)
Platform	Adobe Experience Manager (AEM) as a Cloud Service with Edge Delivery Services
Authoring	Universal Editor (AEM authoring for Edge Delivery Services, "xwalk")
Content source	The AEM as a Cloud Service author, through the xwalk delivery bridge configured in fstab.yaml (type `markup`, suffix `.html`)
Code repository	GitHub `sradha-tcs/tcs-corp`, default branch `main`
Edge Delivery org / site	`sradha-tcs` / `tcs-corp`
Boilerplate	`adobe-rnd/aem-boilerplate-xwalk` (`package.json` still carries the boilerplate name `@adobe/aem-boilerplate` and version `1.3.0`)
Sites	13 locale sites, plus the language master published temporarily for review
Hosts in use	`main--tcs-corp--sradha-tcs.aem.page` (preview) and `main--tcs-corp--sradha-tcs.aem.live` (live)
Production domain and CDN	Not recorded (handover open item O-03, "To be confirmed")

Attribute	Value
Handover	TCS-CORP-HANDOVER.md version 2.37 (PDF: TCS-CORP-HANDOVER.pdf), status "Draft for review"; approvals pending

2. Business and Program Objectives

The handover states that its objectives are derived from the delivered implementation; business KPIs and target values are not recorded in the repository (handover 1.2, BO-9 and SC-9 "To be confirmed").

Id	Objective	How the implementation supports it
BO-1	One corporate web platform for all markets	13 locale sites served by one code base; locale-aware navigation, footer, error pages, UI text, taxonomy and data (utils/site.js)
BO-2	Fast, search-friendly pages	Three-phase loading, image optimisation, self-hosted Inter variable font, module preloading, PageSpeed target of 100
BO-3	Autonomous, efficient authoring	Universal Editor component models for every block, the Additional tab, Site Global Templates, a block library per locale
BO-4	Consistent brand experience	TCS.com Design System (Figma) patterns implemented as blocks; design tokens in styles/styles.css
BO-5	Content discovery	List, Filter List, Cards and Conversation Canvas over the query index and taxonomy
BO-6	Measurable digital experience	Adobe Client Data Layer for every page and block (decorators/data-layer-decorator.js)
BO-7	Inclusive experience	WCAG 2.1 AA target in blocks and shared decorators
BO-8	Low operational overhead	Serverless delivery, Git-based deployment through AEM Code Sync, no build pipeline
BO-9	Business KPIs	Not recorded

Program goals (handover 1.3), with their state on `main` at commit 2d06316:

Id	Goal	State
PG-1	Launch on Edge Delivery Services with Universal Editor authoring	Running on the `aem.page` / `aem.live` hosts; production domain not recorded
PG-2	Block library covering the TCS.com Design System patterns	30 block folders documented in blocks/README.md; Interactive Dashboard is a placeholder (Experimental)
PG-3	13 locale sites with their own references, error pages, sheets and data	Implemented in code and `paths.json`; content status per locale not recorded
PG-4	Templates for recurring page types	Home, Article, Blog, Press Release (templates/README.md)
PG-5	Analytics readiness	Data layer implemented; Adobe Tags, Web SDK, Analytics, CJA, Target and AJO not configured
PG-6	Accessibility (WCAG 2.1 AA)	Implemented in blocks and decorators; a formal third-party audit is not recorded
PG-7	Performance (Lighthouse / PageSpeed 100)	Handover revision 2.36 records Lighthouse 99–100 mobile and 100 desktop on the measured pages; no PageSpeed check runs in CI
PG-8	Handover to the run organisation (AMS)	The handover document; AMS acceptance not recorded

3. Business Value

Summarized from handover 1.5 (Business Benefits); each benefit is traceable to code:

- **Performance and SEO**: pages are rendered by Edge Delivery Services from authored HTML and served from its CDN; only the code of the blocks on the page loads; the first section is rendered first for LCP (`scripts/scripts.js`, `scripts/aem.js`).
- **Faster time to market**: authors build and publish pages from ready blocks; code ships by merging to `main` (AEM Code Sync), without a build or deployment window.
- **Lower cost of ownership**: no delivery application servers, no front-end framework or build chain, one code base for 13 locales.
- **Global consistency with local autonomy**: shared design and blocks; each locale owns its navigation, footer, 404 page, UI text, taxonomy, metadata, redirects and data sheets.
- **Data-driven optimisation, once tags are deployed**: every block and interaction is described in the Adobe Client Data Layer, so Adobe Tags rules could be configured without block changes.
- **Inclusive reach and reuse**: accessibility built into shared decorators; templates add structure (Page Header, About the author, Media contacts) from metadata and sheets.

Why Edge Delivery Services, xwalk and Universal Editor. No verified selection rationale was identified: the handover records the platform as given (Document Control, 1.1) and lists its benefits (1.5), but does not state why it was chosen over alternatives. The handover does state that authors use the Universal Editor and that documents (Google Drive / SharePoint) are not a content source (assumption 5 in 1.8).

4. Target Users

Group	What they do with the solution	Source
Site visitors	Browse the 13 locale sites	Handover 2.2 (Solution Context)
Search engines	Crawl the pages and `/sitemap.xml`	Handover 2.2, helix-sitemap.yaml
Content authors (13 locale teams)	Create, edit, preview and publish pages and sheets in Universal Editor	Handover 2.2, Part 5
Developers	Change code through pull requests, preview on feature-branch hosts	Handover 2.2, AGENTS.md
Site and AEM platform administrators	Edge Delivery site configuration, Admin API, AEM users and groups	Handover 2.2, Part 14
AMS / support	Run support, troubleshooting, monitoring	Handover Parts 13, 14, 17

Visitor segments or personas are not defined: a `persona` metadata column is indexed (helix-query.yaml) and an optional `tcs-corp:audience` tag tree (`cxo`, `it-leaders`, `investors`, `analysts`, `job-seekers`, `media`) is only **proposed** in handover 7.3.2; the published namespace has only `content-type`, `industry` (empty) and `region` (handover 7.3.1).

5. Brands

Brand element	Where it appears	Source
TCS (Tata Consultancy Services)	The site's brand: the TCS logo in the header brand area, linked to its authored TCS Logo Link (usually the locale Home); accessible name from its alt text, else the `tcsLogo` placeholder ("Tata Consultancy Services")	blocks/tcs-header/tcs-header.js, blocks/tcs-header/README.md
TATA Group	The TATA logo in the header tools area (TATA Logo Link, usually `https://www.tata.com/`); accessible name from its alt text, else the `tataLogo` placeholder ("TATA Group")	Same

Brand element	Where it appears	Source
TCS.com Design System	The Figma design the blocks implement ("New Screens from New DS" in the handover revisions)	Handover 1.2 (BO-4), revision history 2.2–2.6; the Figma files are not in the repository

No multi-brand or sub-brand mechanism exists in the code: one brand, one design token set (styles/styles.css) and one header for all locales.

6. Locales

One code base serves every locale; the locale is resolved from the AEM path when editing in Universal Editor, else from the published path (`getLocale` in utils/site.js). Pages outside a locale (Site Global Templates, drafts) use the default locale `en`. paths.json is generated from `LOCALES` by `npm run build:paths` (tools/paths/generate-paths.mjs).

#	AEM path (below /content/tcs-corp)	Published at	<html lang>
1	us/en	/en	en-US
2	de/de-de	/de-de	de-DE
3	fr/fr-fr	/fr-fr	fr-FR
4	ca/fr-ca	/fr-ca	fr-CA
5	es/es-es	/es-es	es-ES
6	pt/pt-pt	/pt-pt	pt-PT
7	nl/nl-nl	/nl-nl	nl-NL
8	se/sv-se	/sv-se	sv-SE
9	dk/da-dk	/da-dk	da-DK
10	no/no-no	/no-no	no-NO
11	cn/zh-cn	/zh-cn	zh-CN
12	jp/ja-jp	/ja-jp	ja-JP
13	kr/ko-kr	/ko-kr	ko-KR

Review roots (KB-D-068). Besides the 13 locales there are three review roots, each published as a site of its own (KB-D-068): `/language-master/en` (testing page functionality with dummy content, content limitations), `/language-masters/en` (the language master: the migrated tcs.com content, validated with its real sheet data, later the source of the global site and the regions and languages; MSM or not is open until the content is final) and `/design-block-master/en` (the block library and the curated design pages); each is published at its own path (`/language-master/en/`, `/language-masters/en/`, `/design-block-master/en/`) as a site of its own. The language master `/content/tcs-corp/language-masters/en`, the rollout source of `/en`, is reviewed this way. Its pages list the language master's own pages from the query index, read its own placeholders, redirects, taxonomy and data sheets and its own references and error pages, and inherit Navigation, Footer and 404 Page up to its root page (`REVIEW_ROOTS`, `REVIEW_SHEETS`, `REVIEW_PAGES`, `indexRoot`, `localeSheetPath`, `localePagePath` in utils/site.js; the `REVIEW` block of tools/paths/generate-paths.mjs: 11 of the 154 mappings and one include). It is to be removed after the rollout (handover 3.9, finding F-88); the rollout date is not recorded. Test: test/site/index-root.test.mjs.

Content status per locale and the launch scope per locale are not recorded (handover PG-3, SC-1 "To be confirmed"). There is no language switcher component and no hreflang (handover 1.6.2).

7. Site Structure

Every locale site has the same layout (details: AGENTS.md, handover Part 3 — Site Structure):

```
/content/tcs-corp/                  site root (not published; "/" needs a redirect to /en/)
├── us/en/                          a locale site; the locale page itself is Home (/en/)
│   ├── …                           content pages
│   ├── references/                 nav (header), footer and other fragments
│   ├── error-pages/                404 (used by 404.html); 403 / 500 for a CDN
│   ├── block-library/              every block variation; not published
│   ├── placeholders, metadata, taxonomy, redirects   sheets / taxonomy page
│   └── data/                       authors, locations, partners, social-links, media-contacts
├── de/de-de/ … kr/ko-kr/           the 12 other locale sites, same layout
├── language-masters/en/             published temporarily at /language-masters/en/ for review
└── site-global-templates/          template pages authors copy; not published
/content/dam/tcs-corp/              AEM Assets (included in paths.json, published with pages)
```

Header, footer and the 404 page are inherited like page properties: the page's own value, else the nearest parent page's up to the locale Home, else the locale's `references/nav`, `references/footer` or `error-pages/404` (utils/inherited-metadata.js). One query index (`/query-index.json`) and one sitemap (`/sitemap.xml`) serve all locales (helix-query.yaml, helix-sitemap.yaml).

8. Delivery Model

Authoring, code and delivery are separate (handover 2.1, 2.6):



Environment	Host	Content	Code
Local development	`http://localhost:3000` (`aem up`)	Previewed content, or `drafts/` with `--html-folder drafts`	Local working copy
Feature preview	`https://{branch}--tcs-corp--sradha-tcs.aem.page/`	Previewed content	Feature branch
Preview (main)	`https://main--tcs-corp--sradha-tcs.aem.page/`	Previewed content	`main`
Live (main)	`https://main--tcs-corp--sradha-tcs.aem.live/`	Published content	`main`
Production	Not recorded (open item O-03)	Published content	`main`

Code is trunk-based: short-lived branches merged into `main` through pull requests; Code Sync deploys every push (handover 12.2, 12.3; AGENTS.md). CI (.github/workflows/main.yaml) runs `npm ci` and `npm run lint` on every push; unit tests and PageSpeed checks are not run by CI. Content is previewed and published from AEM. The only Git tags (`home-content-package_*`) mark earlier states of the `home-content-package` downloads branch (deliverables such as the handover PDF), not code releases. Branch protection, approval workflows and SLAs are not recorded (open items O-05, O-06, O-13).

9. Technology Stack

Layer	Technology	Source
Authoring	AEM as a Cloud Service, Universal Editor, component definitions / models / filters	component-definition.json, component-models.json, component-filters.json, `blocks/*/_*.json`, `models/_*.json`
Delivery	Edge Delivery Services (aem.page / aem.live), path mapping, query index, sitemap	fstab.yaml, paths.json, helix-query.yaml, helix-sitemap.yaml

Layer	Technology	Source
Front end	Vanilla JavaScript (ES modules), CSS3, semantic HTML5; no framework, no transpiling, no build step	scripts/scripts.js, styles/styles.css
Core library	`scripts/aem.js` (never modified), including `sampleRUM` (RUM)	scripts/aem.js
Vendored libraries	DOMPurify 3.2.4 (editor updates), Lottie light player 5.13.0 (Lotto animations)	scripts/dompurify.min.js, scripts/lib/lottie/
Fonts	Self-hosted Inter variable font (Latin `unicode-range`), scaled fallback faces	styles/fonts.css
Analytics	Adobe Client Data Layer, populated centrally; Adobe Tags and the Web SDK not configured	decorators/data-layer-decorator.js, utils/analytics.js
Tooling	Node.js (README: 20 or newer; CI: 24), npm, AEM CLI, ESLint 8 (Airbnb, `eslint-plugin-xwalk`), Stylelint (standard), Husky, Renovate, Node test runner	package.json, .eslintrc.js, .renovaterc.json

No runtime npm dependencies exist; only dev dependencies (package.json).

10. Scope

In scope (implemented on `main`). 13 locale sites and the language master review; 30 block folders (blocks/README.md: 28 Active, Columns Compatibility-only, Interactive Dashboard Experimental); default content through AEM's built-in Text, Title and Image (the built-in Button is "Button (basic)", no longer offered); three section containers (Columns Container, Tabs Container, Gen Carousel); four templates; the Additional tab, animations and skeletons; the data layer; placeholders, metadata, taxonomy, redirects and data sheets per locale; SEO head mapping (decorators/seo-decorator.js); the 404 page with inherited error pages (404.html); WCAG 2.1 AA practices.

Out of scope or not implemented (handover 1.6.2 and 16.4, checked against the code):

Item	State
Adobe Tags, AEP Web SDK, Adobe Analytics, CJA	Not configured; scripts/delayed.js holds only a comment (blocked: measurement plan)
Adobe Target, Adobe Journey Optimizer	No integration in the repository
Consent management	Not implemented
Form submission back end	Not implemented: Form Variations dispatches `form-variations:submit` (finding F-01)
Search service	Not implemented: Conversation Canvas dispatches `conversation-canvas:search`
Production CDN and domain	Not configured; the sitemap origin is the `aem.live` host
Per-locale redirects at the edge, `/` → `/en/`	Not implemented (the platform applies only a root `/redirects.json`)
403 / 500 / maintenance pages served by the platform	Not implemented (only `404.html`)
hreflang, language switcher, breadcrumb, authentication	Not implemented
Interactive Dashboard	Placeholder that renders nothing

11. Assumptions

From handover 1.8 (each still "To be confirmed" where marked there): AEM as a Cloud Service stays the single content source (one mount point in `fstab.yaml`); the site configuration's `public.json` paths match the

repository `paths.json`; bulk metadata applies only once all 13 sheets are listed in `metadata.source`; each locale publishes at least `references/nav`, `references/footer` and `error-pages/404`; authors use Universal Editor, not documents; the site is public and stores no personal data itself; a production domain and CDN will be configured before go-live; analytics, personalisation and form processing come later through the data layer and DOM events; translations are managed in AEM; visitors use current evergreen browsers.

12. Dependencies

Dependency	Used for
AEM as a Cloud Service author, Universal Editor, AEM Assets, Adobe IMS / Admin Console	Authoring, assets, sign-in
Edge Delivery Services (content, media and code bus; Admin API; configuration service)	Rendering and delivery
AEM Code Sync (GitHub App), GitHub `sradha-tcs/tcs-corp`	Code deployment, pull requests, CI
YouTube, Vimeo, X (Twitter)	Embeds and videos, only where authored
Browser Web Speech API	Conversation Canvas speech input, where supported
Node.js, npm, AEM CLI	Developer tooling only
Production CDN / DNS	Not recorded (O-03)

Details and impact if unavailable: handover 1.9.

13. Constraints

Verified constraints the code and content work under (rules in AGENTS.md):

- **No build step**: modules load as authored, so `head.html` modulepreload links and `scripts/module-graph.js` must be regenerated (`npm run build:modules`) after import changes; `npm test` checks them.
- **`scripts/aem.js` is never modified** (boilerplate rule).
- **AEM delivery rules**: empty rows per dialog tab, item first cells, collapsed field suffixes, no grouped fields in key-value blocks, no `classes_*` on sections (container sections use `container*` fields), template values stored on new components, item names in class-name form, non-image DAM files served only once published.
- **Universal Editor**: at most three short dialog tabs (the last one Additional); `multi: true` on `aem-content` fields isn't saved; author has no query index (List and Filter List show an author message).
- **Platform**: only a root `/redirects.json` is applied; bulk metadata needs the site configuration's `metadata.source`; only `404.html` is served as an error page.
- **CSS**: sizes in rem, mobile-first `min-width` breakpoints at 600 / 900 / 1200 px, selectors scoped to the block.
- **Tooling**: ESLint pinned to v8 until airbnb-base supports v9 (.renovaterc.json).
- **Typography**: the Inter font covers Latin only; Chinese, Japanese and Korean text uses the system sans-serif font.

14. Key Terminology

Term	Short meaning
xwalk	AEM authoring (Universal Editor) for Edge Delivery Services
Locale site	One of the 13 country / language sites
Language master	`/content/tcs-corp/language-masters/en`, temporarily published at `/language-masters/en`
Block / section container / item	A decorated component / a section turned into a block / a row added to a block with Add

Term	Short meaning
Additional tab	The last dialog tab of every block except header and footer
Query index	`/query-index.json`, the published pages and their metadata columns
Placeholders	UI text per locale, with English fallbacks in code
Data layer	`window.adobeDataLayer` (Adobe Client Data Layer)

Full definitions: glossary.md.

15. Related

- repository-structure.md: folders and files, purpose and rules
- glossary.md: project, Adobe and handover terms
- references.md: repository documents and cited Adobe documentation
- architecture.md, blocks.md, content-architecture.md, data-layer-and-analytics.md, design-system.md, personalization.md: the other knowledge base topics
- AGENTS.md: verified project rules
- blocks/README.md: block inventory
- utils/site.js: locales and the language master review
- Handover Part 1 — Project Overview, Part 3 — Site Structure, Part 16 — Handover Findings and Open Items Register: TCS-CORP-HANDOVER.md

16. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

17. Known limitations

- Business objectives are carried from the project overview; they are not validated with the business in this program.

18. Open items

- None.

19. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages

Technical Architecture

Attribute	Value
Document ID	ARC-003
Audience	Solution architects, technical leads, developers, AI agents
Purpose	Runtime, block loading, module preloading, templates, authoring and delivery architecture, environments, data and rendering flows, error handling and caching
Scope	—
Prerequisites	—
Owner	Solution Architect (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	8 November 2026
Source files	docs/handover/knowledge-base/architecture.md, scripts/scripts.js, scripts/aem.js, head.html
Related documents	—
Related AEM paths	—

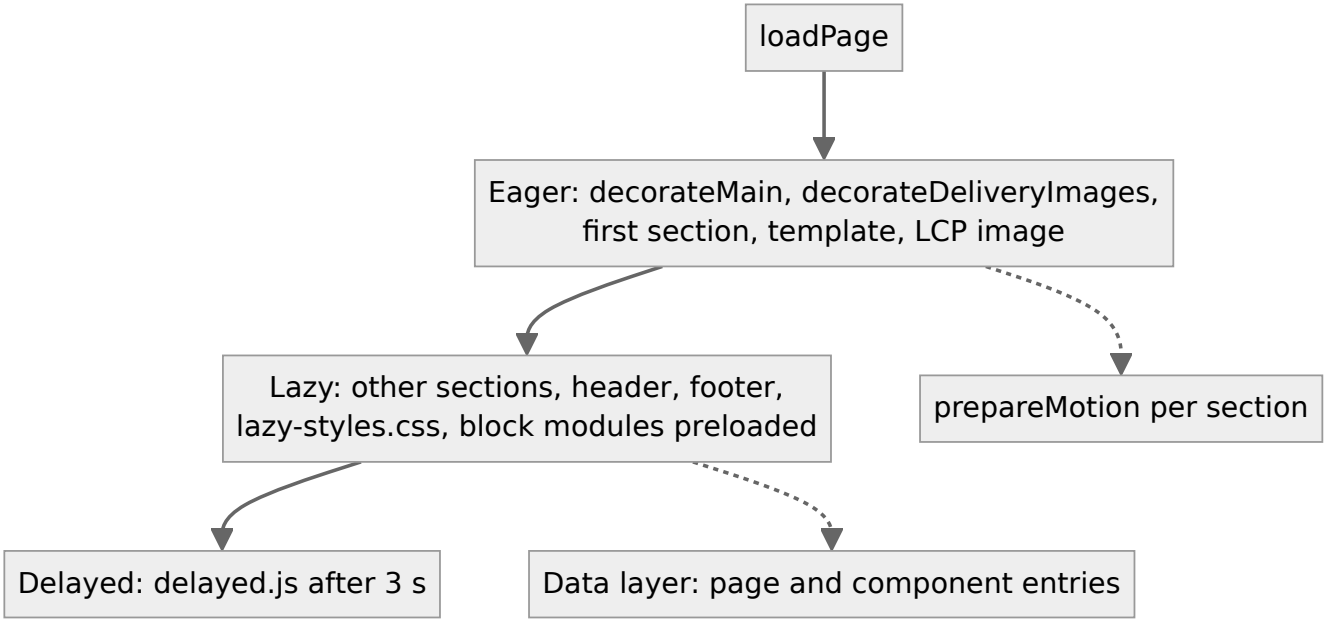


Diagram: three-phase page loading (source: diagrams/page-loading.mmd).

1. Solution summary

The site separates three concerns (AGENTS.md, handover Part 2.1):

Concern	System	Repository artefacts
Authoring	AEM Sites as a Cloud Service author with the Universal Editor (xwalk)	`models/`, `blocks/*/_*.json`, `component-definition.json`, `component-models.json`, `component-filters.json`, `xwalk.json`, `scripts/editor-support.js`
Code	GitHub repository `sradha-tcs/tcs-corp` (based on `adobe-rnd/aem-boilerplate-xwalk`), deployed by AEM Code Sync	`blocks/`, `templates/`, `decorators/`, `utils/`, `scripts/`, `styles/`, `head.html`, `404.html`
Delivery	Edge Delivery Services (content bus, media bus, code bus; preview `*.aem.page`, live `*.aem.live`)	`fstab.yaml`, `paths.json`, `helix-query.yaml`, `helix-sitemap.yaml`, `.hlxignore`

Key characteristics verified in the repository:

- **No build step.** Vanilla ES modules and CSS are served as committed; `.hlxignore` keeps dot files, Markdown, `package*.json`, `test/*`, `docs/*`, `_*` model partials, `LICENSE` and `karma.config.js` off the code bus.
- **Client-side decoration.** Edge Delivery delivers semantic HTML; `scripts/scripts.js` decorates sections and blocks and loads each block's JS / CSS on demand.
- **One site, 13 locales by path.** Locales are defined once in `LOCALES` (`utils/site.js`); `paths.json` is generated from them (`npm run build:paths`). See content-architecture.md.
- **Site data from published sheets.** Placeholders, taxonomy, redirects, data sheets and the query index are fetched as JSON by single-purpose utils, cached per page view.
- **Data layer only.** `window.adobeDataLayer` is populated centrally; Adobe Tags and the Web SDK are not configured.

2. System context



Actor / system	Interaction	Status
Content authors	Edit, preview, publish pages and sheets in AEM with the Universal Editor	Active
Developers	Change code through pull requests; feature-branch preview hosts	Active
Site administrator	Site configuration (`public.json` paths, `metadata.source`), Admin API operations	Active (roles: Not recorded)
Visitors	Browse the 13 locale sites	Active
Search engines	Crawl pages and `/sitemap.xml`	Active (sitemap origin is the `main` live host until a production domain exists)
Production CDN / domain	Production hostname, redirects, 403 / 500 pages	Requires validation (not configured in the repository; handover open item O-03)

3. Application architecture

The code is layered (decorators/README.md, utils/README.md, handover Part 2.3):

Layer	Location	Responsibility
Global head	`head.html`	CSP, viewport, `aem.js`, `scripts.js`, `styles.css`, generated `modulepreload` links

Layer	Location	Responsibility
Core library	`scripts/aem.js`	Edge Delivery boilerplate library: sections, blocks, `loadBlock`, `loadCSS`, `loadScript`, `getMetadata`, `sampleRUM`. **Never modified** (AGENTS.md)
Page lifecycle	`scripts/scripts.js`	`loadEager` / `loadLazy` / `loadDelayed`, `decorateMain`, section options and container sections, templates, 404 loading, preloading, prefetch
Templates	`templates/{home,article,blog,press-release}/`	Structure and styles per Page Type (templates/README.md)
Blocks	`blocks/{name}/`	`{name}.js`, `{name}.css`, `_{name}.json` (models); inventory in blocks/README.md
Decorators	`decorators/`	Create or decorate elements (buttons, links, images, media, data layer, SEO, block options, motion, skeletons)
Utils	`utils/`	No-UI helpers (site / locale, metadata, placeholders, taxonomy, data sheets, query index, redirects, ids, environment)
Styles	`styles/`	`styles.css` (tokens, LCP), `lazy-styles.css`, `fonts.css`, `card-list.css`, `motion.css`
Editor support	`scripts/editor-support.js`, `editor-support-rte.js`, `dompurify.min.js`	Universal Editor live updates (editor context only)

Layering rules (AGENTS.md "Shared code: decorators and utils"): blocks and templates import decorators and utils; decorators may import utils; **utils never import decorators**; one purpose per file; shared behaviour goes into these folders, not into a block.

4. Runtime architecture

Three-phase loading

`loadPage()` in `scripts/scripts.js` runs `loadEager`, then `loadLazy`, then `loadDelayed`.

Phase	Steps (in code order)
Eager (`loadEager`)	`decorateHead()` (SEO head tags and `<html lang>`, `decorators/seo-decorator.js`); `trackInteractions()` (page entry and capture listeners, `decorators/data-layer-decorator.js`); `decorateTemplateAndTheme()`; `decoratePageType()` (Page Type body class); on the 404 page `loadErrorPage(main)`, else `loadTemplate(main)`; `decorateMain(main)`; `reserveHeaderSpace()`; `body.appear`; `preloadBlockModules(first section)`; `loadSection(first)` with `prepareMotion` and `waitForFirstImage` (LCP); `fonts.css` only at 900px and wider or when the session flag `fonts-loaded` is set
Lazy (`loadLazy`)	`autolinkModals` (links containing `/modals/` open the Modal block); `preloadBlockModules(main)` and of the chrome blocks (`header`, `footer`, `fragment`, `tcs-header`, `conversation-canvas`); `prefetchData(main)`; `loadHeader`; `loadFooter` (hidden until the sections have loaded); `loadMainSections` (each section with `prepareMotion`, RUM enhancement after the first); footer shown; `trackBlocks(main)`; `addErrorBackButton` on 404 pages; scroll to the URL hash; `lazy-styles.css`; `loadFonts()`
Delayed (`loadDelayed`)	After 3 seconds: `trackBlocks(document)` (header, footer, late blocks such as modals) and `import('./delayed.js')`. `scripts/delayed.js` contains only a comment (`// add delayed functionality here`)

`decorateMain(main)` runs `decorateIcons` → `buildAutoBlocks` (an empty placeholder) → `decorateSectionsWithOptions` (section options, then the container sections Gen Carousel, Tabs Container and Columns Container wrapped in synthetic blocks) → `decorateBlocks` → `applyBlockOptions` (Additional tab) → `decorateButtons`.

`prefetchData(main)` starts the query index (`getPages`) and the taxonomy (`loadTaxonomy`) at the lazy phase only when the page has a `.list`, `.filter-list` or `.conversation-canvas`, and the redirects sheet (`loadRedirects`) only

with a `.list`. It isn't run in the eager phase so it doesn't compete with the LCP image (code comment). See indexing-and-search.md.

Block loading

From `scripts/aem.js`:

- `decorateBlock(block)` adds `.block`, `data-block-name` (the first class), `data-block-status="initialized"`, wraps text nodes and adds `{name}-wrapper` to the parent.
- `loadBlock(block)` sets `loading`, loads `/blocks/{name}/{name}.css` and imports `/blocks/{name}/{name}.js` in parallel, awaits the default export, catches and logs errors (`failed to load module for {name}`, `failed to load block {name}`) and always ends with `data-block-status="loaded"`.
- `loadSection(section, callback)` loads the section's blocks one after another, runs the callback (here `prepareMotion`), then shows the section.

Module preloading

There is no bundler, so a module's imports are only discovered after it loads (AGENTS.md "Module preloading"). Two generated lists remove these round trips (`npm run build:modules`, `tools/modules/generate-module-graph.mjs`):

- `head.html` has `modulepreload` links for every module `scripts.js` imports (between `<!-- modulepreload -->` markers).
- `scripts/module-graph.js` lists, per block, the modules it imports beyond `scripts.js` and whether it has a stylesheet; `preloadBlockModules()` adds `modulepreload` links (head) and stylesheet `preload` links (body, because `loadCSS` skips a stylesheet already linked in the head).

`npm test` includes `test/modules/module-graph.test.mjs`; AGENTS.md states the tests fail when the lists are out of date.

Templates

`loadTemplate(main)` reads the Page Type (`page-type` metadata, else `template`) and, when it is one of `TEMPLATES = ['home', 'article', 'blog', 'press-release']`, imports `templates/{name}/{name}.js` and loads its CSS, then calls the default export with `main` before decoration. Failures are logged (`failed to load template {name}`) and the page continues. What each template adds: templates/README.md and content-architecture.md.

5. Authoring architecture

Element	Implementation	Source
Editor	AEM Universal Editor on the AEM author; pages under `/content/tcs-corp`, assets under `/content/dam/tcs-corp`	handover Part 2.9.1–2.9.2, Part 5
Component palette	`component-definition.json`: groups Default Content (4), Sections (1), Blocks (57 components)	generated by `npm run build:json` from `models/_component-definition.json` and `blocks/*/_*.json`
Dialogs	`component-models.json` (fields per model)	generated from `models/_component-models.json` and partials
Allowed children	`component-filters.json`: `main` allows `section`, `gen-carousel`, `tabs`, `columns-container`; `section` allows 26 components	generated from `models/_component-filters.json` and partials
Page properties	`models/_page.json` (model `page-metadata`): Title, Description, Keywords, Page Type, Navigation, Footer, 404 Page	content-architecture.md
Section properties	`models/_section.json`: Section Name, Style, Width Type, Rounded edge, Background Image, Image alt text, Anchor name	`scripts.js` `decorateSectionsWithOptions`

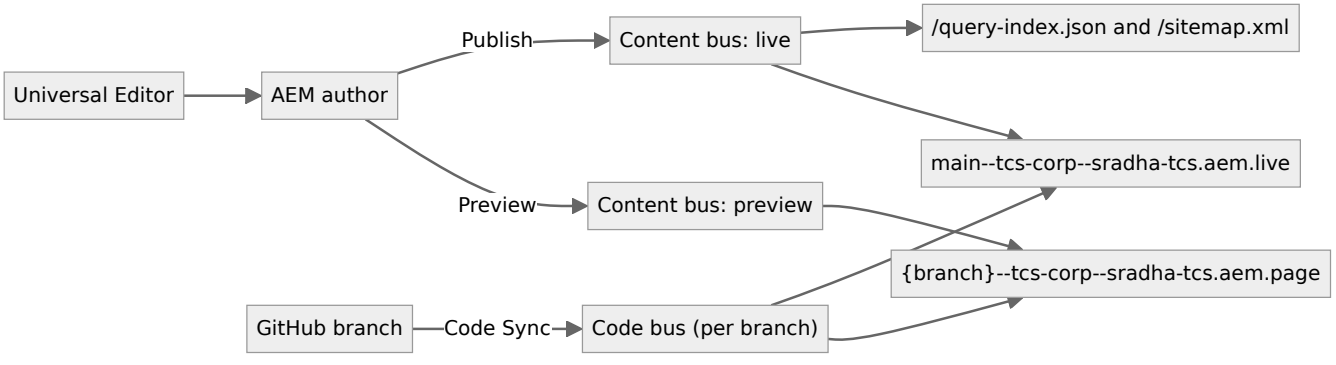
Element	Implementation	Source
xwalk features	`xwalk.json`: `public.xwalk.multi-field.enabled: true`	AGENTS.md: repeating content uses item children, not composite multifields
Live editing	`scripts/editor-support.js`	below

`scripts/editor-support.js` listens on `main` for `aue:content-patch`, `-update`, `-add`, `-move`, `-remove` and `-copy`. For each change it loads `scripts/dompurify.min.js`, sanitises the returned HTML (`USE_PROFILES: { html: true }`) and re-decorates only the changed `main`, block, section or default content (`decorateMain`, `decorateBlock`, `applyBlockOptions`, `decorateSectionsWithOptions`, `loadBlock`, `loadSections`); when an update can't be applied in place, the page reloads. `editor-support-rte.js` handles rich-text instrumentation, re-run by a `MutationObserver`. The file isn't referenced by `head.html` or `scripts.js`; handover Part 2.3.6 states it is loaded only in the Universal Editor context (the loading mechanism is outside this repository: Requires validation).

Authoring rules that shape code (block rows, field collapse, item rows, sections without `classes_*`): AGENTS.md "AEM delivery rules", the block READMEs and docs/audit/README.md.

6. Publishing and delivery architecture

Element	What it does	Source
`fstab.yaml`	One mount point `/` of type `markup`, suffix `.html`, pointing to the AEM author's `bin/franklin.delivery/sradha-tcs/tcs-corp/main` bridge (host not reproduced here)	`fstab.yaml`
`paths.json`	42 mappings (39 for the 13 locales: folder, Home page, taxonomy page each; 3 temporary for the language master under review; sheets are covered by the folder mappings), 15 includes (13 locale pages, `/content/dam/tcs-corp/`, `/content/tcs-corp/language-masters/en`), 28 excludes (13 × 2 block-library entries, `site-global-templates` and its children); generated by `npm run build:paths`	`paths.json`, `tools/paths/generate-paths.mjs`
Site configuration	AEM uses the configuration's `public.json` → `paths`, not the repository file; `metadata.source` lists the bulk metadata sheets	AGENTS.md "Notes"; handover Part 2.10.13
Preview / live	Preview writes to the preview partition (`*.aem.page`), publish to the live partition (`*.aem.live`); images go to the media bus (`/media_<hash>`)	handover Part 2.5.2, 2.10.3
Code Sync	Every push of every branch updates the code bus for that branch	handover Part 2.5.1, 2.10.5
CI	`.github/workflows/main.yaml` (Build): `npm ci`, `npm run lint` on every push; unit tests are not run in CI (handover F-37)	`.github/workflows/main.yaml`



Publishing details (what must be published separately, sheets, rollback): handover Part 12 — Publishing, Deployment & Release.

7. Content architecture (summary)

Pages consist of sections holding default content and blocks; page and bulk metadata arrive as `<meta>` tags; site data (placeholders, taxonomy, redirects, data sheets) is published per locale as JSON; header, footer and 404 pages are reference pages resolved by inherited metadata. Full description: content-architecture.md. Query index and listings: indexing-and-search.md.

8. Component architecture

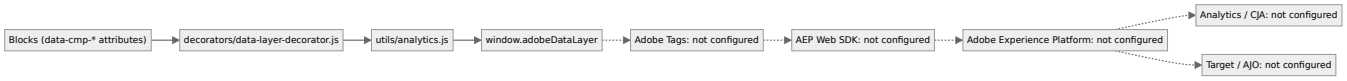
Kind	Examples	How it runs
Authored blocks	Hero, Teaser, List, Cards, Filter List, Accordion, Table, Video, Form Variations, Conversation Canvas	`decorateBlocks` + `loadBlock`; inventory and status in blocks/README.md
Structural blocks	`header`, `footer`	built by `loadHeader` / `loadFooter` (aem.js) in the lazy phase; content from reference pages (header, footer)
Section containers	Gen Carousel, Tabs Container, Columns Container	the section is wrapped in a synthetic block by `buildCarouselSections`, `buildTabsSections`, `buildColumnsSections`; the container fields arrive as section data attributes and `applySectionBlockOptions` turns `container*` fields into block classes
Fragments	Fragment block, nav and footer pages, modals, tab fragments	`loadFragment(path, from)` in `blocks/fragment/fragment.js` fetches `{path}.plain.html` and decorates it; recursion stopped by the chain of including fragments (fragment)
Additional tab	Anchor ID, Hide block, spacing, animation	`classes_*` fields as block classes; `decorators/block-options-decorator.js`, `decorators/motion-decorator.js`, `styles/styles.css`
Shared modules	28 decorators, 26 utils	decorators/README.md, utils/README.md

Shared events (AGENTS.md "Shared events and helpers"): `tabs:reveal` (opens the tab holding its target), `header:menu` (header menu state for the Canvas journey), `form-variations:submit` (with `detail.respondWith(promise)`), `conversation-canvas:search` (cancelable search hand-off).

9. Integration architecture

Integration	Mechanism	Status
AEM author → Edge Delivery	franklin.delivery bridge (`fstab.yaml`), preview / publish	Active
GitHub → Edge Delivery	AEM Code Sync	Active
AEM Assets	Asset picker; images on the media bus; `/content/dam/tcs-corp/` in `paths.json` includes; non-image DAM files are served only once published (AGENTS.md)	Active
AEM Assets, separate environment (EDAM)	Dynamic Media with OpenAPI delivery URLs in the content; standalone image delivery links rendered as responsive pictures (`decorateDeliveryImages`, `decorators/responsive-image-decorator.js`); the Universal Editor Custom Content Advisor for picking (remote-assets-edam.md)	Not adopted (KB-D-064, 8 November 2026): no separate Assets environment, the assets stay in the Sites author's DAM; the rendering of delivery URLs stays in the code (no effect on DAM paths)
Query index and sitemap	`helix-query.yaml`, `helix-sitemap.yaml`	Active (indexing-and-search.md)

Integration	Mechanism	Status
Adobe Client Data Layer	`window.adobeDataLayer` populated by `decorators/data-layer-decorator.js` and `utils/analytics.js` (Core Components schema: page, components, `cmp:*` events)	Active
Adobe Tags (Data Collection)	Would be embedded from `scripts/delayed.js`	Not configured (blocked: measurement plan)
AEP Web SDK	Through Adobe Tags	Not configured (blocked: measurement plan)
Adobe Analytics, CJA, Target, AJO	—	No integration exists in the repository
Consent	—	No consent mechanism is implemented
Form back end	`form-variations:submit` event; no endpoint in code	No verified back end was identified
Search service	`conversation-canvas:search` event, else the authored Search page with `?q=`	No verified search service was identified
Third-party media	YouTube (incl. `youtube-nocookie.com`), Vimeo, X widgets, only for authored embeds	Active
Operational telemetry	`sampleRUM` in `scripts/aem.js`; `404.html` sends a 404 checkpoint	Active



Details: AGENTS.md "Data layer", handover Part 11 — Analytics & Data Layer.

10. Deployment

1. Develop locally with `aem up` (`http://localhost:3000`, local code with previewed content; `--html-folder drafts` for static drafts).
2. Run `npm run lint` and `npm test`; after model changes `npm run build:json`, after import changes `npm run build:modules`, after locale changes `npm run build:paths`.
3. Push a feature branch: Code Sync serves it at `https://{branch}--tcs-corp--sradha-tcs.aem.page/`; the `Build` workflow lints it.
4. Open a pull request with a feature preview URL (AGENTS.md "Publishing Process"); a reviewer merges to `main`.
5. Code Sync updates `main`; preview and live hosts serve the new code (code isn't versioned per content publication).
6. Configuration that lives outside the repository (`public.json` paths, `metadata.source`) is applied by the site administrator.

Rollback: revert the commit on `main` (handover Part 12.6).

11. Environment topology

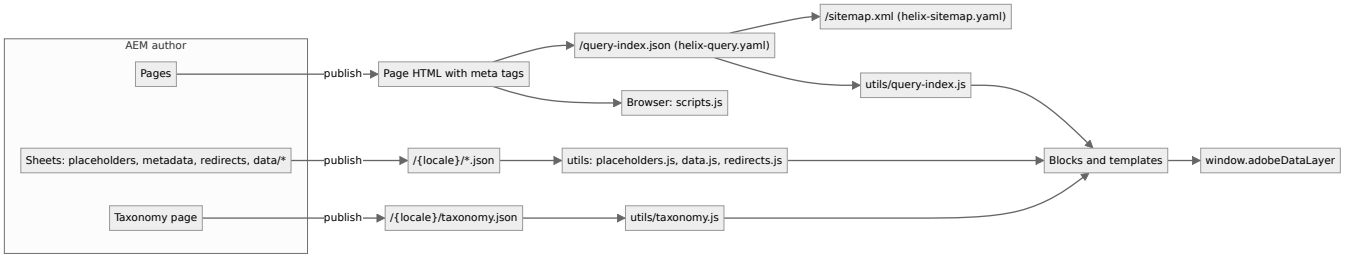
Environment	Host pattern	Content	Code
Local	`http://localhost:3000`	Previewed content, or `drafts/`	Working copy

Environment	Host pattern	Content	Code
Feature preview	`https://{branch}--tcs-corp--sradha-tcs.aem.page/`	Previewed	Feature branch
Preview (main)	`https://main--tcs-corp--sradha-tcs.aem.page/`	Previewed	`main`
Live (main)	`https://main--tcs-corp--sradha-tcs.aem.live/`	Published	`main`
Production domain	Not recorded (not configured; `helix-sitemap.yaml` uses the live host as origin until `cdn.prod.host` is set)	Published	`main`
AEM author	Not reproduced (confidential host)	Source content	—
Other AEM environments	Not recorded	—	—

12. Security boundaries

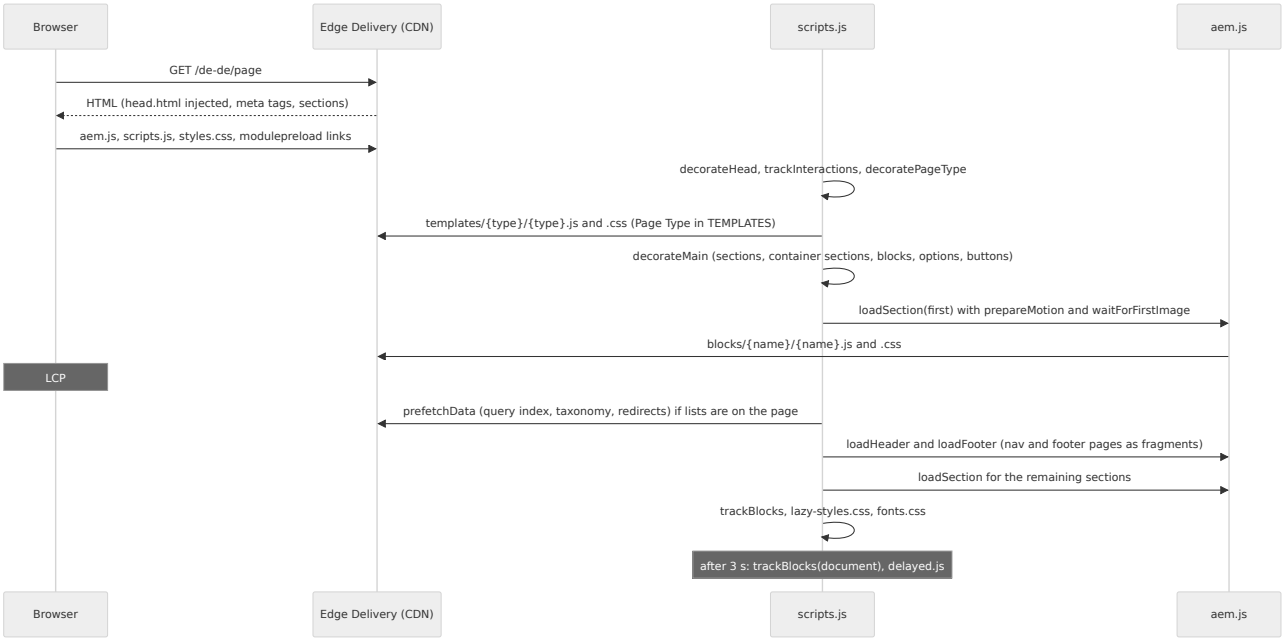
Boundary	Control in the repository	Source
Script execution	CSP `script-src 'nonce-aem' 'strict-dynamic' 'unsafe-inline' http: https:; base-uri 'self'; object-src 'none'` with `move-to-http-header="true"` in `head.html` and `404.html`; only nonce scripts and what they load run	`head.html`, `404.html`
Not restricted by CSP	`style-src`, `img-src`, `frame-src`, `connect-src` aren't set	`head.html`; handover Part 2.7.1
Public code	Everything not in `.hlxignore` is public on the code bus; no secrets in code (AGENTS.md "Security Considerations")	`.hlxignore`
Public data	Every published sheet and `/query-index.json` is public (Authors and Media Contacts sheets hold names, e-mail addresses and phone numbers by design, handover Part 2.10.3)	content-architecture.md
Editor updates	Universal Editor HTML is sanitised with DOMPurify before re-rendering; DOMPurify is loaded only by `editor-support.js`	`scripts/editor-support.js`
Fragments and error pages	`loadFragment` fetches only paths starting with `/` and not `//`; `loadPageContent` (404) fetches same-origin `.plain.html`; `innerHTML` is used only for those responses	`blocks/fragment/fragment.js`, `scripts/scripts.js`
Links	`isSafeUrl` (`utils/card-list-config.js`) allows `http:`, `https:`, `mailto:`, `tel:` and relative paths in List and Cards; new-window links get `rel="noopener noreferrer"` (`decorators/link-decorator.js`)	block READMEs
Analytics privacy	`cmp:submit` carries field names only; search terms pass `redactSearchTerm` (`utils/privacy.js`: e-mail and phone patterns replaced, 100 characters)	`decorators/data-layer-decorator.js`
Unpublished areas	Block libraries and Site Global Templates are excluded in `paths.json`	`paths.json`
Visitor identity	No login, no gated content, no consent mechanism, no cookies set by project code	handover Part 2.7.3
Admin operations	Admin API calls require authorised users; roles are configured outside the repository	Not recorded (handover open item)
Preview / live access protection	—	Not recorded

13. Data flow



Flow	Source → consumer	Detail
Page metadata	Page properties and bulk metadata → `<meta>` → `getMetadata` / `utils/metadata.js` → SEO decorator, data layer, templates, Page Header	content-architecture.md
Inherited metadata	Parent pages' `nav`, `footer`, `error-page` metas → `utils/inherited-metadata.js` → header, footer, 404	parent heads loaded in parallel, cached in `sessionStorage`
UI text	`/{locale}/placeholders.json` → `getPlaceholders()` → every block with UI text	English fallback in code
Tag labels	`/{locale}/taxonomy.json` → `resolveTags()`	ID-derived labels as fallback
Listings	`/query-index.json` → `getPages()` → List, Filter List, Conversation Canvas menu	indexing-and-search.md
Analytics	Blocks' `data-cmp-*` attributes → data layer decorator → `window.adobeDataLayer`	no outbound collection configured
Telemetry	`sampleRUM` checkpoints → operational telemetry collector	platform

14. Rendering flow



15. Error handling

Failure	Behaviour	Where
Block module fails to load or `decorate()` throws	Logged (`failed to load module for {name}`); block marked `loaded`; the section and page continue (block isolation)	`loadBlock`, `scripts/aem.js`
Template fails	Logged (`failed to load template {name}`); the page renders without it	`loadTemplate`, `scripts/scripts.js`
Missing page	Edge Delivery serves `404.html` (`window.isErrorPage`, `window.errorCode = '404'`); `loadErrorPage` tries the inherited 404 Page, then `/{locale}/error-pages/404`, then `/en/error-pages/404`; a page without content is skipped; else the built-in "404 / Page Not Found / Go home" stays; content from another locale gets its `lang` on `<main>`; "Go back" is added for same-origin referrers	`scripts/scripts.js` `loadErrorPage`, `loadPageContent`, `addErrorBackButton`
403 / 500	Not served by Edge Delivery; authored `error-pages/403` and `500` exist for a CDN	Requires validation (CDN not configured)
Sheet or index request fails	Each util resolves to empty data: placeholders → English fallbacks, taxonomy → ID-derived labels, data sheets → empty lists, redirects → none, query index → no pages	`utils/*.js` (`.catch(() => …)`)
Navigation page fails	Falls back to the locale's `references/nav`; none published: header left empty	`blocks/header/header.js`
Fragment can't load (unpublished, path not starting with `/`, self-inclusion)	The Fragment block keeps its authored content (the reference link); in Universal Editor edit mode it shows the hint `fragmentEmpty`	`blocks/fragment/fragment.js`
Slow data	`renderWhenLoaded` shows a skeleton after 150ms (at least 250ms) and removes it on success or failure	`decorators/skeleton-decorator.js`
Universal Editor update can't be applied in place	Page reloads	`scripts/editor-support.js`
Analytics push error	Swallowed (`pushToDataLayer` in `try / catch`)	`utils/analytics.js`
Storage unavailable	Ignored (`sessionStorage`, `localStorage` in `try / catch`)	`scripts.js`, `utils/inherited-metadata.js`, `utils/header-space.js`

Troubleshooting by symptom: handover Part 13 — Global Troubleshooting Guide.

16. Caching

Cache	Holds	Lifetime / invalidation	Source
Edge Delivery CDN	HTML, `.plain.html`, sheet JSON, media, code	Purged for the path on preview / publish and on code sync; manual purge with the Admin API cache operation	handover Part 12.1.8 (platform behaviour)
Production CDN	Live responses	Not recorded (not configured)	—
Query index rows	`/query-index.json`, all locales	One promise per page view (module variable `pages`)	`utils/query-index.js` `getPages`
Pages read from HTML	Rows for pages missing from the index (AEM author)	Map per page view	`utils/query-index.js` `getPagesAt`

Cache	Holds	Lifetime / invalidation	Source
Placeholders, taxonomy, redirects	Locale sheets	One promise per page view	`utils/placeholders.js`, `utils/taxonomy.js`, `utils/redirects.js`
Data sheets	Authors, locations, partners, social links, media contacts	Map per sheet per page view	`utils/data.js`
Inherited metadata	`nav`, `footer`, `error-page` of parent pages	In memory per page view and `sessionStorage` key `tcs-inherited-metadata` for the browser session (a changed parent value shows in a new session)	`utils/inherited-metadata.js`
Font flag	`fonts-loaded`	`sessionStorage`, browser session; not set on `localhost`	`scripts/scripts.js`
Header height	`header-height-v2-{breakpoint}` (rem)	`localStorage`, no expiry; refreshed when the header loads	`utils/header-space.js`
Header overlay mode	`header-overlay-v1` (`always` / `never`; removed for Auto)	`localStorage`, no expiry; set when the header's TCS Header loads, applied in the eager phase (`<html data-header-overlay>`)	`utils/header-space.js`
Preloads	Module and stylesheet paths already preloaded	Set per page view	`scripts/scripts.js` `preloadedModules`

17. Related

- content-architecture.md — sites, locales, page structure, metadata, sheets, references
- indexing-and-search.md — query index, listings, sitemap, reindexing
- data-layer-and-analytics.md, blocks.md, repository-structure.md, project-overview.md — related knowledge base files
- AGENTS.md — project rules (delivery rules, three-phase loading, module preloading, data layer)
- blocks/README.md, decorators/README.md, utils/README.md, templates/README.md
- docs/audit/README.md, docs/animation/README.md
- Handover (TCS-CORP-HANDOVER.md): Part 2 — Solution Architecture, Part 11 — Analytics & Data Layer, Part 12 — Publishing, Deployment & Release, Part 13 — Global Troubleshooting Guide, Part 16 — Handover Findings and Open Items Register

18. Screenshots

No screenshot belongs to this document (it describes code, configuration or process). Evidence screenshots are listed in the screenshot register.

19. Known limitations

- None recorded for this document.

20. Open items

- None.

21. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages